Dermatology - Previous Year Questions

Marrow Q-Bank Collection • NEET PG / INI-CET / AIIMS (2017-2025)

Total Questions: 152
High-Yield Study Edition

Q1. Dermatology INI-CET 2021 ID: MF0837

A forest officer develops the lesion as shown in the image. Which of the following is not a differential to consider?

A Scrub typhus

B ✓ KFD

C Cutaneous anthrax

D Healing brown recluse spider bite

Correct Answer: KFD

The given image shows the presence of **eschar** and **Kyasanur Forest Disease (KFD)** is not a differential diagnosis for the presence of eschar.

Kyasanur Forest Disease is a **hemorrhagic fever** found in the state of **Karnataka**, India. KFD has an abrupt onset of fever, headache, conjunctivitis, myalgia, and severe prostration. Characteristic skin lesions that are associated with KFD are **petechial lesions.** Some cases may develop hemorrhages in the skin, mucosa, and viscera, but there is **no eschar formation**.

Infection is transmitted by the bite of **ticks**, the principal vector being ***Haemaphysalis spinigera***

Other options:

A) Scrub typhus:

- **Scrub Typhus** is also known as chigger-borne typhus.
- Scrub typhus is caused by ***Orientia tsutsugamushi***.
- The vector is **trombiculid mite**.
- Humans are infected when they are bitten by the mite larvae known as **chiggers.**

The incubation period is 1- 3 weeks. The patient typically develops a characteristic **eschar** at the site of the mite bite as shown in the image below, with **regional lymphadenopathy** and maculopapular rash.

C) Cutaneous anthrax:

- In **cutaneous anthrax**, the lesion starts as a **papule** 1-3 days after infection and becomes **vesicular**, containing fluid that may be clear or blood-stained.
- The whole area is congested and edematous and several **satellite lesions** filled with serum or yellow fluid are arranged around a **central necrotic lesion** which is covered by a **black eschar**.
- This lesion is called a **malignant pustule**.
- The disease used to be common in dock workers, carrying loads of hides and skins on their bare backs and hence known as **hide porter's disease.**

D) Healing brown recluse spider bite:

- The bite of the **Loxosceles reclusa**, the brown recluse spider is usually relatively painless.
- After minutes or hours, severe pain develops at the site, accompanied by erythema, edema, and a central bulla.
- In severe envenomation, a '**target lesion**' is seen- central blue/ purple discoloration surrounded by an ischemic halo and an outer ring of erythema (the '**red, white and blue' sign**).
- After 3-4 days, the central area becomes necrotic and an **eschar** develops.
- The eschar sheds eventually, leaving behind an ulcer, which takes time to heal.

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References: Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition, Page no: 530, 415, 249, 250
Rook's Textbook of Dermatology 9th edition, Page no: 34.33, 34.34

Q2. Dermatology INI-CET 2021 ID: MF0861

A patient presents with following lesions on the genitals. Scrapings from the lesion would show?

A ✓ Tzanck cells

B Cells with clear halo around the nucleus.

C Molluscum bodies.

D HP bodies.

Correct Answer: Tzanck cells

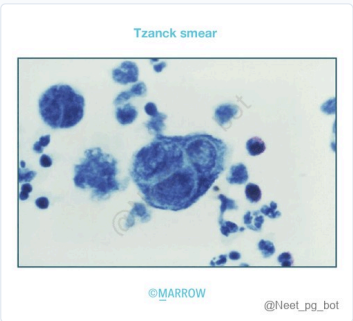
The given image shows **clusters** of **vesicles** seen on an **erythematous base** suggesting a diagnosis of **genital herpes. Tzanck cells** or multinucleated giant cells with intranuclear bodies are present on microscopy. Intranuclear inclusion bodies are seen when the smears are stained with Giemsa stain.

The typical herpes lesions are **thin-walled, umbilicated vesicles,** the roof of which breaks down, leaving tiny superficial **ulcers.** They heal without scarring. The most common site of cutaneous lesions in the face - on the cheeks, chin, around the mouth, or on the forehead.

The Tzanck smear technique is a rapid and inexpensive diagnostic method. Scrapings (preferably from the **base of vesicles**) are obtained from the lesions and smears are prepared. When stained with 1% aqueous solution of **O-toluidine blue** for 15 seconds, **multinucleated giant cells** with faceted nuclei and homogenously stained ground glass chromatin known as ' **Tzanck cells**' are seen.

Genital herpes:

Herpes Simplex Virus (HSV) 2 is more responsible for genital herpes than HSV 1. The vesiculo-ulcerative lesions are very painful. In men, it occurs mainly on the penis or urethra. In women, the cervix, vagina, vulva and perineum can all be involved. The primary infection causes serious symptoms accompanied by fever and malaise, followed by recurrent episodes.



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References: Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition, Page no: 474- 476

Q3. Dermatology INI-CET 2021 ID: MF0860

A young girl presents with leukotrichia and lesions as shown in the image. What is the most likely diagnosis?

A Focal vitiligo

B ✓ Segmental vitiligo

C Piebaldism

D Nevus depigmentosus

Correct Answer: Segmental vitiligo

The image shows areas of **localized, unilateral depigmented macules.** This feature, along with a history of leukotrichia suggest that the patient is suffering from **segmental vitiligo**. It occurs due to a **lack of melanocytes** in the basal layer of the epidermis.

It is a localised depigmentation that occurs due to a **progressive loss in the melanocytes.** It is classified as

- **Segmental vitiligo**- which appears as **unilateral** maculae in a segmented or band-shaped pattern
- **Non-segmental vitiligo**- which appears as **bilateral** maculae distributed in an acrofacial pattern or scattered symmetrically over the entire body

The lesions become apparent between 20-30 years of age.

Hypopigmented macules in vitiligo are commonly found in the **hips, dorsum of the hands/fingers**, **feet, elbows, knees, and ankle areas.** These are the areas of repeated friction. Rarely there may be complete vitiligo (vitiligo universalis), or trichrome vitiligo where there are areas of depigmentation, hyperpigmentation, along with the normal skin are present. The **Koebner phenomenon** is the development of **lesions at the site of trauma** on the uninvolved skin.

Several autoimmune conditions can be associated with vitiligo. They are:

- Thyroid disease (hyperthyroidism and hypothyroidism)
- Pernicious anemia
- Addison disease
- Diabetes
- Hypoparathyroidism

Under the Wood's lamp examination the depigmented areas emit a bright **blue-white fluorescence** and **appear sharply demarcated.**

Treatment involves cosmetic camouflage and the use of sunscreen. The topical application of **corticosteroids** is the **first line of treatment**. Topical calcineurin inhibitors like pimecrolimus, tacrolimus have also shown to be successful. **Psoralen photochemotherapy,** khellin, selective **ultraviolet B therapy** are used as **second-line treatment**. In patients who do not respond to this treatment, surgical grafting techniques can be used, where the melanocytes from a normal pigmented area are transferred to a non pigmented area. In patients with extensive vitiligo where only a few pigmented areas are left, depigmentation can be done.

Other options:

A) **Focal vitiligo**:

These are small **isolated** depigmented lesions that are neither segmentally distributed nor evolved into non-segmental lesions after 1-2 years. The focal vitiligo along with isolated mucosal lesions on one site are considered as **undetermined /unclassified vitiligo.**

C) **Piebaldism:**

It commonly presents with a white **V-shaped forelock.** it is caused due to the **impaired migration of melanocytes** to the skin, resulting in hypopigmented patches.

D) **Nevus depigmentosus:**

It is a **congenital** pigmentary disorder that presents with a non-progressive **hypopigmented macule**. It usually is present from birth or early childhood and remains stable throughout life.

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References: Rook's Textbook of Dermatology 9th edition, P - 88.34

Q4. Dermatology INI-CET 2021 ID: MF0871

The most specific and typical sign of juvenile dermatomyositis in a 17-year-old patient is

A Photosensitive skin rash.

B Nail bed capillary changes.

C ✓ Gottron's papules.

D Malar rash.

Correct Answer: Gottron's papules.

Gottron papules are typical and specific sign of **juvenile dermatomyositis**.

Gottron papules and a heliotropic rash of the eyelids are the classical criteria that are required for the diagnosis of juvenile dermatomyositis along with three of the following:

Weakness	Symmetric Proximal
Muscle enzyme elevation (≥1)	Creatine kinase Aspartate aminotransferase Lactate dehydrogenase Aldolase
Electromyographic changes	Short, small polyphasic motor unit potentials Fibrillations Positive sharp waves Insertional irritability Bizarre, high-frequency repetitive discharges
Muscle biopsy	Necrosis Inflammation

Gottron's papules are **violaceous** (of violet color), **lichenoid** (red-purple), inflammatory, flat-topped **papules** on proximal interphalangeal joints, distal interphalangeal joints, and metacarpophalangeal joints.

This should not be confused with Gottron's sign.

Gottron's sign is erythematous or violaceous, **macules** or **patches** in the same symmetric distribution pattern but **sparing** the **interphalangeal spaces** and sometimes involving **elbows and knees**.

The **heliotrope rash** is a violet discoloration of the eyelids that may be associated with periorbital edema. Rarely, an erythematous, scaly rash develops on the palms of the children called '**mechanic hands**'. Nail bed capillary changes are visible as individual thickened and telengiectatic capillary loops, which can be well seen with the help of capillaroscopy. Facial erythema mimicking **malar rash** of systemic lupus erythematosus, but with the involvement of **nasolabial folds** are also common.

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References: Nelson textbook of pediatrics 21st edition, Page no: 1182, 1183

Q5. Dermatology INI-CET 2021 ID: MF0841

A child presents to you with itchy lesions on the scalp with hair loss as shown in the image below. What would be the next best step?

A ✓ KOH mount

B Slit skin mount

C Gram staining

D Biopsy

Correct Answer: KOH mount

The above image shows **tinea capitis,** which presents with a well-demarcated scaly patch in which hair shafts are broken off just above the skin resulting in alopecia. **KOH mount** is useful for the depicted condition.

The scrapings should be taken from the edge of a lesion and transferred to a slide, to which **KOH** is added and examined under the microscope for the presence of hyphae. Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area.

The clinical variants of tinea capitis are:

- **Endothrix** - a non-inflammatory type in which **multiple black dots** are present within the areas of alopecia. It is most commonly caused by **T. tonsurans and T. violaceum.**
- **Ectothrix** - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. **Microsporum audouinii**, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.
- **Kerion** - a severe inflammatory type that presents with an inflammatory mass with **thick crusting** and **matting of hair** with pus and sinus formation. It is commonly caused by **T. verrucosum and T. mentagrophytes**.
- **Favus** - is characterized by the presence of yellowish, cup-shaped crusts known as **scutula** that develop around hair follicles. It is caused by **T. schoenleinii**.

Treatment:

- **Terbinafine**: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.
- **Itraconazole**: 2-4 mg/kg/day for 4-6 weeks.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 32.38- 32.40

Q6. Dermatology INI-CET 2021 ID: MF0855

The below-given image is characteristically seen in which of the following type of leprosy?

A Borderline lepromatous leprosy

B Lepromatous leprosy

C ✓ Borderline leprosy

D Histoid leprosy

Correct Answer: Borderline leprosy

The given image shows a case of borderline tuberculous leprosy, or dimorphous leprosy, as evidenced by the cutaneous, rash-like discoloration of the patient's skin. This generally overlies the back, and crosses the midline. So, the best choice would be Borderline leprosy.

Microscopically, there is diffuse epithelioid cell granuloma with very scanty lymphocytes and no giant cells; the papillary zone is clear and the nerves are swollen slightly by cellular infiltrate. Acid-fast bacilli are present in **moderate** numbers.

The borderline leprosy patients are **immunologically unstable** and are at an increased risk of developing immune-mediated reactions. **Type 1 lepra reactions** that occur most commonly in mid-borderline leprosy patients, are **delayed hypersensitivity** reactions caused by increased recognition of mycobacterial leprae antigens in skin and nerve sites.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 28.3, 28.5, 28.10

Q7. Dermatology INI-CET 2021 ID: MF0848

A child presented with itchy plaques over the neck, the bilateral popliteal and cubital fossa. What could be the diagnosis?

A ✓ Atopic dermatitis

B Dermatitis herpetiformis

C Pemphigus vegetans

D Psoriasis

Correct Answer: Atopic dermatitis

The given clinical history of itchy plaques over the neck, cubital fossa and popliteal fossa suggests the diagnosis of '**atopic dermatitis**'.

Atopic dermatitis is the cutaneous expression of the atopic state, characterised by a family history of asthma, allergic rhinitis or eczema.

The etiology is only partially defined, but there is a clear genetic predisposition. A characteristic defect is impaired epidermal barrier. In many patients a mutation in gene encoding **filaggrin** (a structural protein in the stratum corneum) is responsible. The lesions are distributed in the **flexural aspects**.(as shown in the image).

Patients may display many immunoregulatory abnormalities, including increased IgE synthesis, increased serum IgE levels, delayed type hypersensitivity reactions.

The cardinal symptom is itching, along with rash.

Atopic Eczema can be divided into 3 phases: Infantile, Childhood, and Adult phases.

Infantile phase: The lesions of Atopic Eczema most frequently start on the **face** but may occur anywhere on the skin surface. Often, the **napkin area** is relatively **spared.**

- When the child begins to crawl, the exposed surfaces, **especially the extensor aspect of the knees and elbows** are most involved.
- The lesions consist of **erythema** and discrete or confluent **oedematous papules**. The papules are **intensely itchy** and may become exudative and crusted as a result of rubbing.
- Secondary infection and lymphadenopathy are common.

Childhood Phase: From 18 to 24 months onwards, the sites most characteristically involved are the **elbow and knee flexures**, a mixture of erythema, crusting, **excoriation**, hyper- and hypopigmentation, and **warty lichenification** may be seen depending on the skin type on the sides of the neck, wrists, and ankles.

- The sides of the neck may show a striking **reticulate pigmentation**, sometimes referred to as **atopic dirty neck**.
- The erythematous and oedematous papules tend to be replaced by **lichenification**.

Adult Phase: The picture is essentially similar to that in later childhood, with lichenification, especially of the **flexures and hands**.

Other options:

B) Dermatitis herpetiformis:

- Dermatitis herpetiformis is a chronic, intensely pruritic, skin condition associated with **gluten-sensitive enteropathy.**
- It typically appears in the 4th decade, and the patients present with itching and rash, most typically over the **extensor surfaces** of the elbows, knees, buttocks, and scalp.
- They may also have symptoms associated with gluten-sensitive enteropathies, like bloating, diarrhea and other gastrointestinal complaints.
- **On examination**, multiple erythematous **papules** and **vesicles** can be seen over the extensor aspects. They appear symmetrical and heal without scarring. The oral cavity may show mucosal abnormalities and enamel pits.

C) Pemphigus vegetans:

It is a **rare form** of the bullous disorder that presents with large, moist, verrucous, vegetating plaques studded with pustules on the groin, axillae and **flexural** surfaces.

Predominant cell	Name of microabscess	Disease
Neutrophilic microabscess	Papillary tip microabscess Munroe's microabscess (superficial) Spongiform pustules of Kogoj (deep)	Dermatitis Herpetiformis Psoriasis Psoriasis
Lymphocytic microabscess	Pautrier's microabscess	Mycosis fungoides
Eosinophillic microabscess		Pemphigus vegetans

D) Psoriasis:

- Plaque psoriasis is the most common type of psoriasis and is seen in 80-90% of patients.
- The peak on occurrence is between 16 to 22 years and between 57 to 62 years. They usually present with **itching** and a **papulosquamous rash**.
- The lesions are red, scaly, and have well demarcation from the normal skin.
- They can also be encircled by a clear peripheral zone which is called the **Woronoff ring**.
- They commonly appear in the scalp, trunk, and **extensor surfaces** of the limbs.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 41.2, 41.5, 41.16, 41.19

Q8. Dermatology INI-CET 2021 ID: MF2121

A 65-year-old man presents with itchy tense bullae over the limbs. The roof of the bullae collapsed after a few days. A clinical image is shown below, what is the likely diagnosis?

A ✓ Bullous pemphigoid

B Pemphigus vulgaris

C Dermatitis herpetiformis

D Linear IgA dermatosis

Correct Answer: Bullous pemphigoid

The given clinical scenario is highly suggestive of **bullous pemphigoid.**

Bullous pemphigoid is characterized by **tense itchy bulla** in the flexor areas and trunk which do not rupture easily. **Elderly aged** people are usually affected. It is a **subepidermal** immunobullous disorder, which occurs due to the production of autoantibodies (IgG) against **BP180 (BPAG2) and BP 230 (BPAG1)** antigens on **hemidesmosomes.**

Mucosal membrane involvement is rare in this disease. Most of the lesions heal without a scar. Since these bullae are subepidermal, pressure over the lesions does not separate the epidermis from the dermis. (**Nikolsky sign is negative**).

Immunofluorescent microscopy will show the presence of subepidermal split and deposition of **IgG antibodies** at the **dermo-epidermal junction.** Histopathology will show the presence of **subepidermal bullae with eosinophils.**

It is mainly treated with corticosteroids. Other drugs used are azathioprine, dapsone, doxycycline, etc.

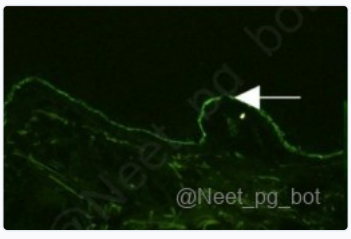
The above image (immunofluorescence microscopy) shows **linear IgG deposits** at the **dermo-epidermal junction** seen in the bullous pemphigoid.

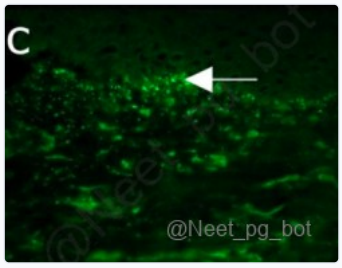
Other options:

Option B: **Pemphigus Vulgaris** is an **intraepidermal** immunobullous disease. It occurs commonly between the **4th to 6th decade.** They present with **flaccid blisters,** unlike bullous pemphigoid**.** Firm pressure over the lesion separates the epidermis from the dermis. **Nikolsky's sign** is **positive**. **Mucosal involvement** is seen in the form of buccal and palatal erosions.

Option C: **Dermatitis herpetiformis** typically appears in the **4th decade,** and the patients present with **itching** and rash, most typically over the **extensor** surfaces. The majority of the patient have symptoms related to **gluten-sensitive enteropathies,** like bloating, diarrhea, and other gastrointestinal complaints. Immunofluorescence microscopy will show the deposition of **IgA antibodies** in the papillary dermis in a **granular or fibrillar pattern**(as shown in the below image).

Option D: **Linear Ig A dermatosis (Chronic bullous dermatosis of childhood)** is an autoimmune blistering disease seen in **infants and children**. The disease can also be seen in adults and is known as Linear IgA disease of adults or IgA bullous pemphigoid. The blisters are arranged in an **annular** pattern commonly known as the **crown of jewels** appearance (image below).





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References: Rook's Textbook of Dermatology 9th edition, Page no: 50.10-50.21, 50.4, 50.53,50.37

Q9. Dermatology INI-CET 2021 ID: MF2130

A young female presented with the following findings on the face. Which among the following will not be useful for its treatment?

A Topical benzoyl peroxide

B ✓ Topical fluocinolone

C Topical tazarotene

D Topical antibiotics

Correct Answer: Topical fluocinolone

The age of the patient and the presence of **papules** and **pustules** with an **erythematous base** on the face are suggestive of **acne.** All the above agents are useful in the treatment of acne except topical fluocinolone (topical steroid). Corticosteroids may provoke an acneform reaction irrespective of their route of administration.

Acne is a chronic inflammatory disease of the pilosebaceous unit. It is commonly seen in **teenagers.** It can vary from very mild **comedonal** acne to severe aggressive **pustular** acne. These eventually may lead to **scarring** due to **sinus formation**.

Acne results from the combination of:

- **Increased sebaceous gland activity** under the influence of androgens - oily skin
- Abnormal follicular differentiation with increased keratinization- **hypercornification** of the **duct** due to increased IL-1
- **Microbial hypercolonization** of the follicular canal leading to its inflammation.

Management-

- Mild comedonal acne is treated with a **topical retinoid**. Mild papular and pustular acne is treated with **topical retinoid and topical antibiotics**.
- Moderate papular/pustular acne and moderate nodular acne are treated with **oral antibiotic + topical retinoid with or without benzoyl peroxide.**
- **Severe** nodular and conglobate acne are treated with **oral isotretinoin.**

Tazarotene is a third-generation retinoid, which is available for the treatment of acne.

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References: Rook's Textbook of Dermatology - 9th Edition - Volume III, Chapter 90 Page nos 90.1,90.2,90.18, 90.37, 90.10

Q10. Dermatology INI-CET 2021 ID: MF2139

A 45-year-old male patient from Bihar presented with normoanesthetic hypopigmented lesion as shown below. He had a history of fever for a long duration in his childhood. No thickened nerves were present. What is the probable diagnosis?

A Tuberculoid leprosy

B Lepromatous leprosy

C Mycosis fungoides

D ✓ Post kala-azar dermal leishmaniasis

Correct Answer: Post kala-azar dermal leishmaniasis

The case presenting with **hypopigmented lesions** with **normal sensations** and a **history of fever in childhood** suggests a diagnosis of **post-kala-azar dermal leishmaniasis**.

Post-kala-azar dermal leishmaniasis (PKDL) is a condition when ***Leishmania donovani*** is found in the skin and manifests as **dermal lesions.** It is usually seen in the kala-azar endemic zones i.e. **Bihar**, Jharkhand, U.P, West Bengal (Northeastern states of India)

These patients present with **erythematous rash** and **hypopigmented macules** similar in appearance and distribution to those of **lepromatous leprosy** but normally less than 1cm. After a variable period of years or months, diffuse **nodulation** begins to develop in these macules.

The rash is **progressive** over many years and seldom heals spontaneously.

The image below shows post-kala-azar cutaneous leishmaniasis.

Other options:

Options A and B: The presence of hypopigmented macules over the trunk and face, suggests a differential of **leprosy**, but the **nerves** and **sensations** are **normal**, hence ruling it out.

Option C: **Mycosis fungoides** is characterized by **polymorphic erythematous patches** and **plaques** which are usually associated with severe **pruritus.** They could be asymptomatic too. The image below shows characteristic lesions in mycosis fungoides.

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References: Rook's Textbook of Dermatology 9th edition, Chapter 33, Page no 33.50, 140.10

Q11. Dermatology INI-CET 2021 ID: MF2128

A child presented with asymptomatic lesions on the forearm and on the shaft of the penis. The lesions on the forearm are shown below. What is the most likely diagnosis?

A Lichen planus

B ✓ Lichen nitidus

C Scrofuloderma

D Scabies

Correct Answer: Lichen nitidus

The image shows an aggregate of **pinhead-sized papules over the forearm** of a child which is suggestive of **lichen nitidus**.

Lichen nitidus presents with tiny **pinhead-sized flesh-colored papules** and has a flat or dome-shaped, shiny surface. They are found on any part of the body but the sites of predilection are the **forearms, penis,** abdomen, chest, and buttocks.

Lichen nitidus is considered to be a **variant of Lichen planus** as both are histologically similar in early stages. It has a "**claw clutching a ball**' appearance on histopathological examination. It refers to a sharply circumscribed inflammatory infiltrate in the dermis, that spans 4-5 dermal papillae.

The diagnosis can be made clinically, but skin biopsy provides a definitive diagnosis. Lesions are usually **asymptomatic** and rarely require any treatment. Symptomatic or generalized lesions can be treated with fluorinated **topical steroids, PUVA, narrow-band UVB phototherapy, astemizole,** or **acitretin.**

Other options:

Option A: Lichen planus is an idiopathic inflammatory disease affecting the skin and mucous membrane. It usually presents with **pruritic, purple, polygonal, papules**. The lesions have **thin white lines** on their surface known as **Wickham's striae.**

Option C: **Scabies** has typical distribution in the **interdigital spaces** and **burrows** are the characteristic lesions. The below image shows a magnified view of a **burrowing trail of the scabies mite.**

Option D: **Scrofuloderma** results from the direct invasion of the **tubercle bacillus** into the skin from **underlying tuberculous foci.**

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References: Rook's Textbook of Dermatology 9th edition, Page no 37.10, 37.3, 34.39
Online Resource, http://www.ijdvl.com/article.asp?issn=0378-6323;year=2013;volume=79;issue=3;spage=338;epage=348;aulast=Madke#
Fitzpatrick's Dermatology, 9th Edition., Page no. 554

Q12. Dermatology INI-CET 2021 ID: MF2643

Which among the following is correct regarding the healing of leprosy if left untreated?

A No spontaneous remission is seen

B Half of the cases resolve spontaneously after bacilli exposure

C Spontaneous remission is more common in children than in adults

D ✓ Indeterminate leprosy have more remission than tuberculoid

Correct Answer: Indeterminate leprosy have more remission than tuberculoid

Among the given options, Indeterminate leprosy have more remission than tuberculoid is the most appropriate answer.

The manifestations of leprosy depend on the immune response of an individual to the bacillus Mycobacterium leprae. About **80-90%** of the people do not develop any disease after bacilli exposure. The remaining patients develop **indeterminate leprosy**. The lesions of indeterminate leprosy tend to be hypopigmented and ill-defined. Among those patients, 75% undergo spontaneous remission. Very few patients present clinically between tuberculoid or lepromatous leprosy. Therefore, more patients with indeterminate leprosy undergo spontaneous remission compared with those having tuberculoid leprosy. **(Option D).**

Leprosy (Hansen's disease) is a chronic infective granulomatous disease caused by the bacillus Mycobacterium leprae.

Leprosy is clinically characterized by one or more of the following features :

- Hypopigmented patches
- Partial or total loss of cutaneous sensation in the affected areas
- Presence of thickened nerves, and
- Presence of acid-fast bacilli in the skin or nasal smears.

Option A is incorrect because **spontaneous remission** is **seen** with leprosy. Even in the absence of specific treatment, a majority of patients **tend to get cured spontaneously**.

Option B is incorrect because around **80-90% of the cases** (not half) resolve spontaneously after bacilli exposure.

Option C is incorrect because spontaneous remission in children is variable. Even though few studies mentions that children have more remission, however, the observation period was not long enough to establish its permanence.

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References: Park's Textbook of Preventive and Social Medicine - 26th Edition, Pg 359
Rook's Textbook of Dermatology 9th edition, Pg 28.7

Q13. Dermatology INI-CET 2021 ID: MF2117

A young man presented with painful lesions as shown in the image below. What is the likely diagnosis?

A ✓ Herpes Zoster

B Molluscum contagiosum

C Herpes Simplex

D Lymphangioma circumscriptum

Correct Answer: Herpes Zoster

The above image shows **unilateral, red, closely grouped vesicles** in a single **dermatome** associated with severe **pain** is suggestive of **herpes zoster** or **shingles.**

It is a segmental eruption due to the **reactivation** of latent varicella-zoster virus from **dorsal root ganglia**. An earlier infection with varicella is essential before zoster can occur.

The first symptom of zoster is usually **pain** and it may be accompanied by fever, headache, malaise, and tenderness localized to areas of one or more dorsal roots. The **mucous membranes** within the affected dermatomes are involved. The lymph nodes draining the affected area are enlarged and tender.

The patients are **infectious**, both from the virus in the lesions and nasopharynx. The most commonly involved dermatomes in decreasing order of frequency are **thoracic** > cervical > **trigeminal** > lumbosacral. The posterior nerve roots and ganglia show inflammatory changes.

Zoster is a **self-limiting infection** but it is painful and carries a risk of secondary infection and post-herpetic neuralgia. **Analgesia** and treatment of secondary infections are sufficient for mild infections.

Indications for antivirals in VZV (**Acyclovir 800 mg**):

- Painful zoster infections in adults
- Facial zoster at any age
- Immunocompromised

Other options:

Option B: The lesions of the **molluscum contagiosum** are shiny, white, and hemispherical. It is **umbilicated** with **a central pore.** It is caused by **poxvirus.**

The below image shows the lesions seen in Molluscum contagiosum

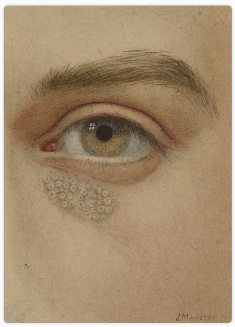
Option C: **Herpes simplex virus** are classified into two major antigenic types,

- Type 1 (HSV-1) which is classically associated with facial infections;
- Type 2 (HSV-2) which is typically genital.

Below image shows small, closely grouped vesicles on an inflamed base and is characteristic of **Herpes Labialis or Cold sore.**

Option D: **Lymphangiomas** are dilated dermal lymphatics that blister onto the skin surface. The fluid is usually clear, but occasionally maybe blood-stained. Lymphangiomas may thrombose and get fibrosed, to form hard nodules in the long term. This may raise concerns about malignancy.

If lymphangiomas are <5 cm across, they are called **lymphangioma circumscriptum** (shown below), and they are called lymphangioma diffusum if they are more widespread.



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References: Rook's Textbook of Dermatology 9th edition, Page nos 25.27-25.28,128.2,109.36

Q14. Dermatology NEET 2018 ID: MC4001

A 15-year-old girl presents with itchy lesions on her arm as shown. Her family history is positive for asthma. What could be the most probable diagnosis?

A Seborrhoeic dermatitis

B ✓ Atopic dermatitis

C Allergic contact dermatitis

D Erysipelas

Correct Answer: Atopic dermatitis

The clinical scenario depicting **pruritic** lesions at a **flexural site** along with a family history of **atopy** (asthma) is suggestive of **atopic dermatitis** (AD) or atopic eczema (AE). The **scratch marks** and **excoriations** shown in the image are due to pruritis, which is the hallmark feature of AD.

AD is a **chronic** relapsing **inflammatory** condition. It presents with **erythematous macules**, papules and **vesicles** that often start in **early** childhood, usually less than 2 years old. The lesions later become excoriated and **lichenified**. It is frequently associated with **dryness** of the skin and **secondary infections.** In infancy, the lesions initially appear on **face** (sparing the napkin area) and later on **extensor** aspect of knees and elbows as the child starts crawling. In later childhood and in adults, the principal site of involvement is **flexures**.

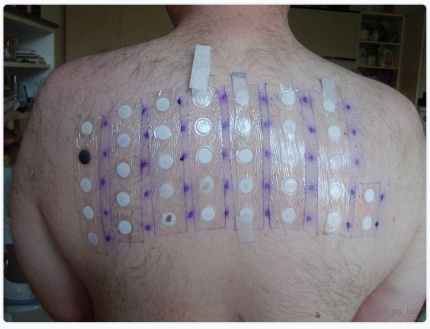
The **filaggrin** monomers in keratinocytes in the epidermis bind and aggregate keratin bundles and intermediate filaments to form the **epidermal barrier**. Any **damage/dysfunction** of this barrier is associated with an **increased risk** of developing AD. Damage to this barrier can be due to **environmental factors** (detergents, hard water, UV light) or **local inflammatory** reactions. Dysfunction of the barrier can be due to **genetic mutation** of the **FLG** gene leading to reduced filaggrin production or **local immunomodulatory effects** leading to reduced filaggrin expression.

Other options:

Option A: **Seborrheic dermatitis** is a chronic inflammatory skin condition characterized by erythema, pruritus, **scaling**, flaking, or crusting of the skin. It is often associated with **Malasezzia** fungal infection. A milder form of seborrheic dermatitis of the scalp is commonly known as **dandruff**.

Option C: **Allergic contact dermatitis (ACD)** is an inflammatory condition caused due to a **delayed hypersensitivity reaction (Type 4 hypersensitivity)** to a specific allergen that comes into direct contact with the skin.Patients usually present with **pruritus, erythema, swelling, papules,** and **papulovesicles.** Some common allergens include **nickel** (found in **artificial jewelry**), **PPD**/Para phenylene diamine (in **hair dye**), **chromate** (in **cement**), rubber or **latex**, cosmetics, etc. **Patch test** (image below) is used for the diagnosis of ACD.

Option D: **Erysipelas** is a bacterial skin infection caused by **Streptococcus pyogenes (Group A Streptococcus)** characterized by raised, erythematous sharply demarcated rapidly spreading skin lesions, along with **systemic symptoms** like fever and chills. The skin may also have a **peau d'orange** or orange peel-like appearance.



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References: Rook's Textbook of Dermatology 9th edition, Page no: 26.18, 40.1, 41.1, 128.1

Q15. Dermatology NEET 2018 ID: MA4071

Which of the following is a primary skin lesion?

A Crust

B Atrophy

C ✓ Purpura

D Induration

Correct Answer: Purpura

Purpura is a **primary** skin lesion. Crust, atrophy, and induration are secondary lesions.

It is a condition with **red** or **purple** discolored **spots** on the skin that **do not blanch** on applying pressure. The spots are caused by **bleeding underneath** the **skin.** It is seen:

- Vasculitis
- Thrombocytopenia
- Vitamin C deficiency

Option A: Crusts or scabs consist of dried-up discharge from the lesions.

Option B: Atrophy refers to loss of tissue. There may be fine wrinkling and increased translucency if the process is superficial.

Option D: Induration refers to dermal thickening that clinically presents as skin that feels thicker and firmer than normal on palpation.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 4.10-4.13

Q16. Dermatology NEET 2018 ID: MC2886

Which of the following statements is false regarding scleredema?

A The dermis is 3–4 times thicker than normal

B Erythema and peau d'orange appearance of the skin can be seen

C Can occur in association with diabetes

D ✓ Associated with sclerodactyly or Raynauds phenomenon

Correct Answer: Associated with sclerodactyly or Raynauds phenomenon

Scleredema is not associated with **sclerodactyly** or **Raynauds Phenomenon. Scleredema** and **scleroderma** are different entities. In **contrast to systemic sclerosis (scleroderma)**, scleredema is **not associated** with sclerodactyly, Raynaud phenomenon, nail fold capillary changes, or serum autoantibodies.

In scleredema, the **dermis** is 3–4 times thicker than normal due to an increase in type 1 collagen synthesis by dysfunctional fibroblast. However, fibroblast proliferation is not seen.

Erythema and a **peau d'orange** appearance of the skin are commonly observed. The **hands and feet** are characteristically **spared**.

Diabetic scleredema is the most common type. In diabetic scleredema, the accumulation of collagen may be due to irreversible non-enzymatic glycosylation of collagen and resistance to degradation by collagenase. It can also occur post-streptococcal infections.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 59.9-59.10

Q17. Dermatology NEET 2018 ID: MC1143

Identify the pigmentation abnormality in the image shown below.

A Mongolian spot

B Naevus of Ito

C Congenital melanocytic naevus

D ✓ Becker's naevus

Correct Answer: Becker's naevus

The given image showing **hyperpigmented** lesion on **shoulder** associated with **hypertrichosis** is suggestive of **Becker's naevus**.

It is an **epidermal melanocytic naevus** with aggregation of melanocyte clusters in the epidermis. It is often **hairy.** The onset is around **adolescence** when there is an **increased sensitivity to androgens** especially in the distribution of outer arms, scapular area and chest.

A triad of features can be seen:

1. Hyperpigmentation
2. Hypertrichosis
3. Acne

Option A: **Mongolian spots** are slate blue areas in the lumbosacral region that spontaneously regress before puberty.

Option B: **Naevus of Ito** is **unilateral** hyperpigmentation in the **acromioclavicular** distribution. It is not known to have hypertrichosis or acne.

Option C: **Congenital melanocytic nevi** are common moles that become raised and have a cerebriform appearance and can be **hairy**. They can occur anywhere. The onset is at **birth** or **soon after birth**

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References: Rook's Textbook of Dermatology 9th edition, Page no: 68.11, 75.19

Q18. Dermatology NEET 2018 ID: MC4453

What is the most common causative organism for verruca vulgaris?

A Human papillomavirus-1

B Human papillomavirus-3

C Human papillomavirus-10

D ✓ Human papillomavirus-2

Correct Answer: Human papillomavirus-2

Verruca vulgaris or common wart is most commonly caused by **human papillomavirus-2** (HPV-2) but is also related to types 1, 4, 27, and 57.

Common warts are usually **asymptomatic.** They show acanthosis, hyperkeratosis, and papillomatosis and present as **firm papules** with a rough horny surface. The most commonly affected sites are on the backs of the **hands** and **fingers**, and, in children on the **knees**. **Pseudo-Koebner phenomenon** may be observed.

Treatment is usually with topical agents such as **salicylic acid**, caustics, glutaraldehyde, formalin, or with **laser, cryotherapy,** surgical removal.

verruca vulgaris.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 25.47, 25.48

Q19. Dermatology NEET 2018 ID: MC3999

Cutis marmorata occurs due to _____

A ✓ Exposure to cold temperature

B Humidity

C Exposure to hot temperature

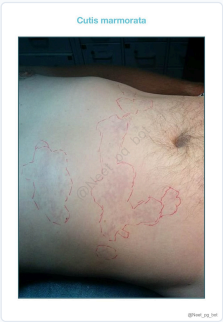
D An adverse reaction to drugs

Correct Answer: Exposure to cold temperature

Cutis marmorata refers to the **pinkish-blue mottled** or **marbled** appearance when subjected to **cold** temperatures.

It is seen throughout **infancy** and in 50% of children. It is caused by simultaneous **dilation and contraction** of the superficial **capillaries**. **Rewarming** restores the skin to **normal**. This is a physiological finding and requires no treatment. However, **persistent** cutis marmorata may be seen in **trisomies 18 and 21**.

Note: Cutis marmorata telangiectatica congenita is a distinct vascular developmental disorder and is easily distinguished as it is fixed (it does not disappear on rewarming), and it may be associated with a variety of other anomalies such as limb growth or renal anomalies.



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References: Rook's Textbook of Dermatology 9th edition, Page no: 116.3

Q20. Dermatology NEET 2018 ID: MC4261

Varicella-zoster virus remains dormant in the:

A Medulla oblongata

B ✓ Dorsal root ganglion

C Skin

D Ventral root

Correct Answer: Dorsal root ganglion

Varicella-zoster virus (VZV) is a human herpesvirus that causes varicella (**chickenpox**) as a primary infection and then remains in a latent form in the **dorsal root ganglia** for a variable period after which it can **reactivate** to cause herpes zoster (**shingles**).

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References: Rook's Textbook of Dermatology 9th edition, Page no: 25.27-25.28

Q21. Dermatology INI-CET 2024 ID: MG5805

Which of the following is not associated with annular lesions?

A Tinea corporis

B Pityriasis rosea

C ✓ Urticaria

D Porokeratosis

Correct Answer: Urticaria

Urticaria is not associated with **annular lesions**. It is characterized by wheals.

Urticaria is a heterogeneous group of disorders characterized by itchy wheals, which develop due to evanescent edema of dermis.

A weal is a descriptive term for **transient, well-demarcated, superficial pink** or **pale swellings** of the dermis due to reversible exudation of plasma in the skin that fade, usually within hours, without leaving a mark . Weals are usually **very itchy** and associated with a **surrounding red flare** when they arise. Lesions begin as **erythematous macules**, which rapidly evolve into pale pink edematous wheals with a surrounding flare.

Option A: **Tinea corporis/ ringworm infection** is caused by the dermatophyte ***Trichophyton rubrum***. It results from invasion and proliferation of the causal fungi in the **stratum corneum** with lesions on the trunk and limbs. **Characteristic lesions** are circular, usually sharply marginated with a raised edge with central resolution.

Option B: **Pityriasis Rosea (PR)** is an unknown etiology, self-limiting condition that typically lasts between 2 to 10 weeks. It is characterized by a herald patch, followed by a secondary eruption with typical lesions that are oval, annular, erythematous, and plaque-like with a collarette of scales. These lesions commonly appear on the trunk in a "fir tree" pattern along lines of cleavage. Differential diagnoses include guttate psoriasis, secondary syphilis, and drug rash. There is no specific treatment for PR; symptomatic treatment is usually sufficient.

Option D: **Porokeratosis** is a **clonal expansion of keratinocytes** which differentiate abnormally. But they are not hyperproliferative. It may present as single or multiple lesions and may be localized or disseminated. Porokeratoses present with **single or multiple papules or plaques** which develop into **annular lesions** with a **thin raised border.**

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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no. 13, 53
Rook's Textbook of Dermatology 9th edition, Page 65.67

Q22. Dermatology INI-CET 2024 ID: MG5799

Match the following

a. Patch test	1. Leprosy
b. KOH mount	2. Lupus vulgaris
c. Biopsy	3. Contact dermatitis
d. Slit skin smear	4. Tinea corporis

A ✓ a-3, b-4, c-2, d-1

B a-1, b-4, c-2, d-3

C a-3, b-2, c-4, d-1

D a-2, b-4, c-3, d-1

Correct Answer: a-3, b-4, c-2, d-1

The correctly matched option is **a-3, b-4, c-2, d-1.**

a. Patch test	3. Contact dermatitis
b. KOH mount	4. Tinea corporis
c. Biopsy	2. Lupus vulgaris
d. Slit skin smear	1. Leprosy

Lupus vulgaris is one of the prevalent forms of **cutaneous tuberculosis** that occurs in previously sensitized individuals with **high immunity**. It is characterized by **reddish-brown papules** and slow-growing plaques with elevated borders and central atrophy. Diagnosis is made by **biopsy.** Caseating granulomas are pathognomonic of tuberculosis.

Tinea corporis/ ringworm infection is caused by the dermatophyte ***Trichophyton rubrum***. It results from invasion and proliferation of the causal fungi in the **stratum corneum** with lesions on the trunk and limbs. Characteristic lesions are circular, usually sharply marginated with a raised edge with central resolution.

KOH mount is used for the diagnosis. **Branching hyphae** are seen in Tinea corporis.

Leprosy is a chronic granulomatous condition caused by ***Mycobacterium leprae***. It affects the skin, cutaneous and peripheral nerves, and mucous membranes.

Cardinal signs for the diagnosis of leprosy:

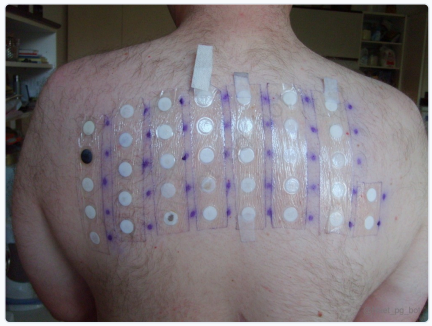
- Characteristic **skin lesions**- hypopigmented or erythematous lesions with definite **impairment of sensation.**
- **Peripheral nerve** involvement- **thickening** of the nerve with sensory impairment. Nerve tenderness can also be seen.
- Detection of **acid-fast bacilli** in **slit-skin smear**, skin biopsy, or positive biopsy PCR.

Allergic **contact dermatitis** is an **eczematous reaction** that occurs as an **immunological response** that occurs due to **exposure** to a substance towards which the immune system has previously been **sensitized**.

The patch test is used to detect **delayed (type IV) hypersensitivity** to antigens that come in contact with the skin, by applying the suspected antigen under occlusive patches to the skin. Patches are **removed after 48 hours** (or sooner if there is itching and burning).

- **48 hours:** Patch removal for **initial reading** to detect **delayed hypersensitivity** reaction.
- **72-96 hours**: Final reading done for confirmation.

An image showing the patch test:



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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no: 16, 222, 256, 288

Q23. Dermatology INI-CET 2024 ID: MG5803

A hypopigmented patch which grows initially and stops progressing after sometime is seen in which type of vitiligo?

A Focal

B ✓ Segmental

C Mixed

D Acrofacial

Correct Answer: Segmental

In segmental vitiligo, **lesions increase initially** (6–12 months) and then **remain static.**

Vitiligo is a localized depigmentation that occurs due to a **progressive loss in the melanocytes** in the **basal layer of skin.** It is classified as

- **Segmental vitiligo**- which appears as **unilateral** maculae in a segmented or band-shaped pattern
- **Non-segmental vitiligo**- which appears as **bilateral** maculae distributed in an acrofacial pattern or scattered symmetrically over the entire body. Non-segmental can either be focal vitiligo or undetermined/unclassified vitiligo.

Acrofacial vitiligo is a type of generalized vitiligo, predominantly seen periorificially (around eyes and on lips on the face) and acral parts (periungual area, palms, and soles). This type of vitiligo is more resistant to therapy due to absence of hair in the affected parts.

Mixed vitiligo: the coexistence of non-segmental and segmental vitiligo in one patient is called mixed vitiligo

According to a recent Vitiligo Global Issue Consensus Conference, the term 'vitiligo' can be used as an umbrella term for all non-segmental forms of vitiligo (including several variants: acrofacial, mucosal, generalized, universal, mixed, and rare variants of vitiligo). The classification is shown below.

Vitiligo (Non segmental)	• Acrofacial • Mucosal (>1 one mucosal site) • Generalized • Universal • Mixed (associated with SV) • Rare variants
Segmental vitiligo	• Uni-, bi-, or pleuri-segmental
Undetermined/ unclassified vitiligo	• Focal • Mucosal (one site in isolation)

Clinical features:

The lesions become apparent between 20-30 years of age.

Hypopigmented macules in vitiligo are commonly found in the **hips, dorsum of the hands/fingers**, **feet, elbows, knees,** and **ankle areas.** These are the areas of repeated friction. **Koebner phenomenon** is seen, which is the development of **lesions at the site of trauma** on the uninvolved skin.

Vitiligo universalis, which is **complete vitiligo,** or trichrome vitiligo where there are areas of depigmentation, hyperpigmentation, as well as areas of normal skin may be seen rarely.

Several autoimmune conditions can be associated with vitiligo. They are:

- Thyroid disease (hyperthyroidism and hypothyroidism)
- Pernicious anemia
- Addison disease
- Diabetes
- Hypoparathyroidism

Examination:

Under the Wood's lamp examination the depigmented areas emit a bright **blue-white fluorescence** and **appear sharply demarcated.**

Treatment:

Treatment involves cosmetic camouflage and the use of sunscreen. The topical application of **corticosteroids** is the **first line of treatment**. Topical calcineurin inhibitors like pimecrolimus, tacrolimus have also shown to be successful. **Psoralen photochemotherapy,** khellin, selective **ultraviolet B therapy** are used as **second-line treatment**. In patients who do not respond to this treatment, surgical grafting techniques can be used, where the melanocytes from a normal pigmented area are transferred to a non pigmented area. In patients with extensive vitiligo where only a few pigmented areas are left, depigmentation can be done.

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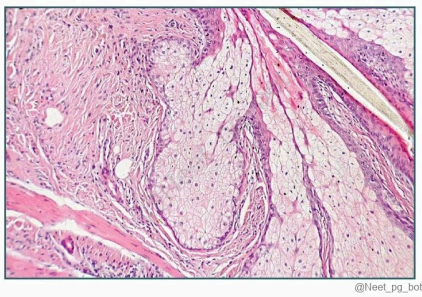
References: Rook's Textbook of Dermatology 9th edition, Page 88.34-88.40
Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no. 149-156

Q24. Dermatology INI-CET 2024 ID: MG7758

What is the mechanism responsible for the release of secretions from the gland seen in the image below?



A Apocrine

B Merocrine

C Eccrine

D ✓ Holocrine

Correct Answer: Holocrine

The gland depicted in the image is a **sebaceous gland**, which releases its secretions through the holocrine mechanism.

Identification features of sebaceous gland:

- **Lobular structure** with clusters of cells.
- **Foamy cytoplasm** due to lipid content.
- **Holocrine secretion** with disintegrated cells.
- Found near **hair follicles** in the dermis.

Sebaceous gland cells synthesize and accumulate **lipid-rich sebum** within their cytoplasm.

Holocrine secretions are produced in the cytoplasm of the cell and released by the **rupture** of the plasma membrane, which **destroys the cell** and results in the secretion of the product into the lumen.

Examples of holocrine glands are

- **Sebaceous glands** of the skin and
- **Meibomian glands** of the eyelid

Other options:

Option A: Apocrine: involves **partial loss** of the **apical** portion of the cell (e.g., mammary glands).

Option B: Merocrine: involves secretion via **exocytosis** (e.g. sweat glands).

Option C: Eccrine glands are a type of sweat gland responsible for thermoregulation through the secretion of sweat. They use the **merocrine** mechanism for secretion.

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References: Gray's anatomy: The anatomical basis of clinical practice - 42nd edition, Page: 33

Q25. Dermatology INI-CET 2024 ID: MG7809

All are a part of NAME syndrome, a subset of Carney's complex, except?

A Atrial myxoma

B Nevi

C ✓ Ebstein anamoly

D Myxoid neurofibroma

Correct Answer: Ebstein anamoly

Ebstein anomaly is not a part of NAME syndrome.

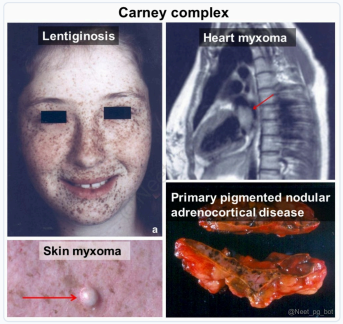
Carney Complex is a rare, **autosomal dominant genetic disorder** characterized by multiple neoplasms and lentigines. It includes features like cardiac myxomas, endocrine tumors, spotty skin pigmentation, and other benign or malignant tumors, caused by mutations in the PRKAR1A gene. It includes:

- **Myxomas** - cardiac, skin, and/or breast
- **Lentigines** and/or pigmented nevi
- **Endocrine overactivity** - primary nodular adrenal cortical disease, testicular tumors, and/or pituitary adenomas with gigantism or acromegaly

NAME syndrome and **LAMB syndrome** are synonymous with the Carney complex:

- **LAMB** - **L**entigines, **A**trial **M**yxoma, and **B**lue nevi.

- **NAME syndrome** is a subset of Carney's complex, which is a rare **autosomal dominant** disorder characterized by multiple neoplasms and pigmentation abnormalities. The acronym NAME stands for:
  - **Nevi**
  - **Atrial myxomas**
  - **Myxoid neurofibromas**
  - **Ephelides** (freckle-like pigmented lesions)



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References: Harrison's Principles of Internal Medicine, 21st Edition, Page no: 2908
Online Resource, https://pubmed.ncbi.nlm.nih.gov/1997386/#:~:text=A%20family%20affected%20in%20two,autosomal%20inheritance%20of%20this%20trait

Q26. Dermatology INI-CET 2024 ID: MG7998

A 60-year-old woman presented with a great toe infection. She had significant discoloration of the toenail with a subungual hyperkeratosis that caused color alteration and a rough surface area. She was suspected to have onychomycosis. Which topical drug is given for toenail onychomycosis?

A Caspofungin

B Itraconazole

C Griseofulvin

D ✓ Tavaborole

Correct Answer: Tavaborole

Tavaborole is the topical drug given in toenail onychomycosis.

Tavaborole is a **topical antifungal**. It is an **oxaborole antifungal** that inhibits fungal protein synthesis by targeting **leucyl-tRNA synthetase.**

Onychomycosis is defined as the infection of the nail plate by fungus.

Most common cause is **Trichophyton rubrum,** but many fungi may be causative.

The clinical scenario of Tinea unguium presents with yellowish discolouration, which spreads proximally as a streak in the nail. Later, subungual hyperkeratosis becomes prominent and spreads until the entire nail is affected. Gradually, the entire nail becomes brittle and separated from its bed.

Diagnosis is made by **KOH mount** examination of dystrophic subungual debris.

Management of onychomycosis involves **Oral terbinafine** as the treatment of choice for Tinea unguium or onychomycosis. It acts as a non-competitive inhibitor of squalene epoxidase, which is essential in the synthesis of fungal ergosterol. Oral terbinafine at 250 mg/day is given for 6 weeks for fingernail infection and 3 months (12 weeks) for toenail infection. An **oral Griseofulvin** dose of 1 g/day for 4–8 months (longer for toenails) is used as a second-line treatment.

Other Options:

Option A: Caspofungin: Caspofungin is an echinocandin antifungal that works by inhibiting fungal cell wall synthesis. It is used in **intravenous form** for systemic fungal infections like candidiasis and aspergillosis.

Option B: Itraconazole: Itraconazole is an **oral antifungal** from the triazole class, used for onychomycosis due to its penetration into keratinised tissues.

Option C: Griseofulvin: Griseofulvin is an **oral antifungal** used for dermatophyte infections, including onychomycosis.

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References: Rook's Textbook of Dermatology 10th edition, Page: 32.33, 32.34

Q27. Dermatology INI-CET 2024 ID: MG7843

A male patient presents with urethritis. It can be caused by all except?

A Neisseria Gonorrhea

B ✓ Gardenella vaginalis

C Trichomonas vaginalis

D Mycoplasma

Correct Answer: Gardenella vaginalis

Gardenella vaginalis does not cause urethritis in male patients.

Urethritis is classified into two types:

- Gonococcal urethritis - caused by **Neisseria gonorrhoeae**(Option A)
- Non-gonococcal urethritis - caused by:
  - Chlamydia trachomatis (most common)
  - Ureaplasma urealyticum
  - **Trichomonas vaginalis** (rare) (Option C)
  - Herpes simplex virus (rare)
  - **Mycoplasma genitalium** (Option D)

The most likely causative organism for acute urethritis in males is Neisseria gonorrhoeae.

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page 1080

Q28. Dermatology NEET 2024 ID: MG6746

A child presents with orangish hue skin with rough and scaly skin as shown in the image. On histopathological examination, orthokeratosis and parakeratosis are seen. What is the most likely diagnosis?

A Follicular psoriasis

B Lichen planus

C ✓ Pityriasis rubra pilaris.

D Keratosis follicularis

Correct Answer: Pityriasis rubra pilaris.

The image shows prominent erythema and scale on the palms of the hands and wrists with **marked orange-yellow palmoplantar keratoderma** which is a feature of **pityriasis rubra pilaris**.

Features of pityriasis rubra pilaris:

- **Salmon red** or **orange red-coloured** dry scaly **plaques** are seen characteristically, which coalesce together, leaving **islands** of **normal** skin, typically known as 'islands of sparing' in between the lesions
- Most common site - **trunk**
- On the elbow and wrist, **nutmeg grater papules** are seen due to **follicular hyperkeratosis**
- **Palmoplantar keratoderma** - keratoderma on the sole is very thick that it appears as a sandal. (**keratotic sandal**)
- Nails - thickened, discoloured distally, showing splinter haemorrhages

Histologically, pityriasis rubra pilaris is characterized by **hyperkeratosis** with **follicular plugging**, alternating **ortho-** and **parakeratosis**, dilated but non-tortuous dermal capillaries, sparse lymphocytic infiltrate, and possible **acantholysis** in the **adnexal epithelium**.

The image below shows islands of sparing in pityriasis rubra pilaris.

The image below shows keratotic sandal.

Other options:

Option A: **Follicular psoriasis** affects **hair follicles** on the **trunk** and **limbs** and can occur alone or **alongside plaque psoriasis**. The lesions are **smaller** than those seen in guttate psoriasis and may appear clustered or spread out. In children, it can be **mistaken** for **pityriasis rubra pilaris**.

Option B: **Lichen planus** is a **papulosquamous** lesion presenting as **pruritic, purple, polygonal, planar,** papules and plaques. These papules are often highlighted by **white dots** or lines called **Wickham striae**, which are created by areas of **hypergranulosis**.

Option D: **Keratosis follicularis** or **Darier's disease** or **dyskeratosis follicularis**, is an **autosomal dominant** condition arising from mutations in the **ATP2A2 gene** which affects the **SERCA2 calcium pump**. It presents with **greasy hyperkeratotic papules** and plaques in seborrheic regions. Neuropsychiatric features such as **depression, psychosis** and **mental retardation** may also occur.

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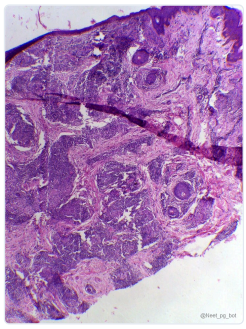
References: Rook's Textbook of Dermatology 9th edition, Chapter 36, Page nos 36.1-36.4

Q29. Dermatology NEET 2024 ID: MG6821

A patient presented with a slight loss of sensation along the ulnar nerve and a solitary nodule in the forearm. Intradermal antigen was injected. The histopathology image is shown below. Which of the following is true regarding this condition?



A Lepromatous leprosy with dermal test positive

B ✓ Tuberculoid leprosy with dermal test positive

C Lepromatous leprosy with dermal test negative

D Erythema nodosum leprosum with negative skin test

Correct Answer: Tuberculoid leprosy with dermal test positive

The patient's **slight loss** of **sensation** along the ulnar nerve, solitary forearm nodule, and histopathology showing **well-defined granulomas** indicate **tuberculoid leprosy** with a **dermal test positive**.

In tuberculoid leprosy, **dermal granulomas,** consisting of groups of **epithelioid cells** with **giant cells**, are seen. The granulomas are elongated and generally run parallel to the surface, following **neurovascular bundles.** The granulomas extend up to the **epidermis**, thereby **obscuring** the **Grenz zone.** It also shows **high positivity** of the **lepromin skin (dermal) test**.

Leprosy is a **chronic granulomatous** condition caused by ***Mycobacterium leprae***. It affects the **skin, cutaneous** and **peripheral nerves**, and mucous membranes. The most commonly affected peripheral nerves are the **posterior tibial nerve**, followed by the **ulnar**, median, and lateral popliteal nerves. The most frequently involved cranial nerve is the **facial nerve**. It has a **long incubation** period, usually 2 to 12 years. It commonly presents as an **area of numbness** on the skin or a visible skin lesion, which could be a **hypopigmented** or **erythematous macule**.

The **Grenz** zone is most prominent in **lepromatous leprosy,** as there are **scarce granulomas** to fill up the space. It is also seen in the **borderline spectrum.**

Cardinal signs for the diagnosis of leprosy:

- Characteristic skin lesions- hypopigmented or erythematous lesions with definite impairment of sensation.
- Peripheral nerve involvement- thickening of the nerve with sensory impairment. Nerve tenderness can also be seen.
- Detection of acid-fast bacilli in slit-skin smear, skin biopsy, or positive biopsy PCR.

The presence of two of the three signs can establish the diagnosis of leprosy. Other tests done for diagnosis of leprosy include nerve biopsy in pure neural leprosy and PGL-1 antibody in cases with low bacterial load.

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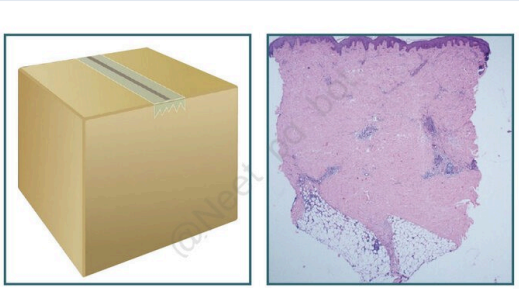
References: Rook's Textbook of Dermatology - 9th Edition - Volume I, 28.1, 28.3,28.7, 28.12-38.13
Harrison's Principles of Internal Medicine, 21st Edition, 1384-1386,1388

Q30. Dermatology NEET 2024 ID: MG6731

Which of the following is associated with the'' box sign''?



A Lichen planus

B Lichen nitidus

C ✓ Morphea

D Amyloidosis

Correct Answer: Morphea

The "**box sign**" is associated with **morphea**, also known as **localized scleroderma**.

Morphea is a condition characterized by **localized skin sclerosis** and **fibrosis** due to **increased collagen deposition**. This can lead to **well-defined, box-like patches** of thickened, hardened skin. The "box sign" refers to the **sharp**, boxy edges of these lesions, which can vary in depth and extent, sometimes involving **deeper tissues** such as **muscle** and **bone**.

Biopsies taken during the indurative stage often display **squared-off edges**, resulting in a "**box-shaped**" or "**boxed dermis**" appearance under the microscope, which can vary in depth and extent, sometimes involving deeper tissues such as **muscle** and **bone**. This characteristic sign occurs because of **extensive collagen sclerosis** throughout the **reticular dermis** and extending into the **subcutaneous fat septa**. The pronounced sclerosis of the collagen fibers causes this distinct boxy appearance. **Similar squared-off edges** may also be observed in **morphea-lichen sclerosus** overlap and scleredema, due to comparable changes in the dermal and subcutaneous tissue structure.

The epidermis in morphea can exhibit varying changes, from normal to **flattened** with **loss of rete ridges**, or slightly acanthotic. In the inflammatory phase, the reticular dermis shows edema and a dense perivascular infiltrate, which gradually gives way to thickened, **horizontally oriented collagen bundles**. The **loss of adnexal structures**, such as hair follicles and eccrine glands, is common, and the subcutaneous fat may be **replaced** by **collagen**, contributing to the appearance of a "box sign."

Management of morphea focuses on **controlling inflammation** and preventing disease progression. Early intervention is crucial and may include **topical treatments** like **corticosteroids** or **tacrolimus** for localized lesions. For more extensive or resistant cases, **phototherapy** and systemic treatments, such as **methotrexate** or **oral corticosteroids,** may be considered. The choice of therapy often depends on the extent and activity of the disease, with the goal of minimizing skin damage and improving functional outcomes.

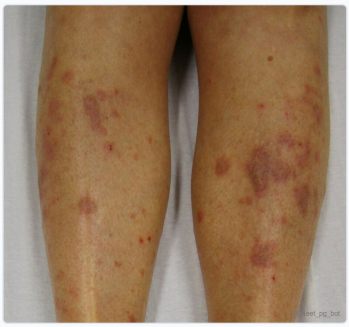
Other options:

Option A: **Lichen planus** is a **papulosquamous** lesion presenting as **pruritic**, **purple,** polygonal, planar, papules and plaques. These papules are often highlighted by **white dots** or lines called **Wickham striae**, which are created by areas of **hypergranulosis**.

The characteristic lesions in lichen planus is shown below:

Option B: **Lichen nitidus** is a **rare variant** of **lichen planus**, where both are histologically similar in early lesions. It is mostly seen in **children** and **young adults**. Lesions of lichen nitidus are typically described as a **minute pinpoint** to pinhead-size, flat or dome-shaped **papules** with a shiny surface. They are most commonly found over the **forearm**, dorsum of the **penis**, abdomen, chest, and buttocks.

Option D: **Amyloidosis** refers to **extracellular deposits** of **fibrillar proteins**. The **kidney** is the **most common organ** involved in amyloidosis. Abnormal fibrils are produced by the aggregation of **misfolded proteins**. The fibrillar deposits bind a wide variety of proteoglycans and glycosaminoglycans, including **heparan sulfate** and **dermatan sulfate**, and plasma proteins, notably serum amyloid P component (SAP). Amyloid appears as an **amorphous, eosinophilic, hyaline, extracellular substance.**



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References: Rook's Textbook of Dermatology 9th edition, Page No: 57.1-57.26, 37.1-37.16
Robbins & Cotran Pathologic Basis of Disease, 9e, Page No: 1166
Robbins and Cotran Pathologic Basis of Disease, 10e, Page No: 1111
Online Resource, https://ijdvl.com/appearances-in-dermatopathology-the-diagnostic-and-the-deceptive/

Q31. Dermatology NEET 2024 ID: MG6744

A 32-year-old man presents with painful vesicular lesions over the genital area. He has a history of multiple sexual partners and inconsistent condom use. What is the most likely diagnosis?

A Chancroid
B ✓ Genital Herpes
C Primary syphillis
D Granuloma inguinale

Correct Answer: Genital Herpes

The most likely diagnosis for this patient with **painful vesicular lesions** over the **genital area** is **genital herpes**.

The diagnosis of genital herpes is suggested by the presence of **painful, multiple vesicular ulcers** clustered on the **genitalia**, along with **inguinal lymphadenopathy**. These symptoms are indicative of an infection caused by the **Herpes Simplex Virus-2 (HSV-2)**, which is the primary cause of genital herpes.

Genital herpes is characterized by **vesiculoulcerative lesions**, which appear as **grouped vesicles** filled with clear fluid on the **penis** in males or on the **cervix, vulva, vagina,** and **perineum** in females. The **painful** nature of these lesions, combined with the history of **multiple sexual partners** and inconsistent condom use, supports the diagnosis. **Recurrences** of genital herpes are **common** and generally tend to be milder compared to the initial outbreak.

Other options:

Option A: **Chancroid** caused by ***Haemophilus ducreyi*** may also present with soft wide **indurated painful genital ulcers** and painful genital **lymphadenopathy**.

Option C: In **primary syphilis,** the typical primary chancre usually begins as a single **painless papule** that rapidly becomes eroded and usually becomes indurated, with a characteristic cartilaginous consistency on palpation of the edge and base of the ulcer.

Option D: **Granuloma inguinale** (also known as **donovanosis**) is caused by ***Klebsiella granulomatis***. It presents with a **painless papule** or **nodule** that erodes into a **beefy red granulomatous ulcer** with rolled edges and often no lymphadenopathy.

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References: Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition, Pg no: 474, 475, 427, 428, 337, 404

Q32. Dermatology NEET 2024 ID: MG6742

An adult male presented with asymptomatic macules over the chest, moist grey, perianal lesions, and findings on the scalp as shown. What is the most likely causative organism of this condition?

A Malassezia

B Trichophyton

C ✓ Treponema pallidum

D Microsporum

Correct Answer: Treponema pallidum

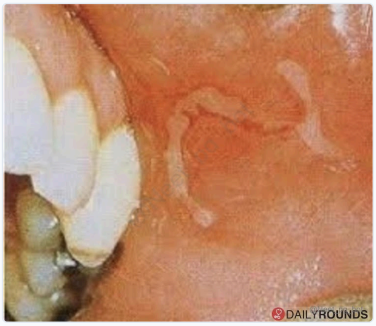
The most likely causative organism for this condition is ***Treponema pallidum***, which caused the asymptomatic macules on the chest, moist grey, perianal lesions, and **moth-eaten pattern alopecia** seen in **secondary syphilis**.

The causative organism for this condition is ***Treponema pallidum***, a **spirochete** bacterium. In secondary syphilis, the **macular** or **roseolar rash** is often the earliest sign, which may fade, leaving hypopigmented areas around the neck, known as **leucoderma syphiliticum** or "**necklace of Venus**." This stage also features a papular rash that can appear on the palms and soles.

Additionally, secondary syphilis can present with **nummular lesions** that resemble psoriasis, often with easily removable scales due to serous discharge. On the **oral mucosa**, papules may form **grey oval patches**, leading to "**snail-track**" **ulcers** (as shown below).

Condyloma lata, which are moist, well-demarcated papules and plaques, are commonly seen in the genital area and are highly infectious. Syphilitic alopecia, characterized by non-scarring patchy hair loss, and generalized lymphadenopathy are also notable features.

The **systemic manifestations** of secondary syphilis include panuveitis, periostitis, joint effusions, and involvement of various organs such as the kidneys, liver, and heart. The lesions in secondary syphilis typically resolve spontaneously, transitioning into the latent stage of the disease.



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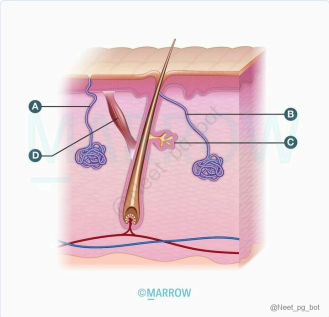
References: Rook's Textbook of Dermatology 9th edition, Chapter 29, Page nos 29.10- 29.13
Online Resource, syphilithttps://www.ncbi.nlm.nih.gov/pmc/articles/PMC3537782/

Q33. Dermatology NEET 2024 ID: MG6732

Match the structures shown in the image:



A A- Apocrine sweat gland; B- Eccrine sweat gland; C- Sebaceous gland; D- Erector pili muscle

B ✓ A- Eccrine sweat gland; B- Apocrine sweat gland; C- Sebaceous gland; D- Erector pili muscle

C A- Sebaceous gland; B- Apocrine sweat gland; C- Eccrine sweat gland; D- Erector pili muscle

D A- Eccrine sweat gland; B- Sebaceous gland; C- Apocrine sweat gland; D- Erector pili muscle

Correct Answer: A- Eccrine sweat gland; B- Apocrine sweat gland; C- Sebaceous gland; D- Erector pili muscle

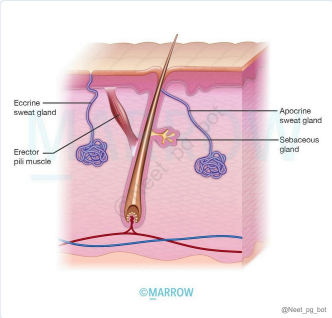
The correctly matched option **(B)- A- Eccrine sweat gland; B- Apocrine sweat gland; C- Sebaceous gland; D- Erector pili muscle.**

Human skin contains various structures, including **sweat** and **sebaceous glands**. **Eccrine sweat glands**, which are crucial for **temperature regulation**, are distributed across most of the body and are identified early in fetal development. These glands have a **coiled secretory portion** in the **lower dermis** and a **duct** that **opens directly** onto the **skin surface**. They are most concentrated on the **palms, soles,** and **forehead**.

Apocrine sweat glands, found in areas like the **axillae, genital,** and **mammary regions**, differ from eccrine glands in that they **release secretion** via **decapitation**, where parts of the cell pinch off. They are **connected** to **hair follicles** and have a larger lumen compared to eccrine glands. Their primary function remains unclear, and they have a lower secretory output.

Sebaceous glands secrete **sebum**, providing **skin lubrication** and waterproofing. These glands are **associated** with **hair follicles** and are **absent** in **non-hairy skin**.

The **erector pili muscle**, a **smooth muscle** bundle, extends from the **dermis** to the **follicle wall** and is involved in **hair erection**.

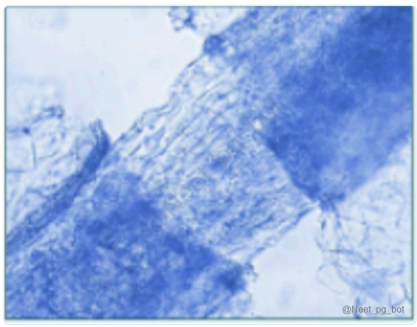


@Neet_pg_bot

References: Rook's Textbook of Dermatology - 9th Edition - Volume I, Page No: 2.2, 2.8, 2.9, 2.43
Inderbir Singh's Textbook of Human Histology- 8th Edition, Page no: 136

Q34. Dermatology NEET 2024 ID: MG7223

Which of the following causes the condition shown in the image below?



A Candida albicans

B ✓ Trichophyton species

C Epidermophyton floccosum

D Microsporum species

Correct Answer: Trichophyton species

The above image shows numerous **fungal spores within** the **hair shaft** suggestive of **endothrix** type of **tinea capitis**, which is caused by ***Trichophyton tonsurans.***

Tinea capitis is of **four** types:

Type of Infection	Causative fungus
Endothrix (infection inside shaft)	• *Trichophyton tonsurans* • *T.violaceum* • *T. schoenleinii* • *T.yaoundei* • *T. soudanense*
Ectothrix (infection outside shaft)	• ***Microsporum*** *audouinii* • *M.canis* • *M.equinum* • *M.ferrugineum*
Favus	• *T. schoenleinii*
Kerion	• *T. verrucosum* • *T. mentagrophytes*

Other options:

Option A: **Candida** is the most common cause of **opportunistic mycoses** in the world. **Immunocompromised** individuals are at an increased risk of **systemic candidiasis**. Clinically, infection with candida can present as cutaneous & mucosal candidiasis, systemic candidiasis, or rarely, as chronic mucocutaneous candidiasis.

Morphologically, they are yeast-like fungi. It appears as budding gram-positive yeast cells with pseudohyphae

Option C: **Epidermophyton floccosum:** Distinguished by **absent microconidia** and **club-shaped macroconidia**, which are **smooth, thick-walled**, and occur in clusters. It is a dermatophyte responsible for **superficial fungal infections** of the skin, particularly tinea pedis (athlete's foot), tinea cruris (jock itch), and tinea unguium (nail infection).

Option D: **Tinea cruris**, a **superficial fungal skin infection** that affects the **groin area**, is primarily caused by ***Epidermophyton floccusum*.** Typical symptoms include redness and inflammation in the groin region, accompanied by sensations of itching, burning, or stinging. The characteristic lesions exhibit **raised borders** surrounding a **central clearing**, thus imparting a ring-like appearance.

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References: Rook's Textbook of Dermatology - Volume II -9th Edition, Chapter 32 Page no 32.38

Q35. Dermatology NEET 2024 ID: MG7048

A 65-year-old treated leprosy patient came with complaints of a well-demarcated, non-tender ulcer on his foot with hyperkeratotic borders as given below for the last 3 months. What should be the appropriate next step?

A Restart MDT

B ✓ Rest the limb, non weight bearing splints and foot care

C Admit the patient and aggressive debridement and antibiotics

D Amputation

Correct Answer: Rest the limb, non weight bearing splints and foot care

The given scenario and the image indicate a neuropathic ulcer. This can be managed by **limb rest**, utilising **non-weight-bearing splints**, and administering **foot care**.

Leprosy, a condition affecting the skin and peripheral nerves, results in **peripheral neuropathy** and subsequent loss of sensation. This can lead to the development of chronic **trophic** or **neuropathic ulcers** characterised by punched-out appearance, regular margins, and **calluses** due to repeated trauma in the presence of anaesthesia of feet.
Management primarily focuses on relieving pressure through the application of non-weight bearing splints for even **pressure distribution** across the entire foot, limb elevation to mitigate **swelling**, and regular **foot inspection** to promptly identify trauma, in addition to the use of microcellular rubber footwear.

Other options:

Option A: Trophic ulcers are not an indication for restarting MDT in an **already treated** leprosy patient.
Option C: Due to the **absence** of debris or signs of **infection** in the ulcer, aggressive debridement and antibiotics are not indicated.
Option D: As there are **no signs** of **vascular compromise** or **gangrene**, amputation of the limb is not required.

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References: Rook's Textbook of Dermatology 9th edition, 28.13, 28.17

Q36. Dermatology NEET 2024 ID: MG6987

A man presents with multiple lesions on his face and upper back, diagnosed as acne vulgaris. Which of the following is the most effective treatment for his symptoms?

A Systemic Minocycline

B ✓ Systemic Isoretinoin

C Systemic Doxycycline

D Topical Dapsone

Correct Answer: Systemic Isoretinoin

The presence of **multiple nodular and cystic lesions** over the **face** and upper back implies that this is a case of **acne vulgaris. Systemic Isotretinoin** is the most effective treatment for acne.

Systemic retinoids are a very efficient treatment for acne vulgaris. Their mechanism of action si by **inhibiting the formation of keratin**, a protein that can clog pores and cause acne. Oral retinoids are usually reserved for **severe cases** of acne that do not respond to topical creams and gels.

Acne is a chronic inflammatory disease of the **pilosebaceous unit.** Comedonal lesions are the earliest lesions of acne. They represent the keratin plugs of the pilosebaceous duct.

When the keratin plug is inside the follicle, it is called a **closed comedone (whitehead)**. When the keratin plug is exposed to the environment, it gets oxidized by the air and turns black which is known as **open comedone (blackhead).**

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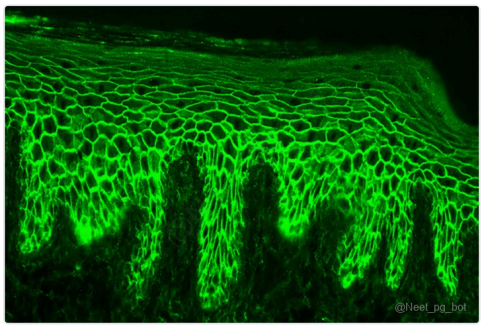
References: Rook's Textbook of Dermatology 9th edition, Page no: 90.37
Online Resource, https://ncbi.nlm.nih.gov/pmc/articles/PMC10207980/

Q37. Dermatology NEET 2024 ID: MG6948

A 42-year-old man comes with vesicular eruptions in the buccal mucosa. Direct immunofluorescence is done and the image is shown below. What is the most likely diagnosis?



A Lupus Vulgaris

B ✓ Pemphigus vulgaris

C Bullous Pemphigoid

D Borderline Leprosy

Correct Answer: Pemphigus vulgaris

The given image shows a characteristic **fish-net** appearance on **direct immunofluorescence,** which is diagnostic of **pemphigus vulgaris**.

This pattern is due to **intraepidermal intercellular** deposition of **IgG** and **C3.**

Direct or indirect immunofluorescence cannot be used to differentiate between pemphigus vulgaris and pemphigus foliaceus, as they present similarly.

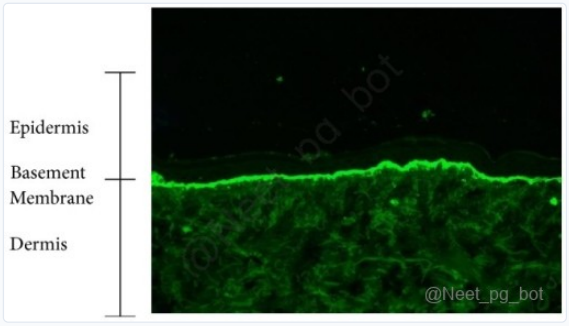
Other options:

Option A: **Lupus vulgaris(Cutaneous TB)** is the most common form of **cutaneous tuberculosis** in adults. The characteristic lesion is a **plaque**, composed of soft, **reddish brown papules**, the appearance on the **diascopy** being said to resemble **apple jelly**. The image below shows the characteristic lesion:

Option C: Bullous pemphigoid

present **linear** or **shoreline** pattern is seen due to **IgG and C3** deposits along the dermo-epidermal junctions, as seen in the following image.

Option D: In **borderline leprosy**, **inverted-saucer-shaped** lesions are seen and are characterized by a **dome-shaped** appearance with **central infiltration** and a gradually **sloping peripheral edge**.



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References: Journal, http://www.ijdvl.com/article.asp?issn=0378-6323;year=2012;volume=78;issue=6;spage=677;epage=691;aulast=Chhabra
Refer Link, www.uptodate.com

Q38. Dermatology NEET 2024 ID: MG6935

A 3-year-old boy had patches of partial hair loss with mild itching and scales. The picture of his scalp is shown below. His sister also has a similar condition. Which of the following is the most likely diagnosis?

A Alopecia areata

B ✓ Tinea capitis

C Pediculis Capitis

D Pyoderma Scalp

Correct Answer: Tinea capitis

Patchy hair loss and **multiple black dots** within the areas of alopecia along with **fine scaling** and itching of the scalp point towards the diagnosis of **tinea capitis.**

Tinea capitis is a **dermatophyte** infection primarily occurring in children. Clinical features of tinea capitis are **patchy hair loss, scaling, tender lymphadenopathy**, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

The clinical variants of tinea capitis are:

Endothrix - a non-inflammatory type in which multiple black dots are present within the areas of alopecia. It is most commonly caused by T. tonsurans and T. violaceum.
Ectothrix - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. Microsporum audouinii, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.
Kerion - a severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation. It is commonly caused by T. verrucosum and T. mentagrophytes.
Favus - is characterized by the presence of yellowish, cup-shaped crusts known as scutula that develop around hair follicles. It is caused by T. schoenleinii.

Other options:

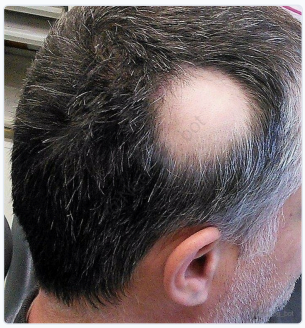
Option A: **Alopecia areata** is a chronic inflammatory disease causing **non-scarring hair loss**. It is characterized by a **circumscribed**, **smooth** patch of hair loss. **Exclamation mark hairs** are often seen at the margins of the bald patches during active phases of the disease.

The image below shows alopecia areata.

Option C : **Pediculis Capitis** (head louse infestation) is very common in **children**. The majority of the transmission is by head-to-head contact, which is frequently observed with overcrowding (as in schools), poverty, and poor hygiene.

Symptoms include **severe itching** of the scalp. On examinations, **nits** are seen, which are firmly attached to hair shafts on which they can be freely glided.

Option D : Pyoderma of the scalp is an **idiopathic cutaneous disorder**. It is characterized by progressive **painful ulceration** which can cause destruction of underlying structures. It is a **benign lesion**, which typically develops **after truama** to that area.



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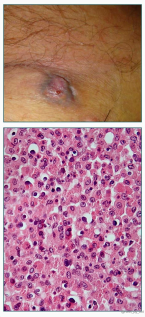
References: Rook's Textbook of Dermatology 9th edition, Page no. 32.38-32.40
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7061570/#:~:text=Pyoderma%20gangrenosum%20(PG)%20is%20a,cases%20reported%20in%20children%20only.

Q39. Dermatology NEET 2024 ID: MG6934

A middle aged man comes with complaints of painless progressively enlarging lesions on his scalp and perianal region for the last 3 months. He has had liver and kidney transplant 2 years back and is on Mycophenolate Mofetil, Tacrolimus and Prednisolone for his immunosupression. Which of the following is the most likely diagnosis?



A Drug induced eruption

B ✓ Malakoplakia

C Cutaneous TB

D Pyoderma Gangrenosum

Correct Answer: Malakoplakia

Clinical scenario of painless progressive **nodular, ulcerated lesions** in an **immuno-compromised** patient with the clinical image showing yellow-erythematous-purple **plaque** and histopathology image showing **Michaelis–Gutmann bodies** are consistent with the diagnosis of **cutaneous malakoplakia.**

Malakoplakia is a rare inflammatory condition that **typically occurs** in the **urinary tract.** The **cutaneous form** of malakoplakia is **less prevalent** and it most commonly occurs in the **perianal or genital regions.** . It is caused due to acquired **defects in phagocyte function,** chronic bacterial infection by ***Escherichia coli*** or ***Proteus*** species, and immunosuppression (renal transplant patients).

The term malakoplakia means **soft plaques**. Grossly, malakoplakia appears as soft, yellow, slightly raised **mucosal plaques.**

Microscopically, they are composed of **large, foamy macrophages** mixed with occasional multinucleate giant cells and lymphocytes. The **macrophages** in malakoplakia have abundant granular cytoplasm due to **phagosomes** stuffed with undigested bacterial products and intracellular deposition of calcium and iron within enlarged lysosomes form laminated basophilic concretions. These are known as **Michaelis–Gutmann bodies**, pathognomonic of malakoplakia (shown below).

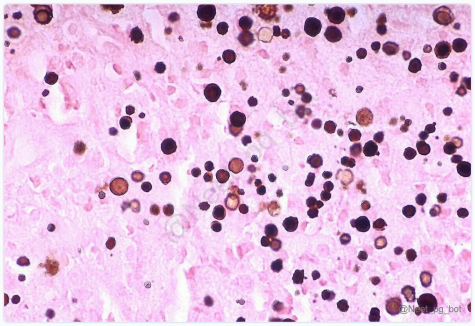
The image given below shows the **Von Kossa stain** demonstrating calcium and iron-containing Michaelis-Gutman bodies (MG bodies) in malakoplakia.

Other options :

Option A : **Drug induced eruption** can mimic a wide range of dermatoses. They can develop due to traditional pharmaceutical agents, biologic agents and targeted immunotherapeutic agents, food additives, and dyes. The morphologies of dryg induced eruptions can be **morbilliform, urticarial, papulosquamous, pustular, and bullous.**

Option C : **Cutaneous TB** (Lupus vulgaris) is the most common form of cutaneous tuberculosis in adults. The characteristic lesion is a **plaque, composed of soft, reddish brown papules**, the appearance **on diascopy** being said to resemble **apple jelly.**

Option D : **Pyoderma gangrenosum** is a **non-infectious neutrophilic dermatosis** commonly associated with **inflammatory bowel disease**. **Central necrosis** and **ulceration of the epidermis** and **dermis** surrounded by a strong inflammatory cell infiltrate are typical findings.



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References: Robbins and Cotran Pathologic Basis of Disease, 10e, Page no: 955-957
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5082045/
Online Resource, https://emedicine.medscape.com/article/1049474-overview?form=fpf

Q40. Dermatology NEET 2023 ID: MF8259

A lady developed a skin reaction as shown in the image after using a hair dye. Which of the following chemicals is responsible for this condition?

A Pollen

B Chromates

C Balsam of Peru

D ✓ p-Phenylenediamine

Correct Answer: p-Phenylenediamine

This is most likely a case of hypersensitivity reaction caused by **p-phenylenediamine** (PPD), which is present in **hair dyes**.

Hair dyes such as PPD have the potential to cause **contact dermatitis**. 10% of PPD users tend to develop contact dermatitis. A **patch test** is recommended 24-48 hours before its usage either on the forearm or behind the ear. The appearance of any **redness**, **blistering**, or **swelling** implies allergy; therefore, the application of dye is not recommended. **Cross-sensitization** with other aromatic benzenes is also noted, which needs to be avoided.

Hair dyes can trigger **acute edema** and **intense pruritus,** followed by desquamation. Hair dye can elicit an **allergic** reaction on the forehead. PPD in the hair dye can also cause an **erythema multiforme**–like reaction.

In PPD-associated allergies, there is **edema** and weeping of the **scalp margins**, eyes, and ears. There may be relative sparing of the scalp. Immediate hypersensitivity reactions present as **urticaria**, and **contact anaphylaxis** have also been noted to occur in patients using PPD. To minimize the risk of an allergic reaction to hair dye, individuals should follow the manufacturer's instructions, perform a patch test, and avoid over-processing the hair. Treatment for PPD-induced contact dermatitis may include topical corticosteroids, oral antihistamines, and in severe cases, oral corticosteroids.

Other options:

Option A: Pollen is not typically found in hair dyes and is not responsible for this condition. Pollen allergies usually present as hay fever or seasonal allergic rhinitis.

Option B: Chromates, which can be found in cement, leather, and paints, are not typically present in hair dyes and are not responsible for this skin reaction.

Option C: Balsam of Peru, a fragrant resin, is found in various cosmetic and food products but is not a common component of hair dyes. It is less likely to be responsible for this condition compared to PPD.

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References: Rook's Textbook of Dermatology 9th edition, 128.15, 89.73,128.60, 128.42

Q41. Dermatology NEET 2023 ID: MF8274

A female patient presented with acne that is not resolving on oral isotretinoin and antibiotics therapy. Which of the following is the next best investigation?

A Look for dietary triggers

B ✓ Evaluate for hyperandrogenism

C Check for antibiotic resistance

D Look for drug triggers

Correct Answer: Evaluate for hyperandrogenism

The next best step for a female patient with acne that is resistant to oral isotretinoin is to **check for hyperandrogenism**.

Hyperandrogenism usually manifests as **hirsutism, acne,** and/or **androgenic alopecia.** Acne that is **persistent, moderate to severe**, or late-onset raises concern for **PCOS** or other causes of **hyperandrogenism.**

The pathogenesis of acne vulgaris involves

- Blockage of the follicular opening by hyperkeratosis
- Sebum overproduction
- The proliferation of commensal *Propionibacterium acnes*
- Inflammation

Management is directed at minimizing inflammation, decreasing keratin production, **antibiotics**, and **antiandrogens** to diminish sebum production.

Other options:

Option A: Looking for dietary triggers could be considered, but there is limited evidence to support a direct relationship between specific food items and acne.

Option C: Since the patient has not shown improvement even with antibiotic therapy, it is reasonable to consider checking for antibiotic resistance. However, evaluating for hyperandrogenism should be prioritized due to its potential role in the patient's condition.

Option D: Looking for drug triggers may be relevant in some cases, but it is less likely to be the primary cause of acne that is not responding to isotretinoin and antibiotics. Drug-induced acne typically resolves once the offending drug is discontinued.

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References: Williams Gynecology 4th edition, Page no. 392-394
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8826298/

Q42. Dermatology NEET 2023 ID: MF8277

A young woman presents with complaints of a painless ulcer in the genital area. It is associated with non-tender inguinal lymphadenopathy. What is the most likely diagnosis?

A Chancroid

B ✓ Syphilis

C Herpes genitalis

D Granuloma inguinale

Correct Answer: Syphilis

The given clinical scenario of a painless ulcer and non-tender lymphadenopathy is suggestive of **primary syphilis.**

The incubation period of syphilis is **9-90 days** and varies inversely with the size of the Spirochaete inoculum.

Stages of syphilis:

- **Primary** chancre: **Painless**, **indurated**, button-like ulcer with **painless lymphadenopathy**.
- **Secondary syphilis** is the stage when generalized manifestations of syphilis appear on the **skin** and **mucous membranes**. A **macular rash** is commonly seen and this stage may be preceded by constitutional symptoms such as fever, sore throat, and malaise.
- **Latent syphilis** is the stage when there are **no clinical features** of the active disease but **serological testing** for the disease is **positive.**
- **Late syphilis (tertiary syphilis)** commonly presents as superficial or nodular **punched-out ulcers** or **gummas** (granulomas appearing as cutaneous plaques or nodules with central ulceration) that appears 3-10 years after infection. It may also present as red nodules (nodular syphilides), neurosyphilis, and cardiovascular syphilis.

Serological testing is done using standard non-treponemal antibody tests like **VDRL** and **rapid plasma reagin** (RPR). Specific treponemal antibody tests like *Treponema pallidum* hemagglutination assay (**TPHA**) and fluorescent treponemal antibody absorption test (**FTA-ABS**) are further required. Microscopically, *T. palladium* is identified with characteristic **corkscrew** morphology and motility on **darkfield microscopy.**

Parenteral penicillin G is the **drug of choice** for **all stages of syphilis**. Treatment of recent sexual contacts is recommended.

Other options:

Option A: Chancroid presents with a painful genital ulcer and tender inguinal lymphadenopathy. It is caused by the bacterium *Haemophilus ducreyi*.

Option C: Herpes genitalis is characterized by painful vesicular lesions that progress to ulcers in the genital region, caused by the herpes simplex virus (HSV). It is usually associated with localized tender lymphadenopathy.

Option D: Granuloma inguinale (also known as donovanosis) is caused by the bacterium *Klebsiella granulomatis*. It presents as a slowly progressive, painless, beefy-red ulcerative lesion in the genital region, often without lymphadenopathy.

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References: Rook's Textbook of Dermatology - Volume II -9th Edition, Page no 29.8-29.10,29.13, 29.15, 29.16
Online Resource, https://pubmed.ncbi.nlm.nih.gov/34055908/
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https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3537782/#:~:text=The%20%E2%80%9Cmoth%2Deaten%E2%80%9D%20pattern,pathognomonic%20manifestation%20of%20secondary%20syphilis.&text=The%20alopecia%2C%
Jawetz, Melnick, & Adelberg's Medical Microbiology- 28th Edition, Page no: 339,341-342
Ananthanarayan and Paniker's Textbook of Microbiology - 9th Edition, Page no 379 - 383

Q43. Dermatology NEET 2023 ID: MF8266

Which of the following is associated with the clinical condition shown in the image?

A Cataract

B Glaucoma

C ✓ Malignant melanoma

D Basal cell carcinoma

Correct Answer: Malignant melanoma

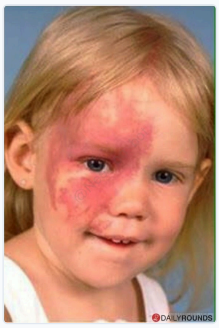
The given image showing a black, heterogeneous, hairy palpable lesion in a child is suggestive of **congenital melanocytic nevus. Malignant melanoma** is more commonly associated with a congenital melanocytic nevus**.**

Congenital melanocytic nevi (CMN) are benign, pigmented, melanocytic nevi present at birth. They are most commonly brown, black, purplish, or red, usually a **palpable lesion** with heterogeneous color and texture. CMN often have **thick hair** at birth, which grows at a significantly greater rate than the surrounding scalp hair.

Congenital melanocytic nevi can be classified based on size: small (<1.5 cm), medium (1.5-19.9 cm), and large or giant (>20 cm). Larger CMN have a higher risk of developing malignant melanoma. Management of CMN depends on the size and location of the lesion. Smaller nevi may be observed, while larger nevi or those at a higher risk for complications may require surgical excision, particularly if there is a concern for malignant transformation. Complications of CMN include neurological abnormalities (**intraparenchymal melanosis** is the commonest finding), **malignant melanoma**, and other tumors like **rhabdomyosarcoma.**

The lifetime risk of malignant transformation in large or giant CMN ranges from 5% to 10%. Regular follow-up and monitoring for changes in size, color, or texture are essential for early detection of potential complications.

Note: More than 50% of patients with **Sturge-Weber syndrome** have **glaucoma** as an ocular complication. The image given below shows port-wine staining in a child with Sturge-Weber syndrome.



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References: Rook's Textbook of Dermatology 9th edition, Page no .75.9 to 75.14, 73.2, 73.3

Q44. Dermatology NEET 2019 ID: MC4001

A 15-year-old girl presents with itchy lesions on her arm as shown. Her family history is positive for asthma. What could be the most probable diagnosis?

A Seborrhoeic dermatitis

B ✓ Atopic dermatitis

C Allergic contact dermatitis

D Erysipelas

Correct Answer: Atopic dermatitis

The clinical scenario depicting **pruritic** lesions at a **flexural site** along with a family history of **atopy** (asthma) is suggestive of **atopic dermatitis** (AD) or atopic eczema (AE). The **scratch marks** and **excoriations** shown in the image are due to pruritis, which is the hallmark feature of AD.

AD is a **chronic** relapsing **inflammatory** condition. It presents with **erythematous macules**, papules and **vesicles** that often start in **early** childhood, usually less than 2 years old. The lesions later become excoriated and **lichenified**. It is frequently associated with **dryness** of the skin and **secondary infections.** In infancy, the lesions initially appear on **face** (sparing the napkin area) and later on **extensor** aspect of knees and elbows as the child starts crawling. In later childhood and in adults, the principal site of involvement is **flexures**.

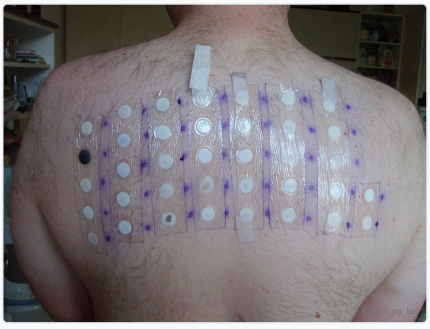
The **filaggrin** monomers in keratinocytes in the epidermis bind and aggregate keratin bundles and intermediate filaments to form the **epidermal barrier**. Any **damage/dysfunction** of this barrier is associated with an **increased risk** of developing AD. Damage to this barrier can be due to **environmental factors** (detergents, hard water, UV light) or **local inflammatory** reactions. Dysfunction of the barrier can be due to **genetic mutation** of the **FLG** gene leading to reduced filaggrin production or **local immunomodulatory effects** leading to reduced filaggrin expression.

Other options:

Option A: **Seborrheic dermatitis** is a chronic inflammatory skin condition characterized by erythema, pruritus, **scaling**, flaking, or crusting of the skin. It is often associated with **Malasezzia** fungal infection. A milder form of seborrheic dermatitis of the scalp is commonly known as **dandruff**.

Option C: **Allergic contact dermatitis (ACD)** is an inflammatory condition caused due to a **delayed hypersensitivity reaction (Type 4 hypersensitivity)** to a specific allergen that comes into direct contact with the skin.Patients usually present with **pruritus, erythema, swelling, papules,** and **papulovesicles.** Some common allergens include **nickel** (found in **artificial jewelry**), **PPD**/Para phenylene diamine (in **hair dye**), **chromate** (in **cement**), rubber or **latex**, cosmetics, etc. **Patch test** (image below) is used for the diagnosis of ACD.

Option D: **Erysipelas** is a bacterial skin infection caused by **Streptococcus pyogenes (Group A Streptococcus)** characterized by raised, erythematous sharply demarcated rapidly spreading skin lesions, along with **systemic symptoms** like fever and chills. The skin may also have a **peau d'orange** or orange peel-like appearance.



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References: Rook's Textbook of Dermatology 9th edition, Page no: 26.18, 40.1, 41.1, 128.1

Q45. Dermatology NEET 2019 ID: MA3381

Bindi Leukoderma is caused by which chemical?

A Mono-benzyl ether of Hydroquinone (MBH)

B Crocein Scarlet MOO and Solvent Yellow 3

C p-phenylenediamine (PPD)

D ✓ Para Tertiary butylphenol (PTBP)

Correct Answer: Para Tertiary butylphenol (PTBP)

Bindi leukoderma is caused by **para tertiary butylphenol (PTBP)** present in adhesive sticker bindi.

- This is called as **Contact Leukoderma** or **Chemical Leukoderma** in which the chemical increases sensitivity and induces apoptosis of the melanocytes in the area of contact.
- There is also evidence that **azo dyes**, by virtue of structural similarity to tyrosine, can **directly inhibit melanogenesis** by competing with that amino acid for **binding with tyrosinase enzyme.**

Common chemicals in the Indian set up which cause Chemical leukoderma are:

Bindi Leukoderma	**Para Tertiary butyl phenol (PTBP)**
Synthetic leather, rubber slippers, Latex condoms, rubber gloves	**Mono-benzyl ether of Hydroquinone (MBH)** -Rubber Antioxidant
Hair dye	**p-phenylenediamine (PPD)**
Alta (red colour like henna)	**Crocein Scarlet MOO and Solvent Yellow 3** (azo dyes)

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References: Journal, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2965911/

Q46. Dermatology NEET 2019 ID: MC1414

The most common triggering factor of the given condition is _____

A Vaccination

B Malignancy

C Drugs

D ✓ Infection

Correct Answer: Infection

The given image shows characteristic **target lesions** of **erythema multiforme**. This most commonly occurs following a **herpes simplex virus infection.**

The common **triggers** are:

- Herpes simplex infection
- Mycoplasma infection
- Bacterial infections
- Fungal infections
- Vaccinations
- Malignancy- carcinoma, lymphoma, leukemia
- Drugs- sulfonamides, rifampicin, penicillins, hydantoin derivatives
- Miscellaneous- lupus erythematosus (Rowell Syndrome), sarcoidosis, polyarteritis nodosa

Erythema multiforme is best regarded as a **self-limiting cytotoxic dermatitis** resulting from **cell-mediated hypersensitivity.**

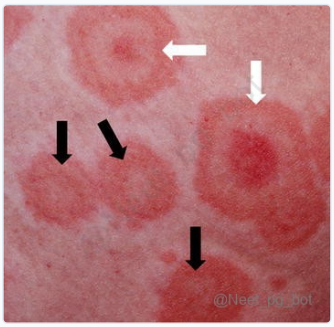
It presents clinically with a spectrum of macular, papular, or urticarial lesions, as well as the classic acral iris or **'target lesions'**.

Target lesions are **less than 3 cm in diameter**, rounded, and have **three** zones:

1. Central area of **dusky erythema** or purpura
2. Middle paler zone of **edema**
3. Outer ring of **erythema** with a well-defined edge

Lesions may involve the palms and the trunk. Oral and genital mucosal membranes lesions are associated with erosions

The image below shows target lesions.



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References: Rook's Textbook of Dermatology 9th edition, Chapter 47, Page no 47.1-5

Q47. Dermatology NEET 2019 ID: MC4908

A young female presented with vaginal itching and green frothy genital discharge. Strawberry cervix is seen on examination. What will be the drug of choice?

A Doxycycline

B Oral fluconazole

C ✓ Metronidazole

D Azithromycin

Correct Answer: Metronidazole

Vaginal itching and green frothy discharge in a young female with strawberry vagina points towards **trichomoniasis**. **Metronidazole** is the drug of choice for trichomoniasis.

It is is the most common parasitic infection in humans in the industrialized world.

It is caused by Trichomonas vaginalis, an anaerobic, flagellated protozoa, exists only as a trophozoite (no cyst stage).

Diagnosis is made by detection of motile flagella during microscopic examination of wet mounts. **Direct immunofluorescent antibody staining** is more sensitive (70-90%) than wet mount.

Culture is done on **Feinberg-Whittington media**.

Treatment: DOC- **Metronidazole 2 gm single oral dose**. All sexual partners must be treated to prevent reinfection, especially from asymptomatic males. Tinidazole can also be used for the treatment of trichomoniasis.

The image below shows strawberry vagina seen in trichomoniasis:

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References: Rook's Textbook of Dermatology 9th edition, Page no 33.36
Current Medical Diagnosis and Treatment - 2019, Page no 1534

Q48. Dermatology NEET 2019 ID: MC2202

What is correct regarding the diagnosis based on the given image?

A The lesions are not infectious

B Trigeminal dermatome is most commonly affected

C Anterior nerve roots are more commonly involved

D ✓ Mucous membranes within the affected dermatomes are involved

Correct Answer: Mucous membranes within the affected dermatomes are involved

The above image shows **unilateral**, **painful**, **grouped vesicles** in a single **dermatome** suggestive of herpes zoster or shingles. The **mucous membranes** within the affected dermatomes are involved in shingles.

Herpes zoster/shingles is an acute cutaneous segmental eruption due to the reactivation of latent varicella-zoster virus from **dorsal root ganglia**. An earlier infection with varicella is essential before zoster can occur.

The first symptom of zoster is usually pain, 1-3 days after which the lesions appear. Occasionally, the pain may not be followed by the eruption ('zoster sine eruptione'). The patients are **infectious**, both from the virus in the lesions and nasopharynx.

The most commonly involved dermatomes in decreasing order of frequency are **thoracic** > cervical > **trigeminal** > lumbosacral. The incidence of ophthalmic zoster increase with age. The posterior nerve roots and ganglia show inflammatory changes.

Herpes zoster ophthalmicus is a variant of zoster involving the ophthalmic division of the trigeminal nerve (V1). Herpes zoster oticus/Ramsay Hunt syndrome is another variant involving the geniculate ganglion and presents with ipsilateral facial nerve palsy, otalgia and vesicles over external ear.

Zoster is a **self-limiting infection** but it is painful and carries a risk of secondary infection and post-herpetic neuralgia. **Analgesia** and treatment of secondary infections are sufficient for mild infections.

Indications for antivirals in VZV (**Acyclovir 800 mg**):
• Painful zoster infections in adults
• Facial zoster at any age
• Immunocompromised

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References: Rook's Textbook of Dermatology 9th edition, Chapter 25, Page nos 25.27-25.28

Q49. Dermatology NEET 2019 ID: MC7576

What is the diagnosis seen in the image below?

A Candidal paronychia

B Staphylococcal scalded skin syndrome

C ✓ Candidal intertrigo

D Diabetic foot ulcer

Correct Answer: Candidal intertrigo

The condition shown in the image is **candidal intertrigo.**

It is a candidal infection of opposing skin surfaces, which presents as **erythema with a little moist exudation** starting deep in a fold. When the web spaces of the toes or fingers are affected, **marked maceration with a thick, white, horny layer** is usually prominent.

As the condition progresses, it spreads beyond the area of contact, usually developing the typical features of candidosis with a **fringed, irregular edge** and **subcorneal pustules** rupturing to give tiny erosions, and then further peeling of the stratum corneum.

Treatment:

Topical antifungal agents such as nystatin, ketoconazole, and clotrimazole are routinely used.

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References: Rook's Textbook of Dermatology 9th edition, Chapter 32, Page nos 32.63-32.64
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5909782/#__sec15title

Q50. Dermatology NEET 2019 ID: MC7578

Which of the following is commonly associated with the condition shown below?

A Oil drop sign

B Leukonychia striata

C Pterygium of nails

D ✓ Pitting of nails

Correct Answer: Pitting of nails

The condition shown in the image above is **alopecia areata**. Classically, it causes fine stippled **pitting of the nails.**

It is a **chronic inflammatory disease** which causes **non-scarring patches** of hair loss, especially in **genetically predisposed.**

The most characteristic lesion is a **smooth circumscribed patch of alopecia,** which has **short extractable broken hairs** called as **exclamation mark hair.** However, there is always **sparing of white hairs** in the region affected.

Other options :

Subungal oil drop sign is a **highly specific** pathognomonic sign seen in Nail **psoriasis.** It is a yellow-red discolouration in the nail bed resembling a drop of oil.

Leukonychia striata are lines that extend across the nail and lie parallel to lunula. They are seen in Hypoalbuminemic states such as nephrotic syndrome .

Nail Pterygium is a wing-shaped deformity of nail due to a **central fibrotic band** which divides the nail into two parts in the proximal.

- **Dorsal pterygium** is seen in **lichen planus.**
- **Ventral pterygium** is seen in trauma, systemic sclerosis, Raynaud phenomenon, lupus erythematosus, familial subungual pterygium, and infections.

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References: Rook's Textbook of Dermatology 9th edition, Chapter 89, Page nos 89.28-33, Chapter 35, Page no 35.9
Harrison's Principles of Internal Medicine, 21st Edition, Page no 384

Q51. Dermatology NEET 2019 ID: MC9586

Identify the condition given in the image :

A Dermatomyositis

B ✓ Acanthosis nigricans

C Pityriasis rotunda

D Melasma

Correct Answer: Acanthosis nigricans

The condition in the given image is **acanthosis nigricans.** It is characterized by **thickened, gray-brown velvety plaques** seen in **flexural** areas such as the back of the neck, axillae, inframammary creases, waist, and groin.

The causal mechanism is due to the secretion of epidermal growth factors in the region involved. It can be benign or malignancy-associated.

- **Benign -** commonly associated with **obesity** or **insulin resistance**, usually mild.
- **Malignancy -** much less common, associated with **gastric, lung,** and **uterine** cancers.

Acanthosis nigricans can also be seen in the following:

- Drugs such as nicotinic acid used to treat hyperlipidemia
- Sudden onset of multiple seborrhoeic keratoses

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References: Rook's Textbook of Dermatology 9th edition, Chapter 147, Page nos 147.14-147.15
Williams Gynecology 3rd edition, Page no 391

Q52. Dermatology NEET 2022 ID: MF5231

A patient presents to you with multiple anogenital warts. The biopsy of these lesions showed squamous atypia. Which of the following human papillomavirus types are considered high-risk?

A HPV 2

B ✓ HPV 18

C HPV 6

D HPV 11

Correct Answer: HPV 18

Human papillomavirus (HPV) **types 16** and **18** are considered **high-risk** types that are commonly implicated in cervical, vaginal and vulval cancer. Other high-risk types include HPV types 31, 33, and 35.

The **E6** protein of high-risk types binds to the **p53** gene while the **E7** protein inactivates the **pRb** gene. p53 and pRb are tumor suppressor genes that when inactivated results in uncontrolled cell proliferation. The most common type of cancer seen is **squamous cell** carcinoma. The image below depicts vulval carcinoma which is also associated with HPV infection.

The image given above shows anogenital warts pointing to **condyloma acuminata** (venereal warts). It is commonly caused by **HPV types 6 and 11.** However, rarely HPV 16 and 18 can cause it too. It is **sexually transmitted**. Patients are advised not to involve in intercourse when active genital warts are present. It can appear discrete or may coalesce to resemble cauliflower-like growths. **20%** of cases may develop into cervical, vaginal, or vulval cancers. Risk factors include **oral contraceptives** and pregnancy. The condition is diagnosed by **colposcopy**.

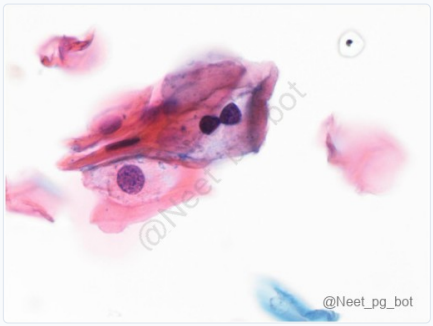
The presence of **koilocytes** on histopathological examination is pathognomonic of the virus. These are cells with irregular and hyperchromatic nuclei, perinuclear halo, and peripheral cytoplasmic condensation. These cells disappear when there are dysplastic changes.

The image below depicts koilocytes.

Other options:

Option A - **HPV type 2** is responsible for **common warts** in non-genital regions.

Options C and D - **HPV types 6** and **11** are responsible for **anogenital warts** and **laryngeal papillomatosis**.



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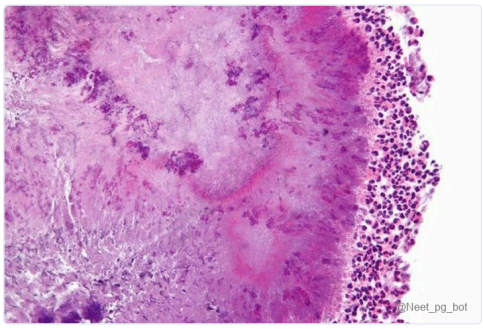
Scott-Brown's Otorhinolaryngology and Head and Neck Surgery 8th Edition: Volume 3: Head and Neck Surgery, Plastic Surgery, 20
Jawetz, Melnick, & Adelberg's Medical Microbiology- 28th Edition, Page no. 650
Rook's Textbook of Dermatology - 9th Edition - Volume I, Page no. 25.55
Online Resource, https://pubmed.ncbi.nlm.nih.gov/9434721/
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK321770/

Q53. Dermatology NEET 2022 ID: MF5280

A farmer presents to you with a swelling on the foot and multiple discharging sinuses in the lower limb. Granules from the discharge were examined under the microscope, which is shown below. Which of the following is true regarding this condition?



A ✓ Both bacteria and fungi can be causative

B Undergoes lymphatic spread

C There is lymphocyte accumulation

D Involves only superficial tissues

Correct Answer: Both bacteria and fungi can be causative

The clinical vignette along with the image showing sulfur granules is consistent with the diagnosis of mycetoma. **Both bacteria and fungi** can be causative agents of the disease.

Mycetoma is a chronic, slow-progressing **subcutaneous** infection that commonly affects the **foot**. Other sites which can be affected include the hands, upper back, and posterior part of the neck. The mode of infection is usually **traumatic**. **Agricultural workers** are more prone to infection.

There are three types - eumycetoma, actinomycetoma, and botryomycosis. **Eumycetoma** is caused by fungal species such as ***Madurella mycetomatis*** and *Madurella grisea*. Adjacent fascia and bones may be involved as the disease progresses.

Actinomyces israelii and *Nocardia asteroides* result in the bacterial form - **actinomycetoma**. It is generally more invasive and can spread via the fascial planes to the underlying **muscle** or bone. Dissemination is rare but can occur in lesions of the head and neck.

Staphylococcus aureus and other pyogenic bacteria can cause botryomycosis.

Initially, it starts off as a small **subcutaneous swelling** of the foot, which then enlarges to form **multiple discharging sinuses**. The discharge contains **granules** which are microcolonies of the causative organism. Different causative organisms produce granules that are characteristic in color, consistency, and size, which helps in delineating the agent in each case.

The granules in eumycetoma are generally **hard,** while that in actinomycetoma is soft in consistency.

Medical management of eumycetoma includes ketoconazole and miconazole, while that of actinomycetoma includes dapsone or rifampicin. It should be continued for **18-24 months** even after symptomatic recovery or the absence of laboratory evidence for infection.

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References: Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition, Page no 401, 604, 605
Jawetz, Melnick, & Adelberg's Medical Microbiology- 28th Edition, Page no. 689
Harrison's Principles of Internal Medicine, 21st Edition, Page no. 1338, 1688

Q54. Dermatology NEET 2022 ID: MF5262

A farmer presents to you with a cauliflower-shaped mass on the foot, which developed after a minor injury. Microscopy shows copper penny bodies. What is the most likely diagnosis?

- A ✓ Chromoblastomycosis
- B Blastomycosis
- C Sporotrichosis
- D Phaeohyphomycosis

Correct Answer: Chromoblastomycosis

The most likely diagnosis based on the clinical vignette is **chromoblastomycosis**.

Chromomycosis is a term used to describe various conditions caused by pigmented (dematiaceous) fungi. Chromoblastomycosis, the most common chromomycosis, is more commonly seen in **tropical** areas among **agricultural workers** and woodcutters who walk barefoot. **Traumatic** implantation is the mode of infection.

Patients present with warty **cauliflower-like** cutaneous nodules on the feet and lower legs. Species of the fungal genus ***Fonsecaea*** are implicated as the causative agents. Diagnosis is made by the demonstration of **sclerotic bodies** (**copper-penny bodies**) in tissue sections. Sclerotic bodies appear as round or irregular, **dark brown** bodies with **septae**.

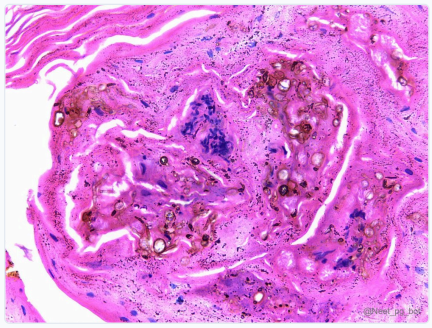
It can be managed with **itraconazole** (100–200 mg daily) or terbinafine (250 mg daily). For small lesions, **surgical excision** is the therapy of choice.

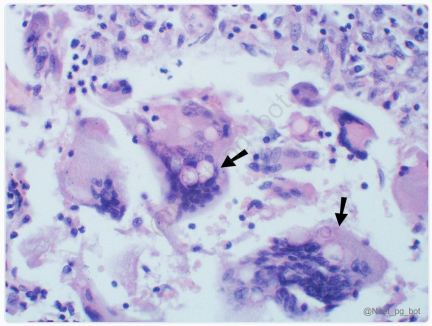
Other options:

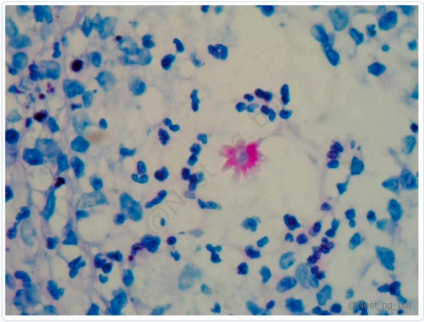
Option B - **Blastomycosis** is an infection caused by ***Blastomyces dermatitidis***. It causes a chronic infection with granulomatous and suppurative lesions, which starts in the lungs but may disseminate to other organs. Symptoms of an **acute lower respiratory tract infection** with pulmonary infiltrates are the most common clinical presentations. Microscopic examination of stained specimens shows **broad-based budding yeast cells**.

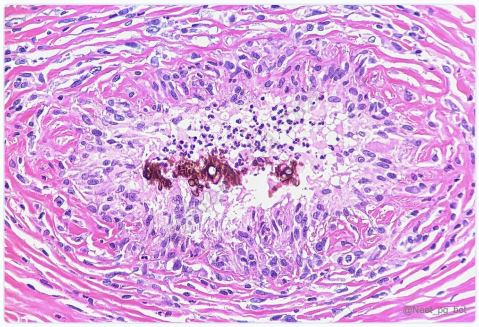
Option C - **Sporotrichosis** caused by the dimorphic fungus ***Sporothrix schenckii***, is commonly seen in gardeners followed by minor **trauma**. Patients present with **skin nodules** or lymph node ulceration. Microscopy shows **asteroid bodies**. It can be seen as an oval, basophilic structure with **eosinophilic** radiation.

Option D - **Phaeohyphomycosis** caused by **dematiaceous fungi** is frequently encountered in **immunodeficient** individuals. Clinical manifestations include **brain abscesses**, **intramuscular abscesses**, and **cysts**. On microscopy, the fungi appear as **distorted hyphal strands**. In contrast to chromoblastomycosis, sclerotic bodies are **not** found.









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References: Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition, Page no. 605, 606, 610, 611
Jawetz, Melnick, & Adelberg's Medical Microbiology- 28th Edition, Page no. 687, 688, 698, 699
Rook's Textbook of Dermatology 9th edition, Page no 32.78

Q55. Dermatology NEET 2022 ID: MF5286

A patient presents with an anesthetic patch in the areas on the face as shown in the image below. Which of the following nerves is the most commonly involved in this condition?

A Abducens nuclei

B ✓ Facial nerve

C Optic nerve

D Trigeminal nerve

Correct Answer: Facial nerve

The clinical scenario points to a case of **leprosy** and the **most commonly involved cranial nerve** is the **facial nerve.**

Leprosy is a chronic granulomatous condition caused by ***Mycobacterium leprae***. It affects the skin, cutaneous and peripheral nerves, and mucous membranes. The most commonly affected peripheral nerves are the **posterior tibial nerve** followed by **ulnar**, median, and lateral popliteal nerves. The most commonly involved cranial nerve is the **facial nerve**.

Leprosy has a **long incubation** period, usually ranging from 2 to 12 years. It commonly presents as an **area of numbness** on the skin or a visible skin lesion which could be a **hypopigmented** or **erythematous macule**. Lesions usually involve the face, extensor aspect of the limbs, trunks, and buttocks. The lesions have **impaired nerve function** and poor hair growth. The patients may also present with burns or ulcers of the anesthetic limb.

Classification systems of leprosy include:

- **Ridley Jopling** classification:
  - Tuberculoid (TT)
  - Borderline tubercular (BT)
  - Borderline (BB)
  - Borderline lepromatous (BL)
  - Lepromatous (LL)
- **WHO** classification:
  - Paucibacillary
  - Multibacillary

The **cardinal signs** for the diagnosis of leprosy include:

- Characteristic **skin lesions**- hypopigmented or erythematous lesions with definite **impairment of sensation.**
- **Peripheral nerve** involvement- **thickening** of the nerve with sensory impairment. Nerve tenderness can also be seen.
- Detection of **acid-fast bacilli** in slit-skin smear, skin biopsy, or positive biopsy PCR.

The presence of two of the three signs can establish the diagnosis of leprosy. Other tests done for diagnosis of leprosy include nerve biopsy in pure neural leprosy and PGL-1 antibody in cases in low bacterial load.

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References: Rook's Textbook of Dermatology - 9th Edition - Volume I, 28.1, 28.3, 28.13
Harrison's Principles of Internal Medicine, 21st Edition, 1385, 1386, 1388

Q56. Dermatology NEET 2022 ID: MF5260

A female patient with a body mass index of 30 kg/m2 presents to you with a lesion on the neck as shown below. Which of the following condition is she most likely to be suffering from?

A Hypothyroidism

B ✓ Metabolic syndrome

C Addison's disease

D Hyperparathyroidism

Correct Answer: Metabolic syndrome

The patient is most likely to be suffering from **metabolic syndrome** based on the clinical scenario given and the image depicting **acanthosis nigricans**.

Metabolic syndrome (syndrome X) is a group of conditions that increases the risk for cardiovascular disease. These include **hypertension**, **hyperglycemia**, **hypertriglyceridemia**, and **central obesity**. Increasing age, genetics, and sedentary lifestyle are considered risk factors.

Insulin resistance is the most accepted hypothesis used to explain the pathophysiology of the condition. An elevated level of circulating fatty acids is considered a major factor in the development of insulin resistance. It is associated with **acanthosis nigricans** which are **velvety hyperpigmented** lesions in **skin flexures** such as the axilla, groin, and back of the **neck**. Acanthosis nigricans can also be seen in **malignancies**, especially those of the gastrointestinal tract, **acromegaly**, and **Cushing's syndrome**.

Other conditions associated with metabolic syndrome include **polycystic ovary syndrome**, **obstructive sleep apnea**, and non-alcoholic fatty liver disease. The condition is managed by dietary and lifestyle modifications. Individual components of the syndrome also need to be addressed by the use of statins, antihypertensives, or hypoglycemic agents.

Other options:

Option A - **Hypothyroidism** is more common in women and can be either congenital or acquired. Autoimmune and iatrogenic hypothyroidism are the common causes of the latter. Symptoms include **dry coarse skin**, **weight gain**, poor appetite, fatigue, and **alopecia**.

Option C - **Addison's disease** (primary adrenal insufficiency), most commonly caused by **autoimmune adrenalitis** presents with **hyperpigmentation** of the skin, especially in areas exposed to **sunlight** or prone to **friction**. This is due to excessive stimulation of **melanocytes** by the raised adrenocorticotropic hormone (**ACTH**). This is seen in chronic cases.

Option D - **Hyperparathyroidism** is a disorder that arises due to excessive secretion of parathyroid hormone. Solitary adenomas are the most common cause. It primarily affects the **renal** and **skeletal** systems. **Sestamibi scan** is the imaging of choice to delineate parathyroid abnormalities.

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page no. 390, 2933, 2934, 2936, 2970, 2971, 3086, 3150, 3151, 3154, 3155, 3156, 3173, 3175
Current Medical Diagnosis and Treatment 2022, Page no. 1157

Q57. Dermatology NEET 2022 ID: MF5248

Irregular pitting of nails with subungual hyperkeratosis is seen in _____.

A Lichen planus

B ✓ Psoriasis

C Atopic dermatitis

D Alopecia areata

Correct Answer: Psoriasis

Irregular pitting of nails with **subungual hyperkeratosis** is seen in **psoriasis**.

Psoriasis is a chronic inflammatory condition of the skin with systemic manifestations. **Plaque psoriasis** is the most common type. The nails are frequently involved, and **punctate pitting** is the most common finding seen. Nail beds can also be affected, giving rise to the **oil-drop sign**—a highly specific sign seen in this condition. Other manifestations include **subungual hyperkeratosis**, **splinter hemorrhages**, and distal onycholysis.

The skin is involved in the form of red/ **salmon pink**, **scaly**, and well-demarcated lesions. A peripheral clearing (**ring of Woronoff**) may be present. It is usually seen on the scalp, trunk, and **extensor surfaces** of the limbs. The lesions are surrounded by **silvery-white scales**. The appearance of a smooth, glossy red membrane with small bleeding points on the removal of the supra-papillary epidermis by gentle scraping (**Auspitz's sign**) is characteristic.

Psoriasis is associated with **Koebner** or **isomorphic phenomenon**. It describes the appearance of new lesions along the site of **trauma** (linear lesions). It is also observed in lichen planus.

Topical application of **steroids**, retinoids, vitamin D, and **emollients** is the mainstay of management. **Phototherapy** and oral **methotrexate**, **acitretin**, and cyclosporine can also be used. Both methotrexate and acitretin are teratogenic drugs. It is recommended that women do not conceive for **three years** after cessation of the acitretin.

Other options:

Option A - **Lichen planus** is an inflammatory disorder affecting the skin and mucosal membranes. Nail involvement includes exaggeration of the longitudinal lines and linear depression. This is due to nail plate thinning. If the nail bed is involved, it can result in **hyperpigmentation**, **subungual hyperkeratosis**, and onycholysis. In some cases, **pterygium unguis** can be seen due to adhesion between the dorsal nail fold and the nail bed.

Option C - **Atopic dermatitis** is a relapsing chronic inflammatory skin disease. Patients present with itchy erythematous papules/papulo-vesicles in **flexural** regions. Nail involvement is in the form of **coarse pitting** and **ridging**.

Option D - **Alopecia areata** is a condition that results in non-scarring hair loss. It can cause fine **stippled pitting** of the nails, roughening of the nail plate (**trachyonychia**), or **atrophic dystrophy**.

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References: Rook's Textbook of Dermatology - Volume II -9th Edition, Page no. 35.1, 35.8, 35.9, 35.10, 35.12, 35.13, 35.14, 35.22, 35.25, 35.26, 35.28, 35.32, 35.33, 35.34, 37.1, 37.12, 41.1, 41.21
Rook's Textbook of Dermatology - 9th Edition - Volume III, Page no. 89.28, 89.32
Harrison's Principles of Internal Medicine, 21st Edition, Page no. 378
Rook's Textbook of Dermatology - 9th Edition - Volume I, Page no. 4.9

Q58. Dermatology NEET 2022 ID: MF5223

A 35-year-old woman presents to you with hair loss for the past three months. She tested positive for COVID-19 eight months ago. What is the most likely diagnosis?

A Tinea capitis

B ✓ Telogen effluvium

C Trichotillomania

D Female-pattern androgenic alopecia

Correct Answer: Telogen effluvium

The most likely diagnosis based on the clinical scenario is **telogen effluvium**.

The hair follicle goes through three phases known as the hair cycle:

- Anagen: Phase of active hair growth
- Catagen: Involutionary phase
- Telogen: Time period between catagen and next anagen phase

Telogen effluvium is a type of **nonscarring alopecia** that is marked by **diffuse shedding** of normal hair. This occurs due to **premature termination** of the anagen phase of the hair cycle. It can be either acute or chronic.

Acute telogen effluvium occurs **2-3 months** after a stressful event such as high **fever**, **surgery**, starvation, and hemorrhage. Hormonal fluctuations during the **postpartum** period can trigger it (**telogen gravidarum**). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally **resolves spontaneously**.

Chronic telogen effluvium is used to describe the diffuse shedding of telogen hair for more than **6 months**. Common causes include **thyroid disorders**, iron-deficiency anemia, and malnutrition.

Other options:

Option A - **Tinea capitis** is used to describe an infection of the scalp by the **dermatophyte** group of fungi. The most common pathogenic species are *Trichophyton tonsurans* and *Trichophyton violaceum*. The presentation can vary from hair breakage associated with scaling to widespread inflammatory mass on the scalp.

Clinical variants of tinea capitis:

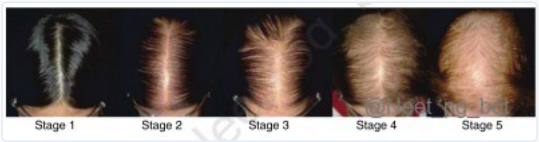
- **Endothrix** - A non-inflammatory type in which **multiple black dots** are present within the areas of alopecia
- **Ectothrix** - A mild inflammatory type in which single or multiple scaly patches with hair loss are seen
- **Kerion** - A severe inflammatory type that presents with an inflammatory mass with **thick crusting** and **matting of hair** with pus and sinus formation
- **Favus** - Characterized by the presence of yellowish, cup-shaped crusts known as **scutula** that develop around hair follicles

Observation under **KOH mount** is used for diagnosis.

Option C - **Trichotillomania** (trichotillosis) is considered an **obsessive-compulsive disorder (OCD)** spectrum disorder, marked by gratification from pulling one's own hair. It could be associated with underlying anxieties or depression. The **vertex** region of the scalp is most commonly involved. Scalp histology will show **perifollicular hemorrhages** and intrafollicular pigment casts.

Option D - **Androgenic hair loss** is used to describe a progressive decrease in hair fiber production. **Dihydrotestosterone**, an androgenic metabolite, is implicated in the etiology. Female-pattern hair loss involves the **mid-frontal scalp** diffusely. There is no or minimal involvement of the temporal region.

Endothrix variant of tenia captitis



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References: Rook's Textbook of Dermatology - 9th Edition - Volume III, Page no. 32.24, 32.39, 32.40, 86.17, 89.6, 89.7, 89.14, 89.17, 89.20, 89.24, 89.25, 89.46
Harrison's Principles of Internal Medicine, 21st Edition, Page no. 387

Q59. Dermatology AIIMS 2018 ID: MC5182

Which of the following nerves is biopsied for the diagnosis of pure neuritic leprosy?

A Median nerve

B ✓ Superficial branch of radial nerve

C Ulnar nerve

D Lateral popliteal nerve

Correct Answer: Superficial branch of radial nerve

Superficial sensory branch of the radial nerve is most commonly used for **nerve biopsy in neuritic leprosy.**

Although any nerve involved would show the pathology, it is prudent to biopsy a branch of a sensory nerve, rather than a mixed nerve or major nerve trunk. So, the **ulnar nerve** will not be the best answer despite it being the **most common peripheral nerve involved** in neuritic leprosy.

The common nerves and sites where nerve biopsy is performed in leprosy are as follows:

- **Superficial sensory radial nerve branch** at the wrist
- **Ulnar cutaneous nerve** near the ulnar border of the base of the hand
- **Sural nerve** near the ankle

Note: Nerve biopsy is the **gold standard** for confirmation of pure neuritic leprosy.

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References: Rook's Textbook of Dermatology 9th edition, Page no.28.14

Q60. Dermatology AIIMS 2018 ID: MC5185

Which component of cement causes allergic dermatitis among construction workers in India?

A Cobalt

B Nickel

C Iron

D ✓ Chromate

Correct Answer: Chromate

Chromate is the **most frequent allergen in cement** causing **occupational contact dermatitis.** The **most common** allergen causing allergic contact dermatitis in India is **nickel**. It is commonly found in **artificial jewelry**. However, among construction workers, chromate is more common than nickel. The reported prevalence of allergic contact dermatitis (ACD) to chromate among construction workers is usually more than 10%.

Allergic contact dermatitis (ACD) is an inflammatory condition caused due to a **delayed hypersensitivity reaction (Type 4 hypersensitivity)** to a specific allergen that comes into direct contact with the skin. Some common allergens include **nickel** (found in **artificial jewelry**), **PPD**/Para phenylene diamine (in **hair dye**), **chromate** (in **cement**), rubber or **latex**, **cosmetics**, etc. **Patch test** is used for the diagnosis of ACD.

Patients usually present with **pruritus, erythema, swelling, papules,** and **papulovesicles.** In severe cases, large vesicles or **blisters** may form due to the disruption of the intercellular bridges. If exposure persists the affected area becomes dry, scaly, and thicker with lichenification and fissuring develops later. There is involvement of the covered areas as well as the exposed areas.

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References: Rook's Textbook of Dermatology 9th edition, Page no 128.2-128.3, 128.11 - 128.14

Q61. Dermatology AIIMS 2018 ID: MC5184

Which of the following is a drug used in the treatment of tinea cruris for which widespread resistance is found to be developing?

A Itraconazole

B Griseofulvin

C ✓ Terbinafine

D Voriconazole

Correct Answer: Terbinafine

The use of **terbinafine** as the agent of choice for **tinea cruris and corporis** has been limited lately due to the **development of resistance.**

Tinea cruris (also known as jock itch) is a dermatophyte infection involving the crural fold. The most common cause is ***Trichophyton rubrum*.**

T. rubrum isolates have been found to exhibit resistance against terbinafine.

Note: The question is based on research papers about *Trichophyton rubrum* strains exhibiting primary resistance to terbinafine.

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC148991/
Online Resource, https://ijdvl.com/the-unprecedented-epidemic-like-scenario-of-dermatophytosis-in-india-iii-antifungal-resistance-and-treatment-options
Online Resource, https://www.cdc.gov/fungal/diseases/ringworm/dermatophyte-resistance.html

Q62. Dermatology AIIMS 2018 ID: MC5183

A patient came with a history of fever, body ache, and joint pain. He was given NSAIDs. A few days later, he developed brownish pigmentation over the nose. What is the diagnosis?

A Melasma

B Fixed drug eruption (FDE)

C Dengue

D ✓ Chikungunya

Correct Answer: Chikungunya

The given history of **fever**, **joint pain**, and centrofacial **hyperpigmentation** points toward a diagnosis of **chikungunya. Post chikungunya pigmentation (PCP)** usually appears **after the fever subsides** and persists for about 3-6 months. Hyperpigmentation associated with chikungunya fever is **macular** and most commonly affects the **nose and cheeks**.

Another close differential is **fixed drug eruptions (FDE).** FDE has a **shorter latency,** usually presents **30 min to 8 h** after drug exposure, and **recurs at the same location** on re-exposure. It presents with **hyperpigmented plaques** that may be preceded by **vesiculobullous lesions**.

Hyperpigmentation of the centrofacial region (nose and cheeks) **mimics melasma** and may be misdiagnosed if proper history is not taken.

Nose pigmentation is striking in cases of chikungunya fever and is termed **chik sign.** The image below shows chik sign on the nose:

FDE is a type of **delayed hypersensitivity reaction.** They occur on **lips**, hands, legs, face, **genitalia**, and oral mucosa, and can cause a **burning sensation**. **NSAIDs, paracetamol and co-trimoxazole** are the most common drugs causing FDE. NSAID-induced FDE mainly affects the genitals and lips.

The image below shows FDE on upper lip area:

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page 141-142, 1633
Rook's Textbook of Dermatology 9th edition, 118.11

Q63. Dermatology AIIMS 2018 ID: MC7112

A mother has a pearly white papule on her forehead (as shown in the image below) which her infant also develops. What is the cause?

A HPV

B ✓ Pox virus

C Adenovirus

D Herpes virus

Correct Answer: Pox virus

The image shows **pearly white umbilicated wart-like lesions** with a characteristic **dimple at the center**, suggestive of **molluscum** contagiosum, which is caused by **pox virus.**

The age of peak incidence is between 2 and 5 years. In children, it is mainly caused by the type I molluscum contagiosum virus (MCV-1). There is a relatively higher incidence of MCV-2 type in adults and HIV infection. Sexually transmitted molluscum contagiosum may be present over the genital area.

It has an incubation period of 2 weeks to 6 months. Widespread and refractory lesions are seen most commonly in HIV or patients on immunosuppressive therapy.

It is characterized by the presence of intracytoplasmic eosinophilic inclusion bodies called **Henderson-Patterson bodies** or molluscum bodies. **Pseudo-Koebner phenomenon** is seen, that is, new lesions develop along lines of trauma.

Treatment is usually not required, and lesions are likely to **resolve spontaneously**. Stimulation of the immune response can be done after destructive or inflammatory therapies with salicylic acid, trichloroacetic acid, and imiquimod. **Cryotherapy** when done needs to be repeated at 3-4 week intervals. Extensive lesions are treated with an antiviral agent like **cidofovir**.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 25.12-25.14

Q64. Dermatology AIIMS 2018 ID: MC7067

A disease associated with the occurrence of blistering and peeling of over >30% of body surface area is usually caused by:

A Viral infection

B ✓ Drugs

C Fungal infection

D Auto-immunity

Correct Answer: Drugs

Blistering and peeling involving greater than 30% of the body surface area points to the diagnosis of **toxic epidermal necrolysis** (TEN). TEN is primarily **drug-induced,** with a culprit drug being demonstrated in approximately **85% of cases.**

Stevens-Johnson syndrome (SJS) and **toxic epidermal necrolysis (TEN)** are both severe mucocutaneous reactions, characterized by blistering and epithelial sloughing. The two terms describe phenotypes within a severity spectrum, where SJS is the less extensive form and TEN is the more extensive

Disease	Epidermal detachment	Purpuric macules and atypical target lesions
SJS	**<10%**	Present
Overlap SJS-TEN	10–30%	Present
TEN with spots	**>30%**	Present
TEN without spots	>30% (large sheets of epidermal loss)	Absent

Common **drugs** that can cause SJS/TEN are:

- Allopurinol
- NSAIDs- oxicam group
- Anti-epileptics - phenobarbital, phenytoin, carbamazepine, lamotrigine
- Antibiotics - penicillins, cephalosporins, sulfamethoxazole, sulfasalazine, nevirapine

@Neet_pg_bot

References: Rook's Textbook of Dermatology 9th edition, Page no: 119.13-119.14

Q65. Dermatology AIIMS 2018 ID: MC7108

A 23 year old patient presents with lesions in the groin as shown in the image below. Which of the following cannot be a cause of these lesions?

A Trichophyton

B Microsporum

C ✓ Aspergillus

D Epidermophyton

Correct Answer: Aspergillus

The image shows a **well-circumscribed**, **annular plaque** in the **groin** region which is characteristic of **tinea cruris**. Tinea cruris is a **dermatophytosis.** *Trichophyton*, *Microsporum* and *Epidermophyton* are dermatophytes whereas **Aspergillus** is not.

Dermatophytes include 3 genera:

1. **Trichophyton** species infect **skin, hair, and nails** and include *T. rubrum, T. mentagrophytes, T. tonsurans, T. schoenleinii, and T. violaceum*.
2. **Microsporum** contains *M. gypseum, M. canis*, and *M. nanum*. These infect only hair and skin (**not nails**).
3. Floccosum is the **only species** in the genus **Epidermophyton**, It infects only skin and nails (**not hair**).

Aspergillus *spp.* produce **large necrotic lesions** like **ecthyma gangrenosum**, and sometimes small papules and cold abscesses.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 32.18-32.19

Q66. Dermatology AIIMS 2018 ID: MA3940

A patient presents with multiple nodulocystic lesions on the face. Which among the following is the drug of choice in this condition?

A Steroids

B ✓ Isotretinoin

C Erythromycin

D Tetracycline

Correct Answer: Isotretinoin

The presence of **multiple nodulocystic lesions** implies that this is a case of **severe acne. Isotretinoin** is the first-line drug for the treatment of **severe acne** (nodular acne and acne conglobata).

Systemic retinoids are **highly teratogenic** and are absolutely contraindicated in pregnant women, those planning to become pregnant, or breastfeeding mothers. Two negative pregnancy tests (on separate occasions) are required for females of childbearing age before starting acitretin or isotretinoin.

Baseline evaluation of serum **lipids**, **transaminases**, and **complete blood counts** should be done before starting any systemic retinoids. These values should be checked **monthly** for the **first 3–6 months** and **once every 3 months** thereafter. **Monthly pregnancy tests** should be done while on the drug.

Note: Female patients should **avoid conception** for **3 years** after discontinuing **acitretin** treatment for cutaneous **psoriasis**.

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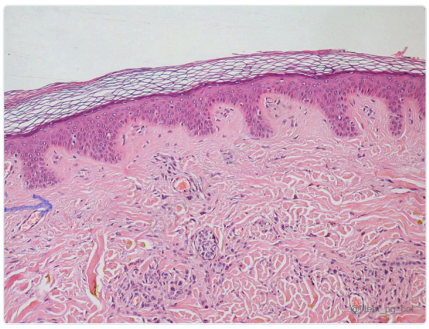
References: Rook's Textbook of Dermatology 9th edition, Page no: 90.37

Q67. Dermatology AIIMS 2018 ID: MC7050

What is the type of connective tissue present in the area marked by the arrow in the given image?



A Dense regular

B ✓ Dense irregular

C Loose irregular

D Specialised

Correct Answer: Dense irregular

The area marked is the **reticular dermis.** It is made up of **dense irregular connective tissue.**

It is found in the regions that are under **considerable mechanical stress** and where protection is given to **ensheathed organs.** It is usually present in:

- Reticular **dermis**
- Superficial **connective tissue sheaths** of the muscle and nerves
- **Adventitia** of large vessels
- **Capsules** of various organs and glands (e.g., testis, sclera, periosteum, perichondrium)

On the other hand, **dense regular collagen** fibers are found in **tendons.**

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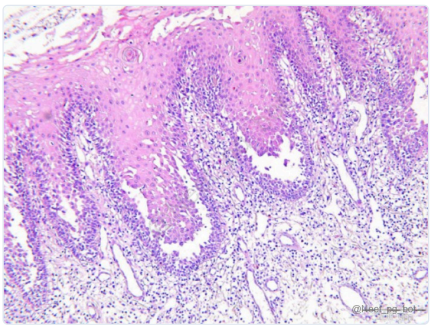
References: Rook's Textbook of Dermatology 9th edition, Page no 2.34

Q68. Dermatology AIIMS 2018 ID: MA0630

The histopathological slide given below is suggestive of:



A Psoriasis vulgaris

B Leishmaniasis

C Mycosis fungoides

D ✓ Pemphigus vulgaris

Correct Answer: Pemphigus vulgaris

The given slide shows a **row of tombstones** appearance, which is classically seen in **pemphigus vulgaris.**

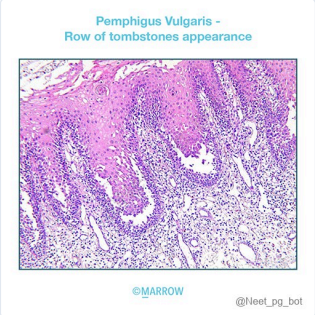
In this condition, there is a **suprabasal splitting** of the epidermis leading to blister formation. The **basal layer** remains **adherent** to the **basement membrane** and gives the resemblance to a row of tombstones as shown in the histology image below.

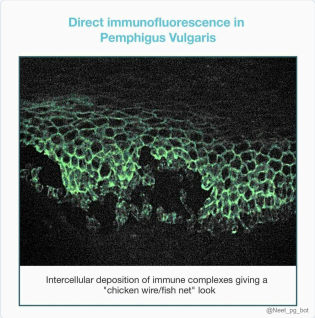
Pemphigus vulgaris is a **chronic immunobullous** disorder characterized by the presence of **antibodies** against **desmosomal proteins**, **desmogleins 1 and 3** expressed in the **skin** and **mucosa.**

Clinically, the patients present with ill-defined, **irregular, slow-to-heal** oral lesions and erosions. **Flaccid blisters** filled with **clear fluid** may arise on normal skin or an erythematous base. The rupture of such blisters produces **painful erosions** and heals **without scarring.**

Nikolsky's sign is positive which is elicited by applying firm **tangential** sliding pressure with a finger on lesional or perilesional skin. This leads to the separation of the normal-looking epidermis from the dermis, producing **erosion.**

Direct immunofluorescence is considered the most accurate **confirmatory test** of this condition. It shows intercellular **IgG** and **C3** deposition on the **surface** of **keratinocytes** throughout the epidermis in **normal, perilesional skin,** in a **characteristic fishnet/chickenwire appearance** as seen below.





@Neet_pg_bot

References: Robbins and Cotran Pathologic Basis of Disease, 10e, Page no. 1160
Rook's Textbook of Dermatology 9th edition, Page no. 50.1

Q69. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4656

A female patient presents with active genital warts with a history of recurrence, but her husband doesn't have it. What advice will you give to prevent transmission?

A Give continuous prophylaxis to the patient

B Give prophylactic anti-virals to the husband

C Give anti-virals whenever active lesions are present

D ✓ Don't involve in intercourse when active lesions are present

Correct Answer: Don't involve in intercourse when active lesions are present

The advice to prevent transmission is to **not involve in intercourse when active genital warts are present**.

Genital warts are caused by HPV and transmitted through the sexual route. Antivirals are not very effective against anogenital warts. Biological treatment options like podophyllin, imiquimod, 5-fluorouracil are given. The best way to prevent transmission of infection is by avoiding sexual intercourse when active lesions are present.

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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no 318
Robbins and Cotran Pathologic Basis of Disease, 10e, Page no 998

Q70. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4633

A male baby born to a non-consanguineous couple develops blisters on areas of friction. Their previous baby also had similar complaints and died at the age of 2 weeks. What is the diagnosis?

- A ✓ Epidermolysis bullosa
- B Chronic bullous disorder of childhood
- C Congenital syphilis
- D Neonatal pemphigus

Correct Answer: Epidermolysis bullosa

The clinical history given in the question points to a diagnosis of **epidermolysis bullosa congenita**.

Epidermolysis bullosa is a **mechanobullous disease** characterized by **blistering** in areas of **mechanical trauma.** Blisters are commonly seen in **trauma-prone areas** like **hands, feet, elbows,** and **knees.** There are three types of epidermolysis bullosa:

- **Epidermolysis bullosa simplex** - defective **K5/K14 protein**
- **Junctional epidermolysis bullosa** - defective **Laminin**
- **Dystrophic epidermolysis bullosa** - defective **Collagen-7**

The image below shows epidermolysis bullosa lesions on the foot:

Other options:

Option B: **Chronic bullous disorder of childhood** - It typically presents with bullous lesions having a **string of pearls/cluster of jewel** appearance (image below). It is an **IgA-mediated** immunobullous disorder. It manifests in early childhood and is not a congenital condition.

Option C: **Congenital Syphilis** - Syphilitic pemphigus presents as a **vesiculobullous disorder**. The investigation of choice is **VDRL**.

Option D: **Neonatal Pemphigus** - It is an **autoimmune** blistering disease caused by the transfer of **maternal IgG autoantibodies** to **desmoglein 3** to the neonate through the placenta when the mother is affected with pemphigus. It is clinically characterized by transient **flaccid blisters** and erosions on the skin and, rarely, the mucous membranes.

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References: Rook's Textbook of Dermatology 9th edition, Page nos 71.4-71.16

Q71. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4538

A rose gardener with a history of thorn prick presents with the following lesions. What is the possible causative agent?

A Streptococcus pyogenes

B ✓ Sporothrix schenkii

C Strongyloides stercoralis

D Staphylococcus aureus

Correct Answer: Sporothrix schenkii

The given image shows the **distribution of lesions along lymphatics**. This distribution along with the history of thorn pick is suggestive of the causative agent **Sporothrix schenkii**.

Sporothrix schenkii causes rose gardener's disease. It is a saprophyte that lives on grass and plants. Hence a history of thorn prick in farmers or gardeners is an important clue.

Other causes of skin lesion distributed along lymphatics is fish tankgranuloma caused by Mycobacterium marinum.

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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no 297
Harrison's Principles of Internal Medicine, 21st Edition, Page no 1542

Q72. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4512

A school-going child who had applied henna comes to dermatology OPD with the following lesions. What is the diagnosis?

A Contact dermatitis due to henna

B ✓ Dermatitis artefacta

C Polymorphous light eruption

D Vitiligo

Correct Answer: Dermatitis artefacta

The likely diagnosis in this case is **dermatitis artefacta** (factitious dermatitis).

Dermatitis artefacta is a **psychodermatological excoriation disorder** with a **female** preponderance, commonly affecting **adolescents and young adults**, where elaborate techniques (using sharp objects, cigarettes, lighters, chemicals etc.) are used to **self induce skin lesions**.

The morphology of the lesions are varied, with oddly linear, clear-cut and angular edges, **surrounded by normal looking skin**. These lesions appear with no prodromal stage or associated symptoms as they are **self inflicted.** They are distributed over easy to reach areas of the body and have a varied appearance based on the mode of injury.

- **Option A: Contact dermatitis** is due to contact with allergic agents such as paraphenylenediamine (PPD). Presentation can vary from erythema, swelling, papules, vesicles and in severe cases, blisters on the areas of contact with the allergen (henna, in this case). There exists a sequence of inflammatory changes here, unlike the lesions in dermatitis artefacta.
- **Option B: Polymorphic light eruptions** as the name suggests are seen over sunlight-exposed areas of the body. The lesions are polymorphic and are symmetrically distributed.
- **Option C: Vitiligo** is characterized by patchy involvement of skin with hypopigmentation.

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References: Rook's Textbook of Dermatology 9th edition, 128.13
Kaplan & Sadock's Concise Textbook of Clinical Psychiatry 11th Edition, 435,436
Fitzpatrick's Dermatology, 9th Edition., 405

Q73. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4174

A 40-year-old woman who presented with complaints of fever and joint pain develops the following lesion on her nose after a few days of taking NSAIDs. What is your diagnosis?

A Dengue

B ✓ Chikungunya

C Melasma

D Fixed drug eruption

Correct Answer: Chikungunya

The given history of fever and joint pain along with centrofacial hyperpigmentation shown in the image point towards a diagnosis of **chikungunya**. Hyperpigmentation associated with chikungunya fever is **macular** and most commonly **affects the nose and cheeks**.

Nose pigmentation is striking in cases of Chikungunya fever and is termed as **Chik sign.** It appears as the fever subsides and persists for about 3-6 months.

Hyperpigmentation of the centrofacial face (nose and cheeks) mimics melasma and may be misdiagnosed if proper history is not taken.

Another close differential is **fixed drug eruptions (FDE)**. FDE is a type of **delayed hypersensitivity reaction.** It is characterized by well-defined brown, **hyperpigmented plaques** that may **follow vesiculobullous lesions.** They occur on **lips**, hands, legs, face, **genitalia**, and oral mucosa, and can cause a **burning sensation**. They usually present 30 min to 8 h after drug exposure and **recur at the same location** on re-exposure. NSAIDs, Paracetamol and Co-trimoxazole are the most common drugs causing FDE.

The image below shows FDE on upper lip area:

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References: Online Resource, http://www.ijdvl.com/article.asp?issn=0378-6323;year=2010;volume=76;issue=6;spage=671;epage=676;aulast=Riyaz
Rook's Textbook of Dermatology 9th edition, Page no 25.75, 25.76
Harrison's Principles of Internal Medicine, 20th Edition, Page no: 346, 367

Q74. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4173

Methylisothiazolinone (MI) in cosmetics is used as?

A Surfactant

B Humectant

C Emulsifier

D ✓ Preservative

Correct Answer: Preservative

Methylisothiazolinone (MI) in cosmetics is used as a **preservative** to inhibit bacterial growth. It is permitted to be used up to the level of **100 ppm**.

Several cases of **allergic contact dermatitis** due to MI were reported due to its presence in wet wipes, hair cosmetics, facial cosmetics, deodorants, and sunscreens.

Furthermore, cases of airborne exposure to MI causing severe airborne and generalized dermatitis from recently painted walls and even from toilet cleaners have been described.

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References: Rook's Textbook of Dermatology 9th edition, Page no 128.36

Q75. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD4172

The given condition is caused by which drug?

A Actinomycin

B ✓ Bleomycin

C Mitomycin C

D Doxorubicin

Correct Answer: Bleomycin

The given image shows **flagellate dermatitis.** It is caused by **bleomycin.**

Flagellate dermatitis starts as bands of pruritic, flagellate, erythema over the trunk, and extremities within few hours to 6 months after administration. The rash heals with persistent flagellate hyperpigmentation.

No treatment is necessary since the lesions usually disappear when bleomycin is discontinued. If the patient is itchy, managing pruritus with an antihistamine may limit trauma to the skin, and thus avoid further hyperpigmentation.

Another side effect that one needs to watch out for with the use of bleomycin is the development of pulmonary fibrosis.

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References: Rook's Textbook of Dermatology 9th edition, Page nos 4.12 and 120.8
Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 13th Edition, Page no 1193
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702947/

Q76. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD0208

Which of the following does not present with maculopapular rash?

- A ✓ Zoster (shingles)
- B Rubella
- C Dengue (during febrile phase)
- D Measles

Correct Answer: Zoster (shingles)

A maculopapular rash is not seen in zoster (shingles).

Shingles cause **vesicular lesions** that are grouped together and specific to a dermatome.

Type of rash	Description
Macule	**Flat** lesion with a well-circumscribed change in skin color < 1 cm
Papule	**Elevated solid** skin lesion < 1 cm
Vesicle	Small **fluid** containing blister

Maculopapular rashes are seen in:

- Rubella
- Dengue (febrile phase)
- Measles
- Chikungunya (acute infectious phase)
- Erythema infectiosum
- Zika virus
- West Nile virus
- Infectious mononucleosis

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References: Harrison's Principles of Internal Medicine 20th Edition, Page no 106
Rook's Textbook of Dermatology 9th edition, Page no 4.13

Q77. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD6694

A child presents to OPD with tense bullae over the torso. Biopsy of the lesion showed a subepidermal level of blistering and neutrophil infiltration. What is the drug of choice?

A Rituximab

B ✓ Dapsone

C Cyclosporine

D Azathioprine

Correct Answer: Dapsone

This clinical scenario with tense blisters showing **subepidermal separation** and **neutrophilic infiltration** at the basement membrane points towards a diagnosis of **IgA pemphigoid** or linear IgA disease or chronic bullous disease of childhood. The drug of choice is **dapsone**. It is bacteriostatic against M. leprae.

It also has an anti-inflammatory action by decreasing the action of neutrophil enzymes. Hence, it is useful in the treatment of IgA pemphigoid, linear IgA disease and chronic bullous disease of childhood.

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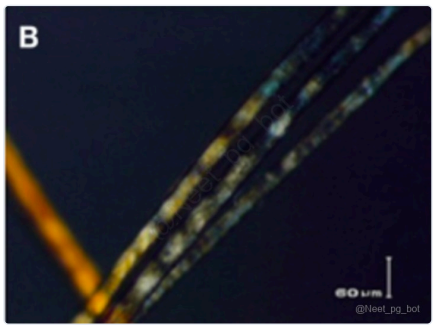
References: Fitzpatrick's Dermatology, 9th Edition., Page no. 989, 998
Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 14th Edition, Page no: 1278

Q78. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD6692

What is the probable diagnosis of this condition affecting hair?



A Trichorhexis nodosa

B Trichorexis invaginata

C ✓ Trichothiodystrophy

D Monilethrix

Correct Answer: Trichothiodystrophy

The given clinical image of hair seen under polarising microscopy reveals **tiger tail banding** (alternating light and dark bands) characteristic of **trichothiodystrophy**.

Trichothiodystrophy is an **autosomal recessive** condition causing **defective DNA repair**. It causes **brittle sulfur-deficient** hair, **photosensitivity**, **ichthyosis** and systemic involvement. Unlike xeroderma pigmentosum (a similar condition with defective DNA repair), there is no risk of skin cancer.

Option A: **Trichorrhexis nodosa** is associated with node formation and fractures. The classic appearance, shown in the image below, is known as **thrust paintbrush** appearance as if two paintbrushes are rubbing against each other. It is associated with trauma to the hair shaft and **arginosuccinic aciduria**.

B- Trichorrhexis invaginata or **bamboo hair** is characterized by invagination of the distal hair shaft into the proximal portion. It is seen in **Netherton's** syndrome. It is also called golf-tee or ball and socket appearance as seen below.

Option D: **Monilethrix**, seen in the image below, is characterized by **beaded hair** and is fragile at the constricted sites. It occurs as an isolated condition or can occur in **Menkes** disease.

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References: Rook's Textbook of Dermatology - 9th Edition - Volume III, Pages 68.19, 78.10, 89.53-54
Fitzpatrick's Dermatology, 9th Edition., Page no. 1537

Q79. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD6689

A patient with a known case of chronic plaque psoriasis with more than 50% skin involvement presents to you. Which of the following will you not recommend?

A Oral methotrexate
B NB UVB
C ✓ Oral steroids
D Oral cyclosporine

Correct Answer: Oral steroids

Oral/systemic steroids are **not used** in the management of **chronic plaque psoriasis**. Topical steroids may be used.
Systemic steroids are avoided because the withdrawal of systemic steroids causes the development of pustular psoriasis.
The **only indication** of using **systemic** steroids in psoriasis is generalized pustular psoriasis in pregnancy (**impetigo herpetiformis**).
Other options maybe used in treatment of extensive (>30%) chronic plaque psoriasis.

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References: Rook's Textbook of Dermatology 9th edition, Page no. 35.22

Q80. Dermatology AIIMS 2020 (May and Nov INI-CET) ID: MD6688

Identify the type of skin lesion.

A Nummular
B ✓ Annular
C Discoid
D Target

Correct Answer: Annular

The given image shows **ring-shaped** lesions with **central clearing**, which are characteristic of **annular** lesions.
In annular lesions, only the **edges** are **active** and appear raised, or in a different color. They are seen in tinea corporis, granuloma annulare, borderline leprosy, and psoriasis.

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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Pg 12,13,284
Rook's Textbook of Dermatology, Pg 4.11-4.12

Q81. Dermatology INI-CET 2022 ID: MF4914

A 10-year-old boy presents with fever, joint pain, and a lesion over his hand as seen in the image below. Which of the following is the clinical finding given and what is the likely diagnosis?

A ✓ Gottron's papules - Juvenile dermatomyositis

B Rheumatoid nodules - Juvenile rheumatic arthritis

C Sclerodactyly - Systemic sclerosis

D Mechanic's hand - Systemic lupus erythematosus

Correct Answer: Gottron's papules - Juvenile dermatomyositis

The given image showing erythematous papules over dorsal metacarpophalangeal and interphalangeal joints (**Gottron's papules**) is a characteristic skin lesion seen in **dermatomyositis (DM)**. Fever and joint pain (arthralgia) are non-specific complaints.

DM is an inflammatory myopathy affecting skin and skeletal muscle. DM can be seen in children (juvenile DM) or in adults. It presents with symmetric, **proximal** more than distal **muscle weakness** associated with characteristic skin manifestations:

- **Heliotrope rash**: Lilac erythema of eyelids with periorbital edema
- **Gottron's papules**: Violaceous, flat-topped papules on the skin overlying the interphalangeal joints and metacarpophalangeal joints
- **Gottron sign**: Erythematous rash over the extensor surfaces of joints such as the knuckles, elbows, knees, and ankles
- **V-sign**: Erythematous macules over anterior chest
- **Shawl sign**: Erythematous macules over posterior neck and upper back
- Mechanic's hand: Roughening and cracking of the skin of the tips and sides of the fingers, resulting in irregular, dirty-appearing lines that resemble those of a mechanic or manual laborer
- **Nail bed telangiectasias**
- Subcutaneous calcium deposits

DM is associated with **interstitial lung diseases** (ILD) and **internal malignancies**, most commonly adenocarcinomas. Adult onset DM has a stronger association with malignancies.

Laboratory evaluation of patients with dermatomyositis will reveal **elevated serum creatinine kinase** and **antinuclear antibodies**. Electromyography will reveal features of myopathy. Histology will show **perifascicular atrophy.** Immunohistochemical staining will reveal **myxovirus resistance protein A**, a sensitive and specific marker.

Treatment of dermatomyositis is include:

- Immunotherapy with **high-dose glucocorticoids** - first-line drug
- **Methotrexate** - **second line** drug. Indications include -
  - When patient presents with severe weakness, or other organ system involvement
  - Patients with risk of steroid complications like patients with diabetes, osteoporosis, or postmenopausal women
  - Patients who do not show improvement after 2-4 months of treatment with prednisone or those who cannot be maintained on a low dose
- **Triple therapy** with **intravenous immunoglobulin** or **rituximab** may be required

The prognosis is favourable in the absence of associated ILD or malignancies.

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References: Rook's Textbook of Dermatology 9th edition, Page 53.1-53.12
Harrison's Principles of Internal Medicine, 21st Edition, Page 2820-2821

Q82. Dermatology INI-CET 2022 ID: MF4962

A 12-week pregnant woman who is already on multidrug therapy for leprosy presents with type 2 lepra reaction. What is the treatment of choice for this patient?

(MDT - multidrug therapy)

A Stop MDT and start oral steroids

B Antibiotics

C Thalidomide

D ✓ Continue MDT and add oral steroids

Correct Answer: Continue MDT and add oral steroids

The treatment of choice for a **pregnant** lady with **type 2 lepra** reaction is to **continue multidrug therapy (MDT)** and **start oral steroids**. MDT is **not stopped** in lepra reactions. MDT is proven to be **safe** in pregnancy. Thalidomide is contraindicated (especially in the 1st trimester) due to the risk of teratogenic changes in the fetus (phocomelia).

Type 2 lepra reaction or **erythema nodosum leprosum** (ENL) is an immune complex-mediated syndrome (**type 3 hypersensitivity**) affecting the skin, nerves, and other organs. It is mainly seen in the multibacillary form of leprosy and is commonly recurrent. It usually occurs during treatment, but can also be seen before or after treatment.

ENL presents with acute onset, painful, erythematous nodules on the face and **extensor surfaces** of limbs that blanch on pressure. **Systemic manifestations** include fever, malaise, anorexia, neuritis, uveitis, dactylitis, arthritis, myositis, etc. can also be seen.

Treatment includes:

- **Thalidomide** is the drug of choice in young men with severe or recurrent ENL. It is **contraindicated** during **pregnancy** due to its high **teratogenicity** causing **phocomelia**. Even in women of reproductive age, it is given with caution.
- **Glucocorticoids** (high doses) are routinely used in all patients and help in **suppressing** the acute signs and symptoms of inflammation.
- **Clofazimine** is useful as long-term maintenance therapy for **preventing recurrent reactions**. Clofazimine is not given to this patient because it is already a part of the MDT regimen containing dapsone, rifampicin, and clofazimine.

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page no 1390, 1387
Rook's Textbook of Dermatology - Volume II -9th Edition, Page no 28.16

Q83. Dermatology INI-CET 2022 ID: MF5358

A patient presents to the outpatient department with a satellite skin lesion. On examination, he has one hypopigmented lesion and one thickened nerve. Images of two blister packs are given below. The treatment best suited for this patient would be:

A Packet 1 for 12 months

B Packet 2 for 12 months

C Packet 1 for 6 months

D ✓ Packet 2 for 6 months

Correct Answer: Packet 2 for 6 months

The treatment best suited for this patient would be **packet 2-maroon** for **6 months**. The clinical scenario of **one hypopigmented satellite skin lesion** and **one thickened nerve** suggests **paucibacillary leprosy**. The treatment of paucibacillary leprosy (PB) is multidrug therapy (MDT) with **rifampicin**, **dapsone,** and **clofazimine** which is the **maroon** blister packet for 6 months.

The **current recommendation by WHO** is to treat **PB** leprosy with a combination of **3 drugs** - rifampicin, dapsone, and clofazimine (maroon blister pack) for **6 months.** Meanwhile, **multibacillary** (MB) leprosy is treated using the same **3 drugs** for a duration of **12 months.** While the NLEP guidelines earlier mentioned a 2-drug MDT for PB leprosy, the **National Strategic Plan and Roadmap for Leprosy (2023-2027)** now recommends a **uniform MDT of 3 drugs** for both PB and MB leprosy.

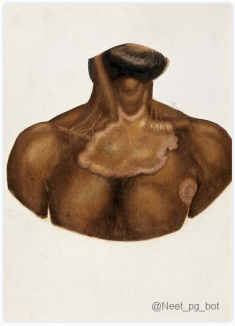
Management of leprosy is based on the bacterial load. WHO defines **PB** disease as the presence of no bacilli on smears or biopsy and **less than or equal to 5 skin lesions** with no or only one peripheral nerve involved. **MB** leprosy is characterized by bacilli on smears or biopsies of **more than 5 skin lesions** and more than 1 peripheral nerve involvement.

The image given below shows a hypopigmented satellite skin lesion seen in leprosy.

Chemoprophylaxis for leprosy should be given in those

- Age more than 2 years
- Close contact for more than 20 hours per week for more than 3 months

A **single dose of rifampicin** is recommended for the close contact of a patient with leprosy provided they do not develop any signs or symptoms pertaining to leprosy.



@Neet_pg_bot

References: Andrew's Diseases of the Skin: Clinical Dermatology - 13th Edition, Page no: 338, 339, 345
Online Resource, https://cltri.gov.in/AboutLeprosy/NLEP_Program%20Management_DLO.pdf
Online Resource, https://dghs.gov.in/WriteReadData/userfiles/file/Leprosy%20New/NSP%20%20Roadmap%20for%20Leprosy%202023-2027.pdf
Harrison's Principles of Internal Medicine 21st Edition, Page No. 1385

Q84. Dermatology INI-CET 2022 ID: MF4940

Chemoprophylaxis for leprosy should be given to which of the following persons?

1. **Contact of age more than 2 years**
2. **Contact for more than 20 hours per week**
3. **Sharing the same clothes, towels**
4. **Living with a patient with leprosy for >6 months**

A ✓ 1, 2, 4

B 1, 3, 4

C 2, 3, 4

D 1, 2, 3

Correct Answer: 1, 2, 4

Chemoprophylaxis for leprosy should be given to **contacts 1,2** and **4.**

Chemoprophylaxis for leprosy should be given to the following contacts:

- A person who has been in close contact with the index case for **20 hours** or more **per week** for **more than 3 months**.
- **Close contact** if the **age** is more than **2 years**

The contact mentioned in option 4, is with the patient for more than 3 months. Therefore, even this contact will require chemoprophylaxis.

Chemoprophylaxis cannot be given if there is/are:

- Possible signs of leprosy
- Signs or confirmed tuberculosis
- Liver or kidney disease
- Pregnancy
- Received rifampicin in the last 2 years

A **single dose** of **rifampicin** is used in chemoprophylaxis. Chemoprophylaxis must be initiated only 4 weeks after treatment for leprosy has begun in the patient so as to ensure the patient is non-infectious.

Dose of rifampicin in chemoprophylaxis of leprosy

Age	Dose
Children 2-5 years	10-15 mg/kg
Children 6-9 years (less than 20kg)	150 mg
Children 6-9 years (more than 20kg)	300 mg
10-14 years	450 mg
15 years and above	600 mg

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References: Online Resource, Page no 23-24 https://www.who.int/publications/i/item/9789290228073

Q85. Dermatology INI-CET 2022 ID: MF4920

Match the following types of scales with their associated conditions.

Scales	Lesion
1. Collarette scales	a. Pityriasis versicolor
2. Silvery scales	b. Pityriasis rosea
3. Mica-like scales	c. Psoriasis
4. Branny scales	d. Pityriasis lichenoides

A ✓ 1-b, 2-c, 3-d, 4-a

B 1-a, 2-b, 3-d, 4-c

C 1-d, 2-c, 3-a, 4-b

D 1-c, 2-b, 3-d, 4-a

Correct Answer: 1-b, 2-c, 3-d, 4-a

The correctly matched option is option A: 1-b, 2-c, 3-d, 4-a.

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References: Rook's Textbook of Dermatology 9th edition, page 40.1 - 40.5,4.13,26.2,29.1550.6,65.2,107.3,109.8

Q86. Dermatology INI-CET 2022 ID: MF4934

A 60-year-old patient presents with the lesion given in the image below. He also describes having the sensation of spiders crawling in the area of the lesion when he woke up. Which of the following is the most likely diagnosis?

A ✓ Herpes zoster infection

B Irritant contact dermatitis

C Allergic contact dermatitis

D Herpes simplex infection

Correct Answer: Herpes zoster infection

The given image showing **unilateral, red, grouped vesicles** in a single **dermatome** associated with **crawling sensation** is highly suggestive of **Herpes zoster infection.**

Herpes zoster infection or **shingles** is a segmental eruption due to the **reactivation** of latent varicella-zoster virus from the **dorsal root ganglia**. An earlier infection with varicella is essential before zoster can occur.

The first symptom of zoster is usually **pain**. It may also be accompanied by fever, headache, malaise, and tenderness localized to areas of one or more dorsal roots. The **mucous membranes** within the affected dermatomes are involved**.**

The patients are **infectious**, both from the virus in the lesions and the nasopharynx. The most commonly involved dermatomes in decreasing order of frequency are **thoracic** > cervical > **trigeminal** > lumbosacral. The posterior nerve roots and ganglia show inflammatory changes.

Zoster is a **self-limiting infection** but it is painful and carries a risk of secondary infection and post-herpetic neuralgia. **Analgesia** and treatment of secondary infections are sufficient for mild infections.

Other options:

Option B: **Irritant contact dermatitis** is the cutaneous response to the physical/toxic effects of a wide range of environmental exposures. The main clinical features are usually **dryness** or **chapping**. The lesions could commonly develop into patchy or diffuse erythema with **scaling** and **fissuring**.

Option C: **Allergic contact dermatitis** is an **eczematous reaction** following **exposure** to a substance to which the immune system has previously been **sensitized**.

Option D: **Herpes simplex lesions** are **thin-walled, umbilicated vesicles,** the roof of which breaks down, leaving tiny superficial **ulcers.** They heal without scarring. The most common site of cutaneous lesions in the face - is on the cheeks, chin, around the mouth, or on the forehead.

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References: Rook's Textbook of Dermatology 9th edition, 128.1, 129.1, 25.1

Q87. Dermatology INI-CET 2022 ID: MF7303

A 60-year-old diabetic female patient presents with progressive skin lesions, as shown in the image. What is the diagnosis?

A Squamous cell carcinoma

B ✓ Basal cell carcinoma

C Discoid lupus eryhtematosus

D Melasma

Correct Answer: Basal cell carcinoma

The image above shows a **central ulcer** with **raised borders** and rolled-out edges (rodent ulcer) suggesting a diagnosis of **basal cell carcinoma**.

Basal cell carcinoma is a **slow-growing malignant tumor** with **locally invasive** characteristics. It affects the pilosebaceous skin.

Ultraviolet radiation exposure is the **strongest predisposing** factor. Other risk factors include exposure to arsenic, coal tar, aromatic hydrocarbons, and ionizing radiation.

Basal cell carcinoma mainly affects **elderly** or **middle-aged men** with excessive skin exposure, commonly in the age group of 40–80 y and in white-skinned people.

Macroscopically, it is categorized as localized and generalized. **Nodular, nodulocystic** (nodular and nodulocystic together account for 90% of basal cell carcinoma), pigmented, and nevoid are the localized cancers, while superficial multifocal and superficial spreading, infiltrative, and cicatrizing are the generalized types. The nodular type is described as **pearly** waxy **nodules** at the margins. A central depression (umbilication) or ulceration and rolled-out edges are characteristic of basal cell carcinoma.

The below gross image shows pearly, rolled-out borders suggestive of basal cell carcinoma.

Sites near the **eye, nose,** and **ear** and of size **>2 cm** are considered to be **high risk**. Direct invasion at these sites can reach the cranium.

Management is by **Mohs microscopic surgery**. Non-surgical measures, such as the administration of **topical 5-fluorouracil** and **imiquimod,** are useful. Metastasis is **extremely rare.**

Other options:

Option A: **Squamous cell carcinoma (SCC)** is the second most common skin cancer. UV radiation exposure is the primary risk factor for SCC. Classically, it appears as a **scaly or ulcerated papule or plaque** that can **bleed with minimal trauma,** but the pain is rare. It can exhibit in situ (confined to the epidermis) or invasive subtypes. Wide surgical excision including subcutaneous fat is the treatment of choice for SCC.

Option C: **Discoid lupus erythematosus (DLE):** DLE is typically located on the nose, cheeks, ear lobe, and concha. It may involve lips, oral mucosa, nose, or eyelids. Initial lesions of DLE are dry red patches, which then progress to **indurated red or hyperpigmented plaques** with adherent scale. Follicular keratosis (plugs of keratin within hair follicles) is observed when the surface scales are removed (carpet-tack sign). Older lesions are **hyperpigmented**, especially on the **plaque edges.** There is **scarring**, which results in **central loss of pigment**, white patches, and **skin atrophy**.

Option D: **Melasma** is the **most common** cause of **facial melanosis** that manifests as hyperpigmented macules on the face. It is exacerbated by **sun exposure** and **oral contraceptive pills**. The condition is predominantly seen in **females**. Melasma is very common in the **3rd trimester** of pregnancy and is most marked in brunettes. It occurs due to increased levels of **estrogen** and **progesterone,** stimulating the activity of melanocytes. It resolves on its own after pregnancy. Hypermelanosis affects mainly the upper lip, the malar regions, forehead, and the chin.

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References: Rook's Textbook of Dermatology - 9th Edition - Volume IV, Page 141.1, 141.2, 141.4, 141.8, 141.11, 141.13 - 141.16, 142.1
Online Resource, https://www.uptodate.com/contents/epidemiology-pathogenesis-and-clinical-features-of-basal-cell-carcinoma?search=basal%20cell%20carcinoma&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
Robbins Basic Pathology, 10th Edition, Page 1148
Bailey & Love's Short Practice of Surgery - 27th Edition, Page no 599, 604-605, 607-609, 611
Rook's Textbook of Dermatology 9th edition, Page No. 4.15, 51.3
Rook's Textbook of Dermatology 9th edition, Page 88.10 - 88.12

Q88. Dermatology INI-CET 2022 ID: MF7299

A diabetic patient presents with the following lesions in the armpit. Which of the following would be seen on Wood's lamp examination?

A Yellow fluorescence

B ✓ Coral red fluorescence

C Blue fluorescence

D Green fluorescence

Correct Answer: Coral red fluorescence

The image showing **brown patches** in **axilla** in a **diabetic** patient is suggestive of **erythrasma**. **Coral-red fluorescence** under Wood's lamp is seen in erythrasma.

Erythrasma is a skin infection caused by ***Corynebacterium minutissimum***. It is common in **diabetics** and **immunodeficient** patients and is associated with **obesity**, **sweating**, and **poor hygiene**. It affects the **axillae**, groins, toe webs, inter-gluteal and submammary regions. Lesions may be symptomless or show mild discomfort and itching.

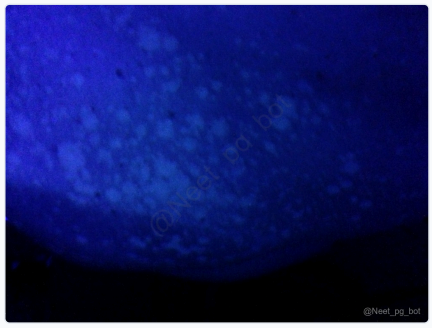
The lesions may be of **irregular shape** with **sharp margins**. They may start as smooth, red patches and then become brown. The lesions are more likely to be **pruritic** in tropical regions and may be asymptomatic in a temperate climate.

Coral-red fluorescence with Wood's light is due to **coproporphyrin III** and suggests erythrasma. The fluorescence may persist after the eradication of the disease.

Treatment of erythrasma is by using topical **clotrimazole** and **miconazole** for two weeks. For more extensive lesions, erythromycin can be administered. Relapse can occur. In such cases, long-term topical povidone-iodine application with drying agents can be used. Photodynamic therapy can also be attempted.

Green and blue-green fluorescence is seen in tinea capitis infections associated with microspora species and favus.

Pityriasis versicolor is caused by ***Malassezia*** species and the typical lesions are asymptomatic **hypopigmented or hyperpigmented** macules and patches and produce **fine scales**. The **color of the scales may vary** from pale ochre to medium brown. **Pale yellow** fluorescence seen in **Wood's lamp** examination is **characteristic** of **tinea versicolor** or **pityriasis versicolor.**



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References: Rook's Textbook of Dermatology - Volume II -9th Edition, Pages 26.38 - 26.41, 26.44
Rook's Textbook of Dermatology 9th edition, Page no 4.20

Q89. Dermatology INI-CET 2022 ID: MF7301

A 10-year-old boy is brought with weakness in his lower limbs and rash over the eyelids. The biopsy shows inflammation of the muscle. What is the finding shown in the image below?

A ✓ Gottron's papules

B Shawl sign

C Heliotrope rash

D Scleroderma

Correct Answer: Gottron's papules

The finding shown in the image is **Gottron's papules.** The given scenario of a patient with weakness, rash, muscle inflammation, and presence of Gottron's papules on the skin is highly suggestive of **dermatomyositis.**

Dermatomyositis (DM) is an autoimmune disorder predominantly affecting the **skin** and **skeletal muscle**. The following are various manifestations of dermatomyositis:

- **Proximal myopathy:** Patients have trouble performing everyday activities like climbing stairs, getting up, and lifting their arms above their heads
- **Gottron's papules**: Violaceous, erythematous papules overlying the dorsal interphalangeal or metacarpophalangeal, elbow, or knee joints, as shown in the image below
- **Gottron's sign**: Symmetric, non-scaling, violaceous, erythematous macules or plaques, often atrophic, in the same distribution as Gottron's papules
- **Shawl sign**/V-sign: Fixed macular erythema, across the shoulders and outer arms (Shawl distribution)
- **Periungual telangiectasias**: Dilated capillaries of the nail folds
- **Mechanic's hand**: Roughening and cracking of the skin of the tips and sides of the fingers, resulting in irregular, dirty-appearing lines that resemble those of a mechanic or manual laborer
- Less common manifestations include: Facial swelling, malignancy, erythroderma, lichen planus

Laboratory evaluation of patients with dermatomyositis will reveal **elevated levels of serum creatine kinase** and **antinuclear antibodies**. Electromyography will reveal features of myopathy. Histology will show **perifascicular atrophy.** Immunohistochemical staining will reveal **myxovirus resistance protein A**, a sensitive and specific marker.

Treatment of dermatomyositis:

- Immunotherapy with **high-dose glucocorticoids** - first-line drug
- **Methotrexate** - **second-line** drug; indications include the following
  - When the patient presents with severe weakness, or other organ system involvement
  - Patients with a risk of steroid complications like patients with diabetes, osteoporosis, or postmenopausal women
  - Patients who do not show improvement after 2–4 months of treatment with prednisone or those who cannot be maintained on a low dose
- **Triple therapy** with **intravenous immunoglobulin** or **rituximab** may be required

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References: Rook's Textbook of Dermatology - Volume II -9th Edition, Chapter 53 - 53.4-53.7

Q90. Dermatology INI-CET 2022 ID: MF7295

Consider the causes of alopecia. Which among the following cause(s) non-scarring alopecia?

1. Alopecia areata
2. Telogen effluvium
3. Androgenic alopecia
4. Frontal fibrosing alopecia

A 3 and 4
B ✓ 1, 2, and 3
C 2, 3, and 4
D Only 4

Correct Answer: 1, 2, and 3

Non-scarring alopecia is caused by androgenic alopecia, alopecia areata, and telogen effluvium (1, 2 and 3). **Frontal fibrosing alopecia** is a type of **primary cicatricial (scarring) alopecia.**

Alopecia areata is a chronic inflammatory disease that causes **non-scarring hair loss**. It is a **T-cell-mediated inflammatory disease.** It is associated with several autoimmune diseases, including **thyroiditis**, lupus erythematosus**,** vitiligo, and psoriasis. It mainly affects the **scalp,** but can also affect the **beard, eyebrows,** and **eyelashes.** In some cases, this progresses to a total loss of scalp hair **(alopecia totalis)** or a loss of all hair on the body **(alopecia universalis)**.

Telogen effluvium is a type of **nonscarring alopecia** that is marked by **diffuse shedding** of normal hair. This occurs due to **premature termination** of the anagen phase of the hair cycle. It can be either acute or chronic.

- **Acute telogen effluvium** occurs **2–3 months** after a stressful event such as high **fever**, **surgery**, starvation, and hemorrhage. Hormonal fluctuations during the **postpartum** period can trigger it (**telogen gravidarum**). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally **resolves spontaneously**.
- **Chronic telogen effluvium** is used to describe the diffuse shedding of telogen hair for more than **6 months**. Common causes include **thyroid disorders**, iron-deficiency anemia, and malnutrition.

Androgenetic alopecia is a disorder characterized by a reduction of hair fiber production by follicles and miniaturization. **Dihydrotestosterone**, an androgenic metabolite, is implicated in the etiology. It is of two types: male-pattern and female-pattern hair loss. Male-pattern hair loss causes **frontal hairline recession** and thinning of the **vertex**. Female-pattern hair loss involves the **mid-frontal scalp** diffusely. There is no or minimal involvement of the temporal region.

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References: Rook's Textbook of Dermatology 9th edition, Page no. 32.24, 32.39, 32.40, 86.17, 89.6, 89.7, 89.14, 89.17, 89.20, 89.24, 89.25, 89.46

Q91. Dermatology INI-CET 2022 ID: MF7290

Match the following therapeutic methods used in the management of psoriasis with their possible side effects.

Column A	Column B
1. PUVA	a. Abortifacient
2. Cyclosporine	b. Effective but is highly nephrotoxic
3. Methotrexate	c. Long-term use increases the risk of skin cancer
4. Acitretin	d. Mucositis, teratogenic

A ✓ 1-c, 2-b, 3-a, 4-d

B 1-c, 2-a, 3-b, 4-d

C 1-b, 2-d, 3-c, 4-a

D 1-b, 2-c, 3-a, 4-d

Correct Answer: 1-c, 2-b, 3-a, 4-d

The correctly matched pairs are as follows:

Column A	Column B
1. PUVA	c. Long-term use increases the risk of skin cancer
2. Cyclosporine	b. Effective but is highly nephrotoxic
3. Methotrexate	a. Abortifacient
4. Acitretin	d. Mucositis, teratogenic

PUVA is highly effective in clearing psoriasis with clearance rates of over 90%. It is associated with both acute (erythema, blistering) and chronic side effects. **Long-term PUVA use** is associated with the development of **non-melanoma skin cancer** (squamous cell cancer and basal cell cancer).

Cyclosporine is associated primarily with **renal impairment** (decreased GFR), metabolic disturbances, and an increased risk of malignancy. Due to its nephrotoxic effects, it is now uncommonly used for the maintenance of psoriasis.

Both **acitretin** and methotrexate are **teratogenic**.

Acitretin is associated with dose-dependent mucocutaneous side effects (**mucositis**) and hyperlipidemia. To prevent retinoid-induced embryopathy, female patients should **avoid conception** for **3 years after discontinuing** acitretin treatment.

Acitretin is approved for use in the **cutaneous** manifestations of **psoriasis.**

Absolute contraindications:

- Pregnant women
- Women who are planning to conceive
- Breastfeeding

Methotrexate is also associated with **mucositis** and **teratogenicity**, it has been used as an **abortifacient** and is used in the management of molar pregnancies. Methotrexate is also associated with **hepatotoxicity**.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 35.26- 35.28
Fitzpatrick's Dermatology, 9th Edition., Page no: 3402

Q92. Dermatology INI-CET 2022 ID: MF7554

Skin manifestations caused by the drug administered for amebic dysentery are due to _____.

A ✓ Fixed drug eruption

B Urticaria

C Maculopapular rash

D Atopic dermatitis

Correct Answer: Fixed drug eruption

The skin manifestations are most commonly due to **metronidazole,** which is associated with **fixed drug eruption (FDE)**.

Although maculopapular rashes can also be seen with metronidazole, **the incidence of a fixed drug eruption is higher with metronidazole.**

Fixed drug eruptions/ reactions are characterized by **sharply demarcated, dull red to brown** lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation. They are usually solitary or few in number and **recur at the exact location**. But, new lesions can occur on different body parts on re-exposure to the offending drug.

FDE is a type of **delayed hypersensitivity reaction**. Paracetamol/NSAIDs are some of the most common drugs causing FDE.

Drug-specific clinical patterns of FDE include:

Drugs	Drug-specific clinical pattern of FDE
NSAIDs	Lips and genitals
Tetracycline/ Trimethoprim-sulfamethoxazole	Genitals
Metamizole	Trunk and extremities
Carbocysteine	Face

Diagnosis is based on history and lesion morphology. **Discontinuation** of the offending drug is the **first step** in **management**.

The image below shows a **well-defined hyperpigmented patch** of a **fixed drug eruption**.

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References: Rook's Textbook of Dermatology 9th edition, Chapter 118, Page nos 118.11-118.12

Q93. Dermatology INI-CET 2022 ID: MF7575

A 40-year-old man presents with erythematous painful lesions on the trunk and face, joint pain, and glove and stocking pattern of anesthesia. Which of the following investigations is used for the diagnosis of this condition?

1. **Nerve examination**
2. **Skin biopsy**
3. **Slit skin smear**
4. **Mantoux test**

A 1, 2

B ✓ 1, 2, 3

C 1, 3, 4

D 1, 2, 3, 4

Correct Answer: 1, 2, 3

The clinical scenario points to a case of **leprosy**. The diagnosis of this condition requires nerve examination, skin biopsy, and a slit skin smear (option B- 1, 2, 3).

Leprosy is a chronic granulomatous condition caused by ***Mycobacterium leprae***. It affects the skin, cutaneous and peripheral nerves, and mucous membranes. The most commonly affected peripheral nerves are the **posterior tibial nerve**, followed by the **ulnar**, median, and lateral popliteal nerves. The most commonly involved cranial nerve is the **facial nerve**.

Leprosy has a **long incubation** period, usually ranging from 2 to 12 years. It commonly presents as an **area of numbness** on the skin or a visible skin lesion which could be a **hypopigmented** or **erythematous macule**. Lesions usually involve the face, extensor aspect of the limbs, trunks, and buttocks. The lesions have **impaired nerve function** and poor hair growth. The patients may also present with burns or ulcers of the anesthetic limb.

Classification systems of leprosy:

- **Ridley-Jopling** classification:
  - Tuberculoid (TT)
  - Borderline tubercular (BT)
  - Borderline (BB)
  - Borderline lepromatous (BL)
  - Lepromatous (LL)
- **WHO** classification:
  - Paucibacillary
  - Multibacillary

Cardinal signs for the diagnosis of leprosy:

- Characteristic **skin lesions**- hypopigmented or erythematous lesions with definite **impairment of sensation.**
- **Peripheral nerve** involvement- **thickening** of the nerve with sensory impairment. Nerve tenderness can also be seen.
- Detection of **acid-fast bacilli** in slit-skin smear, skin biopsy, or positive biopsy PCR.

The presence of two of the three signs can establish the diagnosis of leprosy. Other tests done for diagnosis of leprosy include nerve biopsy in pure neural leprosy and PGL-1 antibody in cases with low bacterial load.

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References: Rook's Textbook of Dermatology - 9th Edition - Volume I, 28.1, 28.3, 28.12
Harrison's Principles of Internal Medicine, 21st Edition, 1385, 1386, 1388

Q94. Dermatology INI-CET 2022 ID: MF7572

Identify the condition shown in the image below.

A Paget disease

B ✓ Lichen sclerosus

C Lichen planus

D Angioma of vulva

Correct Answer: Lichen sclerosus

The given image shows **lichen sclerosus.**

Lichen sclerosus usually presents in **postmenopausal** women. Multiple etiologies have been suggested (infectious, hormonal, genetic, and autoimmune). Symptoms include **itching,** excoriations, **vulval skin thinning** and crinkling, dysuria, and dyspareunia. **Perianal** involvement is frequently seen in the form **figure of 8** appearance. The typical appearance is **white, atrophic papules** may coalesce into plaques and distort normal anatomy. **Labia minora regression**, clitoral concealment, urethral obstruction, and introital stenosis can develop. A **biopsy is needed to exclude malignancy**.

Topical corticosteroids remain the first line of treatment.

Premalignant lesions of vulva:

- **Vulvar intraepithelial neoplasia (VIN)**
- **Paget's disease**
- Lichen sclerosus
- Squamous cell hyperplasia
- **Condyloma accuminata**
- **Bowen's disease** (erythroplasia of Queyrat)

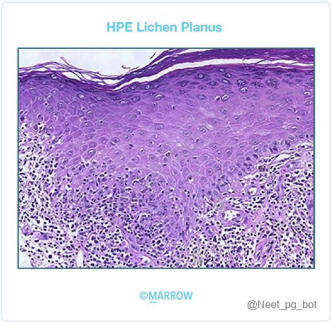
Lichen planus is an inflammatory disorder affecting the skin and mucosal membranes. Nail involvement includes exaggeration of the longitudinal lines and linear depression. It manifests as **6Ps** pruritic, purple, polygonal, planar, **papules,** and **plaques.** It also exhibits **Wickham striae** (thin white lacy lines on the lesion)**,** and the **Koebner phenomenon** (new lesions at sites of skin trauma). **Mucosal involvemen**t is common including mouth and tongue, rarely it can also be found on the anus and genitalia

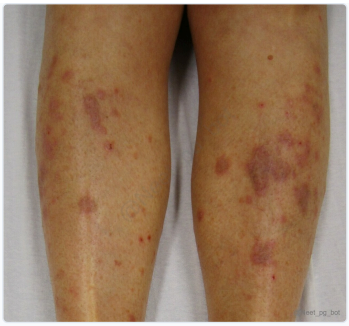
The histological changes in **lichen planus** include increased Langerhans cells in the epidermis, **spongiosis**, and **sawtooth appearance of Rete ridges**. A fully developed lichen planus papule shows **acanthosis** of the epidermis, thickening of the granular layer, and hyperkeratosis. The image below shows the histopathological changes in lichen planus.

Vulvar Paget disease is an intraepithelial neoplasia that presents as an **eczematoid, red, weeping area** on the vulva. It is associated with invasive Paget disease or adenocarcinoma of the vulva or carcinoma at another non-vulvar location. It can be managed by local excision or vulvectomy. This condition needs long-term surveillance as recurrence is very common.

Angioma of vulva - Hemangiomas and angiokeratomas are benign lesions of the vulva. Angiokeratomas are benign tumors characterized by hyperplasia of the epidermis along with multiple dilated vessels in the superficial dermis. They present as well-defined, **erythematous, hyperkeratotic papules**. Senile (cherry) hemangiomas usually present as red macules that rapidly progress to well-circumscribed, raised, red, and soft lesions. Excision biopsy of bleeding hemangiomas, cryosurgery, or carbon dioxide laser can be used for management.

The image below shows a cherry angioma:





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References: Williams Gynecology 4th edition, Page no: 92-100, 688
Shaw's Textbook of Gynaecology, 17th edition, Page no: 323, 473-475
Online Resource, https://ijdvl.com/lichenoid-tissue-reaction-interface-dermatitis-recognition-classification-etiology-and-clinicopathological-overtones/
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6298145

Q95. Dermatology INI-CET 2022 ID: MF7547

A patient comes to the OPD with the following skin manifestation of a disease associated with Raynaud phenomenon. What is the manifestation?

A Poikiloderma

B ✓ Salt and pepper pigmentation

C Leukotrichia

D Vitiligo

Correct Answer: Salt and pepper pigmentation

The clinical scenario is suggestive of **systemic sclerosis**, and the skin manifestation shown is a **salt-and-pepper appearance**.

Systemic sclerosis is an **autoimmune disease** of unknown etiology, complex pathogenesis, and variable clinical presentations. It is associated with significant **disability** and **mortality.** Virtually every organ is affected. **Thickened** and **indurated** skin, called **scleroderma**, is the distinguishing hallmark of systemic sclerosis.

Cutaneous features of systemic sclerosis:

- **Raynaud phenomenon**, with the typical white discoloration of the fingers caused by **vasospasm** and bluish discoloration of the fingertips
- Loss of pigmentation except in the perifollicular areas is known as the salt and pepper appearance of the skin, which is mostly seen in the upper back, chest, and scalp
- Puffy fingers in early disease
- **Sclerodactyly** and **sclerosis** extending proximal to the metacarpophalangeal joints with or without contractures
- **Vasculopathic ulcers** over the bony prominences
- Loss of fingertip pulp
- Severe flexion contractures
- Facial features include **expressionless facies,** mat-like telangiectases, microstomia, perioral furrowing, and a beak-like nose

Other options:

Option A: **Poikiloderma** is a skin disorder characterized by hypopigmentation and hyperpigmentation, wrinkling following atrophy, and telangiectasias. It is seen in ionizing radiation-damaged skin and dermatomyositis.

Option C: The hair in the amelanotic lesions of vitiligo can sometimes lose color and become white. This is called **leukotrichia.**

Option D: **Vitiligo** is characterized by depigmented macules that have sharp demarcation. It is due to the loss of melanocytes.

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References: Rook's Textbook of Dermatology 9th edition, Page 56.4
Harrison's Principles of Internal Medicine, 21st Edition, Page 2779

Q96. Dermatology INI-CET 2022 ID: MF7556

A female patient came to the hospital with a history of relapsing, remitting, mildly itchy erythematous plaques over the scalp, sacral area, and knees. She has a history of joint pain and her nail changes are shown in the image below. What is your diagnosis?

A Adult onset Still's disease

B Pachyonychia congenita

C ✓ Psoriasis

D Lichen planus

Correct Answer: Psoriasis

The diagnosis in this patient with the above clinical features is **psoriasis.**

The image shows **nail pitting, oil drop sign,** and **salmon patches** in fingernails

Psoriasis is a chronic inflammatory condition of the skin with systemic manifestations. The skin lesions can have **exacerbations** and **relapse.** They include **red**, **silvery-white well-demarcated scales.** A peripheral clearing known as the **ring of Woronoff** may be present. It is usually seen on the **scalp**, **trunk**, and **extensor surfaces** of the limbs. The appearance of a smooth, glossy red membrane with small bleeding points on the removal of the supra-papillary epidermis by gentle scraping is known as **Auspitz's sign.**

Psoriatic arthritis presents as early morning stiffness and joint swelling, heel pain due to enthesitis of the Achilles tendon, or plantar fasciitis. The clinical examination may reveal evidence of dactylitis (sausage fingers) or swollen or tender joints. The distal interphalangeal joint (DIP) is classically involved in causing **pencil-in cup** deformity.

Punctate pitting is the most common nail finding and others include the **oil-drop sign**, subungual hyperkeratosis, splinter haemorrhages, and **distal onycholysis**.

The topical application of **steroids**, retinoids, vitamin D, and **emollients** is the mainstay of management. **Phototherapy** and oral **methotrexate**, **acitretin**, and cyclosporine are used for more extensive and systemic diseases.

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References: Rook's Textbook of Dermatology - Volume II -9th Edition, Page nos 35.8-35.10, 35.13-35.14, 35.22-35.30

Q97. Dermatology INI-CET 2023 ID: MF9144

A patient on Anti PD1 immunotherapy(Nivolumab) comes with pruritus, eczema, and urticarial rashes. Histopathological examination reveals a subepidermal blister. The direct Immunofluorescence of the lesions showed Linear C3 and IgG on the basement membrane. What is the diagnosis?

A Pemphigus vulgaris

B ✓ Bullous pemphigoid

C Dermatitis herpetiformis

D Pemphigus foliaceous

Correct Answer: Bullous pemphigoid

In the given clinical scenario history of use of Anti-programmed cell death 1 (PD1) targeted immune checkpoint inhibitors with HPE showing subepidermal split and direct Immunofluorescence of the lesions showed Linear C3 and IgG on basement membrane is suggestive of **bullous pemphigoid.**

Anti-programmed cell death 1 (PD1) targeted immune checkpoint inhibitors such as nivolumab and pembrolizumab are associated with an increased risk of development of bullous pemphigoid. **Histopathology** will show the presence of **subepidermal bullae with eosinophils**. **Immunofluorescent** microscopy will show the presence of subepidermal split and deposition of **IgG antibodies** and **C3** at the **dermo-epidermal junction.**

Bullous pemphigoid is one of the most common **subepidermal** immunobullous disorders. It is mostly a disease of the elderly. It is characterised by a subepidermal split, with tissue-bound and serum IgG antibodies against **BP180, more than BP 230** antigens on **hemidesmosomes,** necessary to maintain the structure of.

Bullous pemphigoid is associated with **psoriasis** and **treatments** of psoriasis, like phototherapy and psoralen. There was an association seen with **debilitating neurological conditions** like **stroke, Parkinsonism**, cognitive impairment and multiple sclerosis, and **haematological malignancies** like **lymphomas** and **myeloid leukaemia**. It is also precipitated by **drugs** like **diuretics, penicillamine** and psycholeptics.

Clinical features include:

- **Prodromal phase** with excoriated papules, eczematous or urticarial skin lesions and haemorrhagic crusts may be seen.
- **Bullous phase** is characterised by:
  - **Intense pruritus** is almost constantly present.
  - **Symmetrically** distributed **tense bullae** and **vesicles** over the **flexural** aspects and **abdomen** of the body.
  - **Partly haemorrhagic crusts**, **erythematous plaques** with an **annular** pattern are seen.
  - Lesions heal **without scarring**, provided there are no superinfections.
  - The bullae contain **clear**, sometimes **haemorrhagic exudates**.
  - **Nikolsky's sign** is **negative**.
  - Oral or **mucosal lesions** and milia formation are **rare**.

The image given below shows the lesions seen in bullous pemphigoid.

Histopathology will show the presence of subepidermal bullae with eosinophils. Immunofluorescent microscopy will show the presence of subepidermal split and deposition of **IgG antibodies** and **C3** at the **dermo-epidermal junction.**

The below image shows linear IgG deposits at dermo-epidermal junction seen in bullous pemphigoid.

Untreated bullous pemphigoid has a **chronic**, **self-limiting** course, with **remissions** and **relapses**. Complete remissions are possible with treatment.

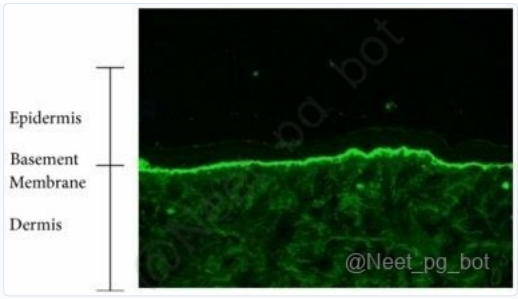
Bullous pemphigoid is **mainly** treated with **topical** and **systemic corticosteroids**. Other drugs used like azathioprine, dapsone, doxycycline and immunomodulatory are also used.

Other options:

Option A-Pemphigus vulagaris is an intraepidermal immunobullous disease. It occurs commonly between the **4th to 6th decade.** They present with **flaccid blisters.** Firm pressure over the lesion separates the epidermis from the dermis. **Nikolsky's sign** is **positive**. **Mucosal involvement** will be seen in the form of buccal and palatal erosions

Option C-Dermatitis herpetiformis typically appears in the **4th decade,** and the patients present with **itching** and **rash**, most typically over the **extensor surfaces of the elbows**, **knees, buttocks**, and scalp. The majority of the patient have symptoms related to **gluten-sensitive enteropathies,** like bloating, diarrhea, and other gastrointestinal complaints. Immunofluorescence microscopy will show the deposition of **IgA antibodies** in the papillary dermis in a **granular or fibrillar pattern**. Complement 3 may also be found along with the IgA deposits.

Option D-Pemphigus foliaceous shows **antidesmoglein-3** antibodies, **anti-desmocollin** antibodies and a **predominance** of **anti-desmoglein-1** antibodies. Lack of mucosal involvement in pemphigus foliaceous, where anti-desmoglein-1 antibodies are prominent, is due to the **low expression of desmoglein 1** in the **mucosa.** In pemphigus vulgaris, where anti-desmoglein-3 antibodies are prominent, mucosal involvement is extensive in early stages as **desmoglein 3** is abundantly present in the mucosa.



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References: Rook's Textbook of Dermatology 9th edition, Page no: 50.1-50.22,50.52-50.55
Online Resource, https://pubmed.ncbi.nlm.nih.gov/32716997/

Q98. Dermatology INI-CET 2023 ID: MF9150

A 24 year old young male presented with asymptomatic scaly lesions over back, symmetrical in distribution. Images of the lesions are given, Identify the disease and the pattern of lesions

A ✓ Collaratte scale-Christmas tree pattern-Pityriasis rosea

B Wickhams striae-Lichen planus

C Christmas tree pattern-Lichen planus

D Wornoff ring-Seborrhic dermatitis

Correct Answer: Collaratte scale-Christmas tree pattern-Pityriasis rosea

The above description with **central detachment** and **peripheral attachment** of scales with the images suggestive of **collarette scales** and lesions in **christmas tree pattern** is suggestive of **pityriasis rosea.**

Pityriasis rosea/pityriasis circinata/roseola annulata is an **acute self-limiting papulosquamous** disorder. It is commonly seen in **children** and **young adults**.

Clinical features include:

- **Prodrome** of a sore throat, gastrointestinal disturbance, fever or arthralgia.
- **Herald patch** - Large and conspicuous eruptions on the thigh, upper arm, trunk or neck covered by **fine scales**

The image given below shows a herald patch.

- After 5-15 days, it turns into discrete oval lesions, dull pink in color and covered by fine dry **silvery-grey scales**.
- These **secondary lesions** occur as macules and papules that are elliptical or ovular, symmetric, and commonly involve the **thorax, back,** and **abdomen**.
- It is followed by a **marginal collarette of scale** with a **central wrinkling** and **atrophy**. The collarettes of the scales are attached peripherally. It is also called a **hanging curtain sign.**

The image given below shows collarette scales.

- The long axes of the lesions characteristically follow the lines of skin tension parallel to the ribs in a **Christmas tree pattern** on the upper chest and back.

The images given below show Christmas tree pattern along the lines of skin tension on the back and the chest.

The rashes of pityriasis rosea usually **resolve** by **8 weeks**.

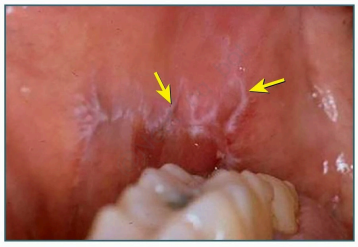
Other options:

Lichen planus is an idiopathic inflammatory skin disease affecting the skin and mucous membranes, often with a chronic course with relapses and periods of remission.Patients present with **pruritic, purple, polygonal, papules.** The lesions have **thin white lines** on their surface known as **Wickham's striae.** They are commonly seen on the volar aspect of the **wrists, the lumbar region**, and **around the ankles.**The **mucosal involvemen**t is seen. The commonest mucosal sites are buccal mucosa and tongue, rarely it can also be found on the anus and genitalia.

A **hypopigmented ring** surrounding individual **psoriatic lesions** is known as the **Woronoff ring**.It is usually associated with treatment, most commonly UV radiation or topical corticosteroids. The pathogenesis of the Woronoff ring is not well understood, but may result from the **inhibition of prostaglandin synthesis.**

Collarette scales in Pityriasis rosea

Wickham's Striae on the buccal mucosa.



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References: Rook's Textbook of Dermatology 9th edition, page 115.9,35.8,37.3

Q99. Dermatology INI-CET 2023 ID: MF9140

Statement 1: Histopathological examination of a patient with Flaccid bulla and painful erosions showing suprabasal blister and inflammatory infiltrate in the blister cleft. Statement 2: The row of tombstones appearance is diagnostic of Pemphigus vulgaris.

A Statement 1 is false and Statement 2 is true

B Statement 1 is true and Statement 2 is false

C ✓ Both the statements are true but 2 is not the conclusion of 1

D Both the statements are true but 2 is the conclusion of 1

Correct Answer: Both the statements are true but 2 is not the conclusion of 1

Both statements 1 and 2 are correct but statement 2 is not the conclusion of statement 1. Statement 2 is not the conclusion of statement 1 because DIF is used for a definitive diagnosis.

Pemphigus vulgaris is a **chronic immunobullous** disorder characterized by the presence of **antibodies** against **desmosomal proteins** expressed in the **skin** and **mucosa.** In the **skin**, **desmoglein 1** is mostly seen in **all layers**, while desmoglein 3 is more in the lower layers. In the **oral mucosa**, **desmoglein 3** is seen in all layers and desmoglein 1 is present in lower layers.

Clinically, the patients present with ill-defined, **irregular, slow-to-heal** oral lesions and erosions. **Flaccid blisters** filled with **clear fluid** may arise on normal skin or an erythematous base. The rupture of such blisters produces **painful erosions** and heals **without scarring.** Lesions may also involve the mucosa of the conjunctiva, nasopharynx, larynx, esophagus, urethra, vulva, or cervix.

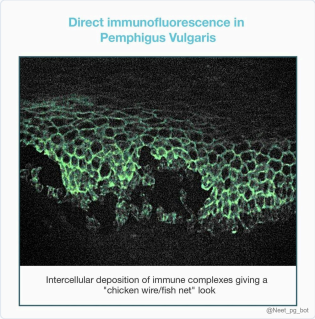
Nikolsky's sign is positive which is elicited by applying firm **tangential** sliding pressure with a finger on lesional or perilesional skin. This leads to the separation of the normal-looking epidermis from the dermis, producing **erosion.**

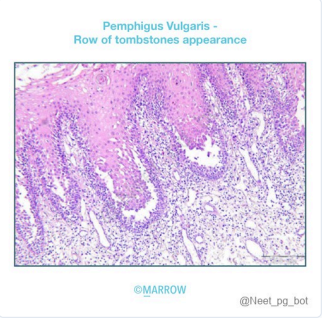
Direct immunofluorescence is considered the most accurate **confirmatory test** of this condition. It shows intercellular **IgG** and **C3** deposition on the **surface** of **keratinocytes** throughout the epidermis in **normal, perilesional skin,** in a **characteristic fishnet/chickenwire appearance** as seen below.

Histology shows a **suprabasal split** of the epidermis with a characteristic **row** of **tombstone** appearance as seen below.

Other investigations include a **Tzanck smear** showing **acantholytic cells**. Anti-Dsg **ELISA assays** can be used to monitor the disease activity.

Systemic corticosteroids are the **mainstay** of treatment for controlling the disease, improving the quality of life and reducing disease activity.





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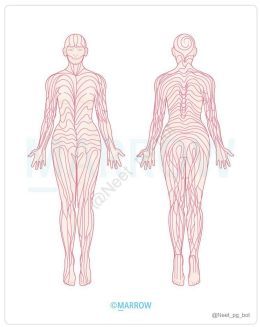
References: Rook's Textbook of Dermatology 9th edition, page 50.1-50.5
Rook's Textbook of Dermatology 10th edition, Page 50.4 - 50.5, 50.7

Q100. Dermatology INI-CET 2023 ID: MF9131

Identify the following lines in dermatology



A Hinderer lines

B Marionettes line

C Relaxed skin tension lines

D ✓ Blaschko lines

Correct Answer: Blaschko lines

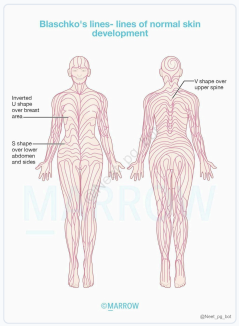
The above image shows **Blaschko lines** as identified by the **spiraling over the scalp**. Blaschko's lines are **lines of normal developmental growth patterns** in the skin. They do not correspond to any known nervous, vascular, or lymphatic structures. Conditions that are distributed along these lines include **epidermal nevi,** and **incontinentia pigmenti.**

Other options:

Option A: **Hinderer's lines** are 2 intersecting lines—one drawn from the ala to the tragus and the other from the oral commissure to the lateral canthus.

Option B: Melomental folds, or **marionette lines,** are one of the consequences of facial aging. The curvilinear wrinkles form because of facial movements and the aging process extend downward from the oral commissures.

Option C: **Relaxed skin tension lines** run perpendicular to the direction of contraction of the underlying muscles and parallel to the dermal collagen bundles.To produce favourable aesthetic results, incisions should generally be designed to follow rhytids or the relaxed skin tension lines as the resulting scars will be stronger and less likely to stretch.



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References: Rook's Textbook of Dermatology 9th edition, Page 7.7,20.1
Online Resource, https://pubmed.ncbi.nlm.nih.gov/19021675/
Online Resource, https://academic.oup.com/asj/article/26/1_Supplement/S22/223368#

Q101. Dermatology INI-CET 2023 ID: MF9134

Match the following wood lamp findings with the disease

1. Erythrasma	a) Pink florescence
2. Pityriasis versicolor	b) Milky white florescence
3. Tinea Capitis	c) Yellow fluorescence
4. Vitiligo	d) Blue fluorescence
	e) Coral red fluorescence

A 1-a,2-c,3-d,4-b

B ✓ 1-e,2-c,3-d,4-b

C 1-e,2-d,3-c,4-b

D 1-a,2-d,3-c,4-b

Correct Answer: 1-e,2-c,3-d,4-b

The correct match is

1. Erythrasma	e)Coral red fluorescence
2. Pityriasis versicolor	c)Yellow fluorescence
3. Tinea Capitis	d)Blue fluorescence
4. Vitiligo	b)Milky white florescence

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References: Rook's Textbook of Dermatology 9th edition, Page no 4.20

Q102. Dermatology INI-CET 2023 ID: MF7295

Consider the causes of alopecia. Which among the following cause(s) non-scarring alopecia?

1. Alopecia areata
2. Telogen effluvium
3. Androgenic alopecia
4. Frontal fibrosing alopecia

A 3 and 4

B ✓ 1, 2, and 3

C 2, 3, and 4

D Only 4

Correct Answer: 1, 2, and 3

Non-scarring alopecia is caused by androgenic alopecia, alopecia areata, and telogen effluvium (1, 2 and 3). **Frontal fibrosing alopecia** is a type of **primary cicatricial (scarring) alopecia.**

Alopecia areata is a chronic inflammatory disease that causes **non-scarring hair loss**. It is a **T-cell-mediated inflammatory disease.** It is associated with several autoimmune diseases, including **thyroiditis**, lupus erythematosus**,** vitiligo, and psoriasis. It mainly affects the **scalp,** but can also affect the **beard, eyebrows,** and **eyelashes.** In some cases, this progresses to a total loss of scalp hair **(alopecia totalis)** or a loss of all hair on the body **(alopecia universalis)**.

Telogen effluvium is a type of **nonscarring alopecia** that is marked by **diffuse shedding** of normal hair. This occurs due to **premature termination** of the anagen phase of the hair cycle. It can be either acute or chronic.

- **Acute telogen effluvium** occurs **2–3 months** after a stressful event such as high **fever**, **surgery**, starvation, and hemorrhage. Hormonal fluctuations during the **postpartum** period can trigger it (**telogen gravidarum**). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally **resolves spontaneously**.
- **Chronic telogen effluvium** is used to describe the diffuse shedding of telogen hair for more than **6 months**. Common causes include **thyroid disorders**, iron-deficiency anemia, and malnutrition.

Androgenetic alopecia is a disorder characterized by a reduction of hair fiber production by follicles and miniaturization. **Dihydrotestosterone**, an androgenic metabolite, is implicated in the etiology. It is of two types: male-pattern and female-pattern hair loss. Male-pattern hair loss causes **frontal hairline recession** and thinning of the **vertex**. Female-pattern hair loss involves the **mid-frontal scalp** diffusely. There is no or minimal involvement of the temporal region.

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References: Rook's Textbook of Dermatology 9th edition, Page no. 32.24, 32.39, 32.40, 86.17, 89.6, 89.7, 89.14, 89.17, 89.20, 89.24, 89.25, 89.46

Q103. Dermatology INI-CET 2023 ID: MF4538

A farmer from Tamil Nadu presents with fever, body ache, and cough for 3 days. Lab studies showed thrombocytopenia. On examination, a black lesion is present, as in the picture below. IgM scrub typhus-positive. What drug will be used in the management?

A Doxycyline

B Ceftriaxone

C Ceftriaxone+Doxycycline

D ✓ Doxycycline + Azithromycin

Correct Answer: Doxycycline + Azithromycin

The above clinical scenario and the image show an **eschar** at the bite site along with positive IgM of **scrub typhus** caused by ***Orientia tsutsugamushi*** and transmitted by **trombiculid mite.** As per a study published in 2023, **Doxycycline** and **Azithromycin combined** therapy was found to be **superior** to **Doxycycline monotherapy.**

Scrub Typhus is also known as chigger-borne typhus, which is caused by **Orientia tsutsugamushi**. The vector is trombiculid mite. Humans are infected when they are bitten by the mite larvae known as **chiggers.** The incubation period is 1- 3 weeks. The patient typically develops a characteristic **eschar** at the site of the mite bite, with **regional lymphadenopathy** and **maculopapular rash**.

Diagnosis is by serologic assays.

Treatment is with a combination therapy of **azithromycin** and **doxycycline.**

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References: Online Resource, https://www.nejm.org/doi/full/10.1056/NEJMoa2208449
Harrison's Principles of Internal Medicine, 21st Edition, page: 1436, 1437

Q104. Dermatology INI-CET 2023 ID: MG1066

A young male complains of an itchy rash over his back as seen below. What is the most likely sign and diagnosis?

A ✓ Collarette scale-pityriasis rosea

B Wickham's striae - lichen planus

C Woronoff's sign - psoriasis

D Christmas tree appearance - psoriasis

Correct Answer: Collarette scale-pityriasis rosea

Pityriasis rosea is the most likely diagnosis in this patient presenting with christmas tree lesions, and will likely have **collarette scales.**

The image given below shows collarette scales.

Pityriasis rosea/pityriasis circinata/roseola annulata is an **acute self-limiting papulosquamous** disorder. It is commonly seen in **children** and **young adults**.

Clinical features include:

- **Prodrome** of a sore throat, gastrointestinal disturbance, fever or arthralgia.
- **Herald patch** - Large and conspicuous eruptions on the thigh, upper arm, trunk or neck covered by **fine scales**

The image given below shows a herald patch.

- After 5-15 days, it turns into discrete oval lesions, dull pink in color and covered by fine dry **silvery-grey scales**.
- These **secondary lesions** occur as macules and papules that are elliptical or ovular, symmetric, and commonly involve the **thorax, back,** and **abdomen**.
- It is followed by a **marginal collarette of scale** with a **central wrinkling** and **atrophy**. The collarettes of the scales are attached peripherally. It is also called a **hanging curtain sign.**
- The long axes of the lesions characteristically follow the lines of skin tension parallel to the ribs in a **Christmas tree pattern** on the upper chest and back.

The images given below show Christmas tree pattern along the lines of skin tension on the back and the chest.

The rashes of pityriasis rosea usually **resolve** by **8 weeks**.

Other options:

Lichen planus is an idiopathic inflammatory skin disease affecting the skin and mucous membranes, often with a chronic course with relapses and periods of remission.Patients present with **pruritic, purple, polygonal, papules.** The lesions have **thin white lines** on their surface known as **Wickham's striae.** They are commonly seen on the volar aspect of the **wrists, the lumbar region**, and **around the ankles.**The **mucosal involvemen**t is seen. The commonest mucosal sites are buccal mucosa and tongue, rarely it can also be found on the anus and genitalia.

A **hypopigmented ring** surrounding individual **psoriatic lesions** is known as the **Woronoff ring**.It is usually associated with treatment, most commonly UV radiation or topical corticosteroids. The pathogenesis of the Woronoff ring is not well understood, but may result from the **inhibition of prostaglandin synthesis.**

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References: Rook's Textbook of Dermatology 9th edition, 25.91

Q105. Dermatology INI-CET 2023 ID: MG1058

An overweight 12-year old girl has the following blackish discolouration on her neck, as shown below. What is this condition and its' association?

A ✓ Acanthosis nigricans - Insulin resistance

B Stria Albicans - Cushing syndrome

C Xeroderma pigmentosum - Malignancy

D Cafe au lait spots - McCune Albright syndrome

Correct Answer: Acanthosis nigricans - Insulin resistance

The condition shown in the above image is likely **acanthosis nigricans** - that is strongly associated with **insulin resistance/hyperinsulinemia**

Acanthosis nigricans is a darkening of the skin of the **intertriginous areas**. The hyperpigmentation is generally velvety with poorly defined borders and thickening of the skin.

It can be grossly classified into benign and malignant.

Benign acanthosis nigricans is associated with **obesity** or **insulin resistance**. **Insulin resistance** is the most accepted hypothesis used to explain the pathophysiology of the condition. **Elevated circulating fatty acids** are considered a major factor in developing **insulin resistance.**

Malignant acanthosis nigricans has a rapid onset and is characterized by symmetrical, hyperpigmented, rugose, velvety plaques. It affects the axillae and flexures, nape of the neck, and mucosal surfaces. If it occurs de novo and is associated with weight loss, malignancy should be suspected. Generalized pruritus and skin changes of tripe palms (acanthosis palmaris) are also associated with malignancy. Associated malignancies are generally **GI or lung carcinomas**. In malignancy, tumor cells are believed to produce transforming growth factor-alpha or cytokines that activate **insulin-like growth factors**.

Acanthosis nigricans can only be treated by treating the underlying disease. In addition, treatment specific to acanthosis nigricans is done for aesthetic reasons. Medications like keratolytics, topical vitamin D, and melatonin can be used. Procedures like laser, dermabrasion, and chemical peels are also available.

Other options:

Option B: **Stria Albicans** is a stretch mark that is white or pearly in color. It is most common in pregnant women and people who have gained or lost weight quickly. Stria Albicans is not associated with Cushing syndrome, which is a hormonal disorder caused by excess cortisol production and produces a strechy, purplish rash most commonly over the abdomen/flanks.

Option C: **Xeroderma pigmentosum - malignancy** is a rare genetic disorder that makes the skin extremely sensitive to ultraviolet radiation from the sun. People with xeroderma pigmentosum have a greatly increased risk of developing skin cancer.

Option D: **Cafe au lait spots - McCune Albright syndrome** is a genetic disorder that causes multiple light brown, oval-shaped birthmarks on the skin. It is also associated with other symptoms, such as precocious puberty and bone abnormalities - polyostotic fibrous dysplasias.

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References: Harrison's Principles Of Internal Medicine - 19th Edition, Page no.2857

Q106. Dermatology INI-CET 2023 ID: MG1055

Which of the following are cause(s) of non-scarring alopecia?

1. Alopecia areata
2. Telogen effluvium
3. Androgenic alopecia
4. Frontal fibrosing alopecia

A 3 and 4

B ✓ 1, 2 and 3

C 2, 3 and 4

D Only 4

Correct Answer: 1, 2 and 3

Non-scarring alopecia is caused by **androgenic alopecia, alopecia areata, and telogen effluvium.**

Alopecia areata

It is a chronic inflammatory disease that causes **non-scarring hair loss**. It is a **T-cell-mediated inflammatory disease.** It is associated with several autoimmune diseases, including **thyroiditis**, lupus erythematosus**,** vitiligo, and psoriasis.

It mainly affects the **scalp,** but can also affect the **beard, eyebrows,** and **eyelashes.** In some cases, this progresses to a total loss of scalp hair **(alopecia totalis)** or a loss of all hair on the body **(alopecia universalis)**.

Telogen effluvium

It is a type of **nonscarring alopecia** that is marked by **diffuse shedding** of normal hair. This occurs due to **premature termination** of the anagen phase of the hair cycle. It can be either acute or chronic.

Acute telogen effluvium occurs **2–3 months** after a stressful event such as high **fever**, **surgery**, starvation, and hemorrhage. Hormonal fluctuations during the **postpartum** period can trigger it (**telogen gravidarum**). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally **resolves spontaneously**.

Chronic telogen effluvium is used to describe the diffuse shedding of telogen hair for more than **6 months**. Common causes include **thyroid disorders**, iron-deficiency anemia, and malnutrition.

Androgenetic alopecia

It is a disorder characterized by a reduction of hair fiber production by follicles and miniaturization. **Dihydrotestosterone**, an androgenic metabolite, is implicated in the etiology. It is of two types: male-pattern and female-pattern hair loss. Male-pattern hair loss causes **frontal hairline recession** and thinning of the **vertex**. Female-pattern hair loss involves the **mid-frontal scalp** diffusely. There is no or minimal involvement of the temporal region.

CICATRICIAL/SCARRING ALOPECIA: Frontal fibrosing alopecia is a primary cicatrical condition that affects **post-menopausal women** in particular and frequently involves the eyebrows.

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References: Rook's Textbook of Dermatology 9th edition, Page no 89.28-89.33, 37.7

Q107. Dermatology INI-CET 2023 ID: MG1449

A middle-aged Indian man came to your OPD with the lesion as shown in the image below. The most likely diagnosis is

A Squamous cell carcinoma

B ✓ Basal cell carcinoma

C Lupus vulgaris

D Nevus

Correct Answer: Basal cell carcinoma

The image above shows a **central ulcer** with **raised borders** and rolled-out edges (rodent ulcer) suggesting a diagnosis of **basal cell carcinoma**.

Basal cell carcinoma is a **slow-growing malignant tumor** with **locally invasive** characteristics. It affects the pilosebaceous skin.

Ultraviolet radiation exposure is the **strongest predisposing** factor. Other risk factors include exposure to arsenic, coal tar, aromatic hydrocarbons, and ionizing radiation.

Basal cell carcinoma mainly affects **elderly** or **middle-aged men** with excessive skin exposure, commonly in the age group of 40–80 y and in white-skinned people.

Macroscopically, it is categorized as localized and generalized. **Nodular, nodulocystic** (nodular and nodulocystic together account for 90% of basal cell carcinoma), pigmented, and nevoid are the localized cancers, while superficial multifocal and superficial spreading, infiltrative, and cicatrizing are the generalized types. The nodular type is described as **pearly** waxy **nodules** at the margins. A central depression (umbilication) or ulceration and rolled-out edges are characteristic of basal cell carcinoma.

Sites near the **eye, nose,** and **ear** and of size **>2 cm** are considered to be **high risk**. Direct invasion at these sites can reach the cranium.

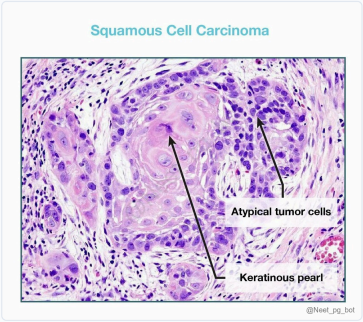
Management is by **Mohs microscopic surgery**. Non-surgical measures, such as the administration of **topical 5-fluorouracil** and **imiquimod,** are useful. Metastasis is **extremely rare.**

Other Options:

Option A: **Squamous cell carcinoma** is the second most common malignant tumor, seen more commonly in men. Microscopically, the proliferation of irregular masses of squamous epithelium is seen. The image is shown below.

Option C- **Lupus vulgaris** originates from an underlying focus of tuberculosis, typically in a bone, joint, or lymph node, or maybe from the hematogenous and lymphatic spread. It presents as a **slowly enlarging plaque** with an **elevated border** and **central atrophy**. The edges of the lesion gradually extend in some areas and **heal with scarring** in others. In India, the buttocks, and trunk are more frequently involved. **Apple jelly nodules** are demonstrated on **diascopy.**

Option D: **Nevus** are aggregates of melanocytes at the dermis or dermo-epidermal junction. It can get transformed into **malignant melanoma** and the features suggestive of malignant transformation are a **change** in **size, shape, color, thickness, satellite lesions,** and **tingling/ itching/serosanguinous discharge.**



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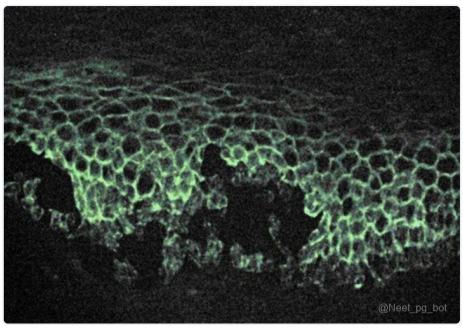
References: Rook's Textbook of Dermatology - 9th Edition - Volume IV, Page 141.1, 141.2, 141.4, 141.8, 141.11, 141.13 - 141.16, 142.1
Bailey & Love's Short Practice of Surgery - 27th Edition, Page no 599, 604-605, 607-609, 611

Q108. Dermatology INI-CET 2023 ID: MG1063

A patient with several skin lesions undergoes an investigation and an image of it is shown below. What is this test?



A Indirect immunofluorescence

B ✓ Direct immunofluorescence

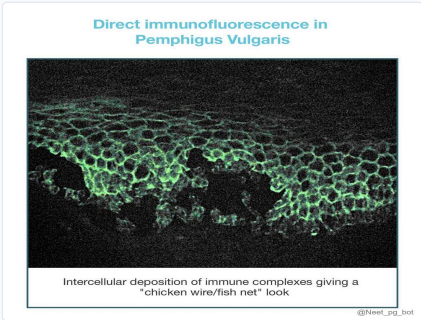
C Immunohistochemistry

D Fluorescent in-situ hybridization

Correct Answer: Direct immunofluorescence

Direct immunofluorescence test, that is shown above, is the most accurate test to diagnose **Pemphigus Vulgaris.**

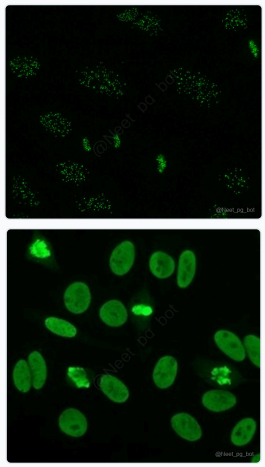
Direct Immunofluorescence uses **anti-IgG antibodies** which detects the **anti-desmoglein IgG antibodies** deposited on the **Desmosomes**. These desmosomes are present at cell junctions; this leads to **the fishnet pattern.**



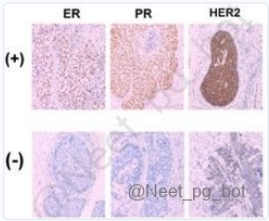
Direct immunofluorescence (DIF) is a type of immunofluorescence that is used to diagnose autoimmune blistering diseases, such as pemphigus. In DIF, antibodies are applied directly to a skin biopsy specimen. If the patient has pemphigus, the antibodies will bind to the desmosomes, which are the proteins that hold skin cells together. This binding will produce a **fish net pattern** of fluorescence under a microscope.

Other options:

Indirect immunofluorescence (IIF) is a type of immunofluorescence that is used to diagnose autoimmune diseases. In IIF, antibodies are first applied to a patient's serum, and then the serum is applied to a skin biopsy specimen. The presence of antibodies in the patient's serum will cause the skin biopsy specimen to fluoresce under a microscope. Seen below is a centromeric pattern on IF in Sjogren's disease -1 and Diffuse/homogenous staining in SLE - 2.

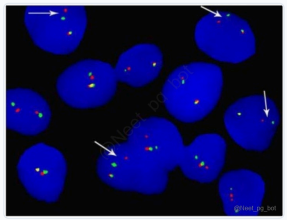


Immunohistochemistry is a type of microscopy technique used to visualize the presence of specific proteins/receptors in cells or tissues. It uses antibodies that are labeled with enzymes or fluorescent dyes to bind to specific antigens. An example is receptor identification in breast cancer specimens.



Fluorescent In Situ Hybridisation (FISH) is a simple technique for the detection of specific DNA sequences in cells/tissues. It uses fluorescent labeled probes to bind to complementary DNA FISH can be performed on **prenatal samples**, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections. It is used to detect

- **numeric abnormalities** of chromosomes (aneuploidy)
- subtle **microdeletions** and **rearrangements**.
- complex **translocations** that are not demonstrable by routine karyotyping
- **gene amplification** (e.g., HER2 in breast cancer or NMYC amplification in neuroblastomas)



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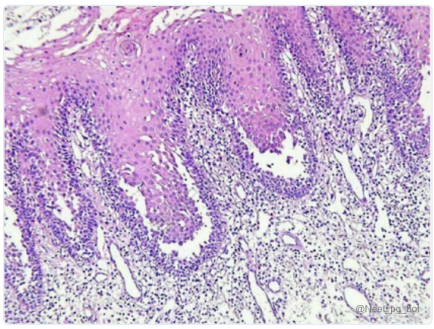
References: Rook's Textbook of Dermatology 9th edition, Chapter 50 Page no 50.8

Q109. Dermatology INI-CET 2023 ID: MG1069

A patient presents with vesicles and oral lesions. Histopathological examination reveals suprabasal split. Which of the following is the most likely diagnosis?



A Varicella

B IgA disease

C ✓ Pemphigus vulgaris

D Pemphigus foliaceous

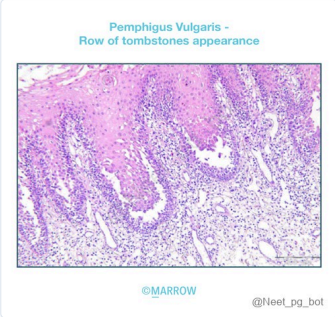
Correct Answer: Pemphigus vulgaris

The image given shows the **row of tombstone** appearance seen in **Pemphigus vulgaris.**

Pemphigus vulgaris is an autoimmune blistering disease caused by the deposition of IgG antibodies against desmoglein 3, a protein that holds skin cells together. It is characterized by widespread flaccid bullae, which are large, fluid-filled blisters that rupture easily. The lesions typically start on the trunk and spread to the extremities. Oral lesions are common in pemphigus vulgaris. Histologically, pemphigus vulgaris is characterized by a suprabasal split with acantholysis (loss of cohesion between skin cells).

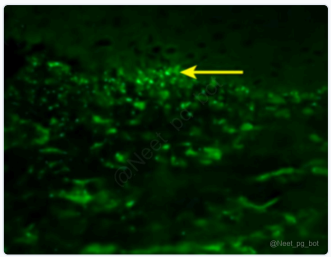
Acantholysis and **cell apoptosis** are the key pathological processes in pemphigus. Histological changes include:

- **Intercellular oedema** with **loss of intercellular attachments** in the basal layer is the **earliest change**
- **Suprabasal epidermal cells separate** from the basal cells to form clefts and blisters.
- **Basal cells remain attached** to the basement membrane, in a '**row of tombstones**' on the floor of the blister.
- Blister cavities contain rounded-up **acantholytic cells,** also seen in **Tzank smears.**



Type of pemphigus	Cutaneous distribution	Mucosal involvement	Lesions
Pemphigus vulgaris	- Scalp, face, chest, upper trunk - May be generalised	- Oro-pharynx, conjunctiva, genital	- Flaccid blisters - Erosions - Flexural vegetations
Pemphigus vegetans	- Flexural aspects	- Oro-pharynx, conjunctiva, genital	- Vegetating plaques - Vesicles - Pustules - Erosions
Pemphigus foliaceous	- Scalp, face, chest, upper back, seborrheic areas - May be generalised	- None	- Scaly papules - Crusted erosions - Erythroderma
Paraneoplastic pemphigus	- Upper body - Palmoplantar	- Severe mucositis	- Polymorphous bullae - Erosions - Target lesions

IgA disease (Linear IgA bullous dermatosis): Autoimmune blistering disease caused by the deposition of IgA antibodies at the dermal-epidermal junction. It is characterized by vesicles and bullae, which are large, fluid-filled blisters. The lesions typically start on the trunk and spread to the extremities. Oral lesions are not common in IgA disease. Histologically, IgA disease is characterized by a suprabasal split with linear IgA deposition at the dermal-epidermal junction. **Anti-transglutaminase** antibodies are positive in **dermatitis herpetiformis**. Patients present with erythematous papules and vesicles on the extensor aspect of elbows, knees, buttocks, and scalp. Histopathology demonstrates **subepidermal blister** formation with neutrophils located at the tips of the dermal papillae. Direct immunofluorescence will reveal the **deposition of IgA** in the papillary dermis in a granular or fibrillar pattern.



Pemphigus foliaceus

Pemphigus foliaceus is an autoimmune blistering disease caused by the deposition of IgG antibodies against **desmoglein 1**, a protein that holds skin cells together. It is characterized by **superficial**, **flaccid** bullae that are prone to rupture and erosion. The lesions typically start on the face and scalp and then spread to the trunk and extremities. Oral lesions are not common in pemphigus foliaceus. Histologically, pemphigus foliaceus is characterized by a **subcorneal split** with acantholysis and minimal inflammation.

The image given below shows the lesions seen in pemphigus foliaceous.

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References: Rook's Textbook of Dermatology 9th edition, 50.7
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK499829/

Q110. Dermatology AIIMS 2019 (May & Nov) ID: MC2623

A 5-year-old boy had painful sores in the mouth with a rash over the hands along with fever. Which of these is the causative agent?

A Pox virus

B ✓ Coxsackievirus

C HHV-7

D Parvovirus

Correct Answer: Coxsackievirus

The presence of **fever**, along with **painful sores** in the **mouth** and **small, tense vescicles** with surrounding erythema on the **palms and soles** in a **child** point towards the diagnosis of **hand-foot-mouth disease.** The most common cause of which worldwide is **coxackievirus A16**.

Spread is by **droplets** or **faecal** contamination and the incubation period is **7 days.** The fever is generally mild, and is followed by the appearance of lesions. The **characteristic lesions** are slightly oval, with a grey blister roof and a narrow rim of erythema. The **oral lesions** are aphthoid and are less painful than aphthous ulcers.

Only **symptomatic treatment** is given. The lesions generally **disappear in 7 - 10 days**.

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References: Rook's Textbook of Dermatology 9th edition, Page no 117.6
Fitzpatrick's Dermatology, 9th Edition., Page no 1591

Q111. Dermatology AIIMS 2019 (May & Nov) ID: MC6129

The following lesion is seen in a 25-year-old man who came to a primary health centre 5 days after he had unprotected sexual intercourse. On palpation the ulcers were painful. Suppurative inguinal lymphadenopathy was present on the right side. The patient has no previous history of drug allergy. Which kit will be used for the management of this particular sexually transmitted disease?

A Kit 6/Yellow

B Kit 2/green

C ✓ Kit 3/white

D Kit 1/grey

Correct Answer: Kit 3/white

The appearance of **painful genital ulcers**, along with **unilateral suppurative** inguinal lymphadenopathy, **5 days** after unprotected sexual intercourse best fits the symptom cluster of a **non - herpetic genital ulcer**. The treatment of which is **kit 3 / white kit** in case of **no penicillin allergy.**

The above-shown clinical image and the findings are consistent with a diagnosis of **chancroid.**

Soft Chancre/Chancroid:

- The causative organism is ***Haemophilus ducreyi*.**
- **The incubation** period is **3 to 10 days.**
- It presents with **multiple painful ulcers** on genitals.
- It may be accompanied by unilateral **painful, suppurative inguinal lymphadenopathy.**
- Treatment: Inj. Benzathine penicillin 2.4 million units + tab. Azithromycin 1 gm single dose

Note: In the syndromic approach to sexually transmitted infections, kit assignment based on the identification of a cluster of symptoms rather than making an actual diagnosis is given emphasis.

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References: Online Resource, http://naco.gov.in/sites/default/files/Syndromic%20Poster_REDUCED.pdf

Q112. Dermatology AIIMS 2019 (May & Nov) ID: MC7531

A patient with fever and joint pain developed pigmentation on the nose after a few days of taking NSAIDs as shown below. What is the diagnosis?

A Melasma

B Fixed drug eruption

C Lichen planus

D ✓ Chikungunya

Correct Answer: Chikungunya

The given clinical history of **fever** and **joint pain** and the image showing hyperpigmented macules on nose (**chik sign**) is suggestive of Chikungunya.

Post-chikungunya pigmentation (PCP) most commonly affects the nose and cheeks. It often appears as the **fever subsides** and may persist for about 3-6 months.

PCP mimics melasma and may be misdiagnosed if proper history is not taken.

Another close differential is **fixed drug eruptions** (FDE). FDE is a type of delayed hypersensitivity reaction. It is characterized by well-defined brown, **hyperpigmented plaques** that may follow **vesiculobullous** lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation.They usually present 30 min to 8 h after drug exposure and **recur at the same location** on re-exposure. NSAIDs, Paracetamol and Co-trimoxazole are the most common drugs causing FDE.

The image below shows FDE on upper lip area:

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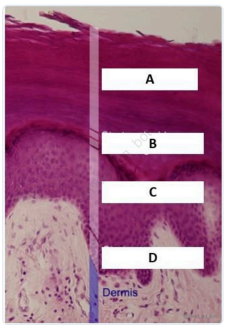
References: Online Resource, http://www.ijdvl.com/article.asp?issn=0378-6323;year=2010;volume=76;issue=6;spage=671;epage=676;aulast=Riyaz
Harrison's Principles of Internal Medicine, 21st Edition, Pages 141-142, 1633
Rook's Textbook of Dermatology 9th edition, Page 118.11

Q113. Dermatology AIIMS 2019 (May & Nov) ID: MC7641

Which cell is not present in the layer marked D?



A Keratinocyte

B Merkel cell

C Melanocyte

D ✓ Langerhans cells

Correct Answer: Langerhans cells

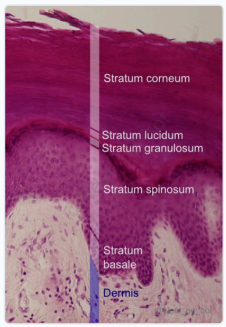
The layer marked as **D** is **stratum basale**. **Langerhan's cells** are not present in stratum basale. Langerhan's cells are predominantly present in stratum spinosum.

Keratinocytes, melanocytes, and Merkel discs are all seen in stratum basale:

(Option A) Keratinocytes are predominant cells of the epidermis and constitute the majority of the stratum basale.

(**Option B) Merkel discs** are type 1 slow-adapting touch receptors.

(Option C) Melanocytes are **dendritic** cells which are derived from the neural crest cells. The **melanosomes** synthesized by the melanocytes migrate to the adjacent keratocytes through the dendritic processes



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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no 13

Q114. Dermatology AIIMS 2019 (May & Nov) ID: MC7640

Topical steroids are most effective in:

A Dermal atrophy

B ✓ Eczematous dermatitis

C Bullous lesions due to HSV

D Rosacea

Correct Answer: Eczematous dermatitis

Topical steroids are the drug of choice in the treatment of **eczematous dermatitis**.

Other options:

- Steroids cause atrophy and hence is contraindicated in patients with dermal atrophy.
- Herpes infection is associated with immunosuppression so steroids are not useful.
- Routine steroid use not recommended in rosea as it can worsen the condition.

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References: Illustrated Synopsis of Dermatology and Sexually Transmitted Diseases, Page no 53
Fitzpatrick's Dermatology, 9th Edition., Page no 377

Q115. Dermatology AIIMS 2019 (May & Nov) ID: MA4061

You have ordered a patch test for an 8-year-old boy with a history of allergic contact dermatitis. After the application of the patch, when should you ask him to return for removal?

A 24 hours

B 72 hours

C ✓ 48 hours

D 96 hours

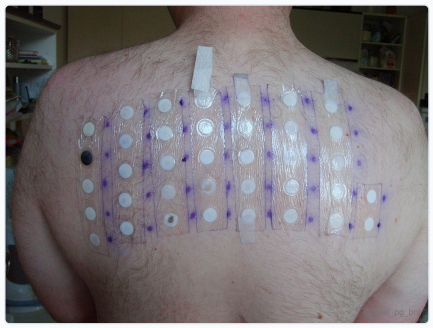
Correct Answer: 48 hours

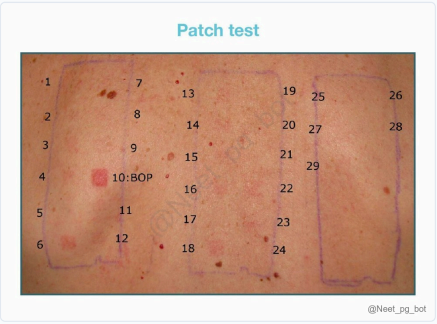
In **patch testing**, the patch is removed after **48 hours** to examine the site of reaction.

Patch testing is a test to detect **sensitivity** to a specific **allergen**. A battery of suspected allergens is applied to the patient's back under occlusive dressings or patches. It is allowed to remain in contact with the skin for 48 hours.

- **48 hours:** Patch removal for **initial reading** to detect **delayed hypersensitivity** reaction.
- **72-96 hours**: Final reading done for confirmation.

Note: Since the question is regarding the timing of patch removal and not regarding the final diagnosis, 48 hours is the answer.





@Neet_pg_bot

References: Rook's Textbook of Dermatology 9th edition, Section 4.23
Fitzpatrick's Dermatology, 9th Edition., Page no 406
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3900336/

Q116. Dermatology AIIMS 2019 (May & Nov) ID: MD0207

A patient was treated with steroids for psoriasis. On stopping the treatment, he develops fever, malaise, and lesion as seen in the image below. What is the most probable diagnosis?

A ✓ Pustular psoriasis

B Staphylococcal infection

C Acute generalised exanthematous pustulosis

D Subcorneal pustulosis

Correct Answer: Pustular psoriasis

The given clinical scenario points towards the diagnosis of **pustular psoriasis.**

Pustular psoriasis is a type of psoriasis characterized by **tiny pus-filled lesions** (pustules) having a characteristic **lakes or sheets of pus appearance** (seen in the image).

The patient presents with a history of **fever** and malaise along with the presence of **pustules** on a **circumscribed, fiery red, edematous or scaling plaque.** The pus is usually **sterile.** Sudden **withdrawal of systemic steroids** is usually the trigger for this condition.

Pustular psoriasis can be divided into:

- **Localized pustular psoriasis**
- **Generalized pustular psoriasis**:
  - Acute generalized pustular psoriasis (**Von Zumbusch**)
  - Subacute annular and circinate pustular psoriasis
  - Acute generalized pustular psoriasis of pregnancy (**Impetigo herpetiformis**)

Treatment: **Acitretin** (oral retinoid) is the **drug of choice** for pustular psoriasis. However, it is **contraindicated in pregnancy. Systemic steroids** are used instead. **Cyclosporine** is the second-line drug.

Note: Systemic steroids are never used for the treatment of psoriasis except in the case of Impetigo herpetiformis where they are first-line agents.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 18.18, 26.22, 35.33

Q117. Dermatology AIIMS 2019 (May & Nov) ID: MD0208

Which of the following does not present with maculopapular rash?

- A ✓ Zoster (shingles)
- B Rubella
- C Dengue (during febrile phase)
- D Measles

Correct Answer: Zoster (shingles)

A maculopapular rash is not seen in zoster (shingles).

Shingles cause **vesicular lesions** that are grouped together and specific to a dermatome.

Type of rash	Description
Macule	**Flat** lesion with a well-circumscribed change in skin color < 1 cm
Papule	**Elevated solid** skin lesion < 1 cm
Vesicle	Small **fluid** containing blister

Maculopapular rashes are seen in:

- Rubella
- Dengue (febrile phase)
- Measles
- Chikungunya (acute infectious phase)
- Erythema infectiosum
- Zika virus
- West Nile virus
- Infectious mononucleosis

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References: Harrison's Principles of Internal Medicine 20th Edition, Page no 106
Rook's Textbook of Dermatology 9th edition, Page no 4.13

Q118. Dermatology AIIMS 2019 (May & Nov) ID: MD0209

A patient presents with flaccid bullous lesions involving the oral cavity and the skin. He has lesions as shown below. Acantholytic cells are seen on Tzanck smear. What is the most probable diagnosis?

A Pemphigus foliaceus

B ✓ Pemphigus vulgaris

C Dermatitis herpetiformis

D Bullous pemphigoid

Correct Answer: Pemphigus vulgaris

Flaccid lesions, **mucosal** involvement, and the presence of **acantholytic cells** are suggestive of **pemphigus vulgaris.**

Option A: There is no mucosal involvement in pemphigus foliaceus.

Option C: Dermatitis herpetiformis presents with multiple small vesicles and no acantholysis.

Option D: Bullous pemphigoid presents with tense bullae and no acantholysis.

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References: Harrison's Principles of Internal Medicine 20th Edition, Page no: 356
Rook's Textbook of Dermatology 9th edition, Page no: 50.4

Q119. Dermatology AIIMS 2019 (May & Nov) ID: MD0210

A 10-year-old child presents with hair loss. The hair loss is patchy as shown in the image below. Black dots are noted in areas of hair loss. Which of the following types of tinea is this child suffering from?

A Ectothrix

B ✓ Endothrix

C Kerion

D Favus

Correct Answer: Endothrix

The given clinical scenario depicting a **young child** with **patchy hair loss** and the image showing **black dots** is suggestive of **endothrix type** of **tinea capitis.**

Tinea capitis is a dermatophyte infection primarily occurring in **children**. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with **sinus** formation.

The **clinical variants** of tinea capitis are:

- **Endothrix** - a non-inflammatory type in which **multiple black dots** are present within the areas of alopecia. It is most commonly caused by **T. tonsurans and T. violaceum.**
- **Ectothrix** -a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. **Microsporum audouinii**, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.
- **Kerion** -a severe inflammatory type that presents with an inflammatory mass with **thick crusting** and **matting of hair** with pus and sinus formation. It is commonly caused by **T. verrucosum and T. mentagrophytes**.
- **Favus** - is characterized by the presence of yellowish, cup-shaped crusts known as **scutula** that develop around hair follicles. It is caused by **T. schoenleinii**

Treatment:

- **Terbinafine**: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.
- **Itraconazole**: 2-4 mg/kg/day for 4-6 weeks.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 32.8, 32.38 - 32.40

Q120. Dermatology NEET 2020 ID: MD0683

Identify the drug used to treat the condition shown below:

A Anti-fungal therapy

B ✓ Anti-tubercular therapy

C Topical steroids therapy

D Anti-leprosy treatment

Correct Answer: Anti-tubercular therapy

The given image shows an **annular plaque** with **central scarring** points to **lupus vulgaris**, which is treated using **anti-tubercular drugs**.

It is the **most common** cutaneous tuberculosis in India. The characteristic lesions are annular, infiltrated plaques with **serpiginous edges**, central atrophy and scarring and are commonly seen on face, buttocks and extremities.

On diascopy, **apple jelly** nodules are seen.

Histologically, tubercles are seen in the superficial dermis showing scanty or absent central caseation with surrounding epithelioid histiocytes and giant cells.

Other differential diagnoses for annular plaque include:

- **Tinea corporis** shows **central clearing**, peripherally raised **scaly margins** and presents with an **itch**.

- **Cutaneous leishmaniasis** lesions have **central crusting** and are seen on the **face** in patients from areas of high leishmaniasis prevalence.

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References: Rook's Textbook of Dermatology 9th edition, Page nos 27.21-27.23, 32.35-32.36, 33.44-33.45

Q121. Dermatology NEET 2020 ID: MD0684

Identify the finding associated with the condition shown in the image below.

A ✓ Isomorphic phenomenon

B Meyerson phenomenon

C Gottron's papule

D Nikolsky's sign

Correct Answer: Isomorphic phenomenon

The image shows **plaque psoriasis** of the dorsum of the hand, and **isomorphic** or **Koebner** phenomenon is associated with the condition.

Option B: **Meyerson** phenomenon refers to the formation of an **eczematous ring** around a **melanocytic nevus.**

Option C: **Gottron's papules** are **violaceous,** erythematous **papules** overlying the dorsal interphalangeal or metacarpophalangeal, elbow, or knee **joints.**

Option D: **Nikolsky sign** refers to easy **peeling** of **skin** on applying **tangential pressure** over a **bony prominence** and is classically seen in pemphigus and staphylococcal scalded skin syndrome.

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References: Rook's Textbook of Dermatology 9th edition, Chapter 123.2

Q122. Dermatology NEET 2020 ID: MD0681

Identify the condition shown in the image below:

A ✓ Herpes labialis

B Herpangina

C Molluscum contagiosum

D Impetigo

Correct Answer: Herpes labialis

The above image shows multiple **vesicles** & **crusts** clustered on an inflamed base around **the lips** which is characteristic of **herpes labialis** or **cold sore**.

It is commonly caused by herpes simplex virus type 1(**HSV-1**) and rarely by herpes simplex virus type 2 (HSV-2). It begins with a prodrome of **pain**, **burning**, and **tingling** followed by the appearance of erythematous papules. The papules then become vesicular and may ulcerate.

It is treated with **oral acyclovir** 200 mg and 5% **topical acyclovir.**

Option B: Herpangina or vesicular pharyngitis is characterized by the presence of **ulcerating small vesicles** on the **oral fauces** and the **posterior pharyngeal wall.** It is commonly caused by the coxsackie virus.

Option C: Molluscum contagiosum is characterized by **pearly white, wart-like umbilicated papules**. It is caused by the pox virus.

Option D: Impetigo is characterized by **golden yellow** or **honey-coloured crusts** that form after the rupture of pustulating vesicles. Non-bullous impetigo is caused by both S. aureus and streptococcus. Bullous impetigo is caused by S. aureus.

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References: Fitzpatrick's Dermatology, 9th Edition., Page no 2721
Online Resource, https://commons.wikimedia.org/wiki/File:Herpes_labialis_-_opryszczka_wargowa.jpg
Rook's Textbook of Dermatology 9th edition, Page 25.12, 25.17-19, 25.82 & 26.14

Q123. Dermatology NEET 2020 ID: MD0753

Which of the following drugs used as nail lacquer belongs to morpholines?

A ✓ Amorolfine
B Oxiconazole
C Ciclopirox olamine
D Tioconazole

Correct Answer: Amorolfine

Amorolfine is a **morpholine** that is available as a 5% **nail lacquer**. It is used in the treatment of **onychomycosis**.

Options B and D: Oxiconazole and tioconazole are classified as **imidazoles**
Option C: Ciclopirox olamine is a **hydroxypyridone** compound.

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pubmed/20737634
Rook's Textbook of Dermatology 9th edition, Page no. 18.12

Q124. Dermatology NEET 2020 ID: MD0866

A child was born with membranes around the body and had ectropion and eclabium. He is brought to the OPD with lesions covering his face, trunk, and extremities. Which of the following is an unlikely diagnosis?

A ✓ Ichthyosis vulgaris

B Lamellar ichthyosis

C Bathing suit ichthyosis

D Harlequin ichthyosis

Correct Answer: Ichthyosis vulgaris

The clinical picture shows a neonate encased in a **shiny, parchment-like membrane** (collodion baby) suggestive of a disorder belonging to the **autosomal recessive congenital ichthyosis spectrum**.

Ichthyosis vulgaris is not a part of this disease spectrum. It **does not present at birth** and is therefore an unlikely diagnosis.

Ichthyosis vulgaris is an **autosomal dominant** condition caused due to a mutation in the **filaggrin gene (FLG).** It develops in the first few months of life. Patients exhibit **accentuated palmar** and **plantar creases** and have **dry scaly skin**. The scales appear light grey and cover the extensor surfaces of the limbs and the trunk. It usually spares the groin and larger flexures. Immunohistochemistry reveals an **absent** or **markedly reduced filaggrin** signal.

Autosomal recessive congenital ichthyosis spectrum includes the following:

- Harlequin ichthyosis (HI)
- Bathing suit ichthyosis (BSI)
- Lamellar ichthyosis (LI)
- Congenital ichthyosiform erythroderma (CIE)
- Self- improving congenital ichthyosis (SICI)
- Transient manifestations, such as collodion baby.

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References: Rook's Textbook of Dermatology 9th edition, Page no. 65.4, 65.6 - 65.10

Q125. Dermatology NEET 2025 ID: MH0163

1. A child presents with trigeminal neuralgia associated with painful vesicular lesions around the mouth. On inoculation of the cell culture, multinucleated giant cells with intranuclear inclusions were seen. What is the causative organism?

A Adenovirus

B ✓ HSV

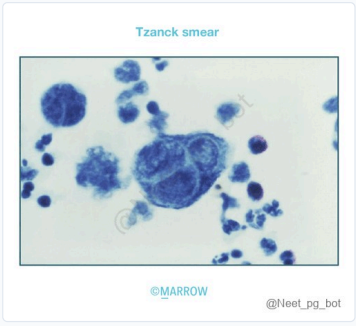
C HPV

D CMV

Correct Answer: HSV

The given clinical scenario of a child complaining of painful vesicular lesions around the mouth, with histopathology showing multinucleated giant cells, suggests the most probable diagnosis to be oral herpes caused by Herpes Simplex Virus (HSV). Herpes infection can be caused by the HSV-1 and HSV-2 viruses. HSV-1 causes orofacial lesions, i.e. herpes labialis, while HSV-2 mainly causes genital herpes which is characterised by vesiculoulcerative lesions of the penis of the male or the cervix, vulva, vagina, and perineum of the female. Genital herpes is often accompanied by symptoms like fever, headache, malaise, and myalgia, pain, itching, dysuria, vaginal and urethral discharge, and tender inguinal lymphadenopathy. Diagnosis of herpes simplex virus: * Most sensitive test- HSV DNA- PCR (can differentiate between HSV 1 and 2) * Virus isolation * Histopathology showing multinucleated giant cells * Microscopy - Tzanck smear, which is a rapid, diagnostic method, shows Tzanck cells. These are multinucleated giant cells with faceted nuclei and homogenously stained ground-glass chromatin. The following image shows the histology of HSV inclusion bodies.

The image given below shows Tzanck cells.



Pearl PM0144: Inclusion bodies seen in various conditions/diseases 1. Bacterial Inclusion Bodies

Disease/Bacteria	Inclusion bodies
Granuloma inguinale, *Klebsiella granulomatis*	Donovan bodies
Psittacosis, *Chlamydia psittacci*	Levinthal-Cole-Lille bodies (LCL bodies)
Trachoma, *Chlamydia trachomatis*	Halberstaedter-Prowazek bodies
Lymphogranuloma venereum, *Chlamydia trachomatis*	Miyagawa's granulocorpusles
Diphtheria, *Corynebacterium diptheriae*	Babes-Ernst granules

2. Viral Inclusion Bodies * Intra-cytoplasmic Inclusion Bodies

Disease/Virus	Inclusion bodies
Rabies	Negri bodies
Vaccinia, variola (smallpox)	Guarnieri bodies
Variola (smallpox)	Paschen bodies
Molluscum contagiosum	Henderson-Patterson bodies
Fowlpox	Bollinger bodies

* Intra-nuclear Inclusion Bodies

	Inclusion bodies	Disease/Virus
Cowdry type A	Lipschutz inclusions	Herpes Virus
	Torres bodies	Yellow fever
Cowdry type B	Basophilic inclusions	Adenovirus
	Acidophilic inclusions	Poliovirus

* Both Intra-nuclear and Intra-cytoplasmic Inclusion Bodies

Disease/Virus	Inclusion bodies
Measles Virus	Warthin-Finkedly giant cells
Cytomegalovirus	Owl-eye inclusions (basophilic, predominantly intra-nuclear, small granular intracytoplasmic inclusions are found)

3. Inclusion Bodies in Parasitic Infections

Disease/Parasite	Inclusion bodies
Trypanosomiasis	Morula cells with IgM Deposits
Kala-azar	Leishman-Donovan Bodies

4. Miscellaneous Inclusion Bodies

Disease/Location	Inclusion bodies
Ovarian cortex	Walthard Inclusions
Spherical globules surrounded by granulosa cells in the ovary	Call-Exner bodies
Parkinsonism, Lewy body dementia	Lewy Bodies
Amyotrophic Lateral Sclerosis	Bunina bodies
Multiple Myeloma	Russell Bodies

Pearl PM0587: Cytopathic effects caused by viruses Morphological changes in the cells that are caused by viruses are known as cytopathic effects (CPE) and the viruses causing these CPEs are known as cytopathogenic viruses. These CPEs can be detected by a light microscope.

Type of cytopathic effect	Virus
Rapid crenation of cells and degeneration of the cells	Enterovirus
Syncytium/multinucleated giant cell formation	Measles, mumps, respiratory syncytial virus (RSV)
Discrete focal degeneration	Herpes virus
Large granular clumps resembling grapes	Adenovirus
Cytoplasmic vacuolation	Simian virus 40 (SV 40)

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References: Rook's Textbook of Dermatology 10th edition • Page no: 3.30
Harrison's Principles of Internal Medicine, 21st Edition • Page no: 1461, 1473, 1477
Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition • Page no: 476

Q126. Dermatology NEET 2025 ID: MH0150

2. A patient presents with an asymptomatic macular lesion since childhood. Diascopy was inconclusive and Wood lamp examination showed white accentuation. What is the likely diagnosis?

A Nevus anemicus

B Vitiligo

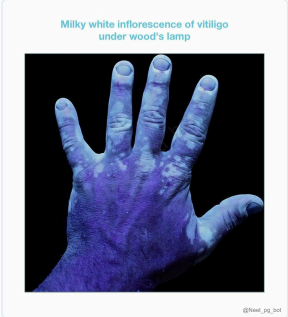
C ✓ Nevus depigmentosus

D Indeterminate leprosy

Correct Answer: Nevus depigmentosus

Based on the given clinical scenario, the most likely diagnosis is nevus depigmentosus. This condition presents as a stable, congenital or early childhood hypopigmented macule. On Wood's lamp examination, the lesion shows white accentuation. The lesion remains non-progressive and asymptomatic throughout life, which fits the case scenario. Nevus depigmentosus is a stable, well-circumscribed hypopigmented patch that is present at birth and may become more noticeable during the first year as the surrounding skin pigmentation develops. There is usually a single oval or rectangular lesion, but the presence of three or more nevus depigmentosus or hypomelanotic patches in a newborn should raise suspicion for possible tuberous sclerosis complex. It is a congenital nonprogressive hypopigmented macule or patch that is stable in its relative size and distribution throughout life. Nevus depigmentosus can be diagnosed using the following criteria: * Leukoderma present at birth or of early onset * No alteration in the distribution of leukoderma throughout life * No alteration in texture or change in sensation in the affected area * Absence of hyperpigmented border. Diascopy test is negative in nevus depigmentosus but positive in nevus anemicus. Wood's lamp examination demonstrates an off-white color. Histopathologic examination shows normal to decreased number of melanocytes with S-100 stain. Mostly it does not require any treatment as it is non-progressive and adequate counseling is sufficient. Other options: Option A: Nevus anemicus are pale macules or patches most often on the trunk. They can be distinguished from other differentials because of their response to firm stroking. Stroking evokes an erythematous line and flare in normal surrounding skin, but the skin of a nevus anemicus does not redden. They can also be diagnosed by diascopy, in which pressure of the skin with a glass slide will obscure the borders of a nevus anemicus. In this case, diascopy was inconclusive, making nevus anemicus less likely. The image below shows nevus anemicus:

Option B: Vitiligo is an acquired condition resulting from the progressive loss of melanocytes.It is characterised by milky-white, sharply demarcated macules. Under Wood's lamp, vitiligo lesions appear chalky white due to the complete absence of melanin. Since the lesion here has been stable since childhood and not progressive, vitiligo is ruled out.



Option D: Indeterminate leprosy presents as hypopigmented macules with reduced sensation and poorly defined margins. The case mentions an asymptomatic lesion with no sensory loss, making indeterminate leprosy unlikely.

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References: Online Resource • https://journals.lww.com/pigi/fulltext/2014/01020/a_brief_review_of_nevus_depigmentosus.4.aspx
Online Resource • https://pubmed.ncbi.nlm.nih.gov/9922008/
Rook's Textbook of Dermatology 9th edition • Page 88.34, 28.7
Nelson Textbook of Pediatrics, 22nd Edition • Page 4054
Fitzpatrick's Dermatology, 9th Edition. • Page 1734

Q127. Dermatology NEET 2025 ID: MH0151

3. A 68-year-old farm labourer presented with a swelling over the right foot for 2 years, which was initially asymptomatic but now has serous discharge and occasional black granules. KOH mount of the discharge will reveal which of the following?

A Arthrospore

B Budding yeast with pseudohyphae

C Hyphae with flower-like conidia arrangement

D ✓ Septate hyphae of 4 to 5 microns

Correct Answer: Septate hyphae of 4 to 5 microns

KOH mount of the discharge is most likely to show septate hyphae of 4 to 5 microns. A chronic foot swelling with draining sinuses with black granules is classic for eumycetoma, mainly caused by Madurella mycetomatis. On KOH mount, the crushed black grains typically show slender, septate hyphae 4–5 microns in diameter. Mycetoma is a localized chronic infection caused by various species of fungi (eumycetoma) and aerobic actinomycetes (actinomycetoma). The causal organisms produce a chronic inflammatory reaction leading to focal neutrophil abscess formation, with scattered giant cells and fibrosis. Grains in the form of white, yellow, red or black granules are found in the centre of the inflammatory response, and may be discharged in pus through multiple sinuses onto the skin surface. A tentative diagnosis can be made on the basis of grain colour, texture and direct microscopic appearance. Pus should be examined under the microscope for the presence of grains, which may then be processed for histology or examined directly in potassium hydroxide. If the infection is eumycotic, when crushed under the cover slip, the grains will be seen to consist of masses of fungal mycelium with septate hyphae of 4 to 5 microns. The final identification of the infecting fungus requires the isolation of the causal agent in culture or by molecular diagnostic techniques. Other options: Option A: Arthrospores are seen in dermatophytes (tinea infections). Option B: Budding yeast with pseudohyphae is typical of Candida infections. Option C: Hyphae with flower-like conidia describe Sporothrix schenckii (sporotrichosis). Pearl PM0568: Differences between Eumycetoma and Actinomycetoma

Feature	Eumycetoma	Actinomycetoma
Etiology	Filamentous fungi	Fungus-like bacteria
Grains	Madurella mycetomatis and grisia- black grains	Actinomadura madurae and Nocardia asteroids- white-yellow grains
Progression	Slow	Rapid
Sites of involvement	Feet and hands	Involvement of covered parts
Swelling	Well circumscribed and marked in the later stage	Ill-defined and merges with surrounding tissue
Tissue destruction	Marked in the later stage	Involvement of bone is earlier and extensive
Filaments	Wide (2–5 μm) and septate hyphae, Periodic Schiff (PAS) positive and gram-negative easily visualized at ×400	Slender (0.5–5 μm) and nonseptate, PAS negative and gram-positive filaments difficult to visualize even at ×800
Response to treatment	Unresponsive and requires surgical debulking	Good response

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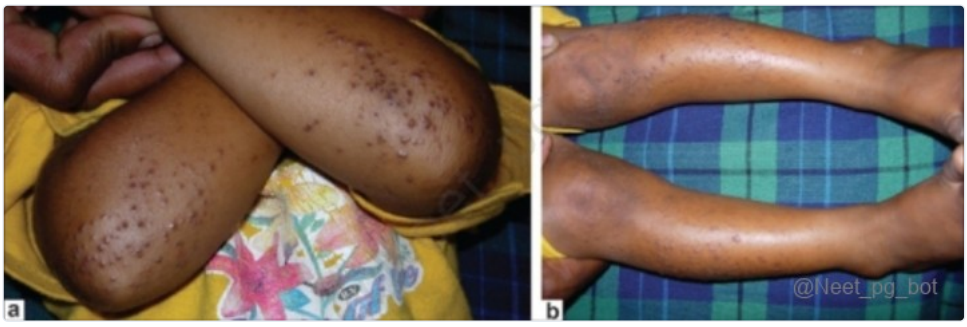
References: Rook's Textbook of Dermatology 9th edition • Page 32.73 - 32.75
Online Resource • https://pathology.medresearch.in/index.php/jopm/article/view/192/380#:~:

Q128. Dermatology NEET 2025 ID: MH0153

4. An 8 year old girl presented with follicular hyperkeratotic papules over the elbow, knees and buttocks. She has dry skin and lusterless hair. Which of the following is the next step in evaluation?



A Vitamin D

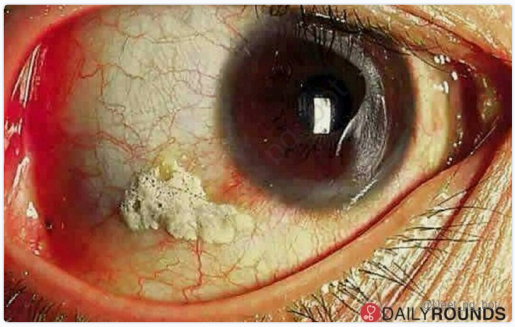
B Deep tendon reflex

C Vitamin B12

D ✓ Ocular examination

Correct Answer: Ocular examination

Ocular examination is the next step in evaluation. This child's presentation with follicular hyperkeratotic papules over extensor surfaces, dry skin, and lusterless hair is characteristic of phrynoderma, which is a manifestation of Vitamin A deficiency. The next important step in evaluation is to assess for ocular involvement (to look for conjunctival xerosis, Bitot's spots, keratomalacia), since Vitamin A deficiency can lead to visual impairment if not detected early. The earliest sign of vitamin A deficiency is defective dark adaptation and night blindness (nyctalopia), followed by xerophthalmia. Aberrant keratinization in the oral and nasal mucosa may result in hyposmia and hypogeusia. Patients also often complain of generalized xerosis. Ocular complications range from night blindness, corneal xerosis and Bitot spots (white keratinized lesions) to corneal ulceration, keratomalacia, corneal perforation, iris prolapse and blindness. The most significant complication of severe vitamin A deficiency is blindness. Cutaneous findings include xerosis and scaling, dermomalacia and phrynoderma. Phrynoderma is characterized by keratotic follicular papules involving the anterolateral thighs and posterolateral upper arms, and can spread to the extensor extremities, shoulders, posterior neck, back, buttocks and abdomen. Treatment is done by administering 50,000–2,00,000 IU of daily vitamin A supplementation until clinical signs and symptoms resolve and serum retinol levels normalize. The image below shows a Bitot's spot seen on the surface of the conjunctiva:



Other options: Option A: Vitamin D deficiency causes rickets/osteomalacia, not follicular keratotic papules. Option B: Deep tendon reflex is relevant for neurological causes (e.g., B12 deficiency), not for keratotic papules. Option C: Vitamin B12 deficiency leads to megaloblastic anemia and neurological signs, not dry skin with keratotic papules. Pearl PM1940: Vitamin-A supplementation & treatment of Vitamin-A deficiency

Vitamin A supplementation

Target group	6 to 11 months age	12 to 59 month of age (1yr to 5yr)
Dose	100,000 IU (oral)	200,000 IU (oral)
Frequency	Once	Every 4–6 months

Note: A total dose of 17 lakh IU given as per the following schedule: * 100,000 IU is given with measles vaccine at 9 months * 200,000 IU is given with DPT booster at 15–18 months * 200,000 IU is given every 6 months (till 5years of age)

Treatment of vitamin A **deficiency**

Age	< 6 months	6-12 months	> 1 year
Dose	50,000 IU	100,000 IU	200,000 IU

Note: Route of administration is oral, in the form of retinyl palmitate. Pearl PM1463: WHO grading of Vitamin A deficiency The term xerophthalmia is reserved to cover all the ocular manifestations of vitamin A deficiency.

Stage	Features	Pathophysiology
XN	Night blindness	Lack of rhodopsin
X1A	Conjunctival Xerosis	Conjunctival epithelial alteration, loss of goblet cells
X1B	**Bitot's spots**	Conjunctival keratinization and plaque formation
X2	Corneal xerosis	Corneal epithelial alteration
X3A	Corneal ulceration/ Keratomalacia affecting <1/3rd of the corneal surface	Liquefaction of corneal stroma
X3B	Corneal ulceration/ Keratomalacia affecting >1/3rd of the corneal surface	-
XS	Corneal scar	-
XF	Xerophthalmic fundus	Degeneration of the retinal pigment epithelium

The first symptom of xerophthalmia is nyctalopia. The first sign of xerophthalmia is conjunctival xerosis. The pathognomonic feature of xerophthalmia is the appearance of Bitot's spots. Note: Up to Stage X2, the manifestations are reversible if treated with vitamin A supplements. Active conjunctival xerosis and Bitot's spots begin to resolve within 2—5 days after supplementation. Most will disappear within 2 weeks, although a significant proportion of temporal lesions may persist in shrunken form and may therefore not be completely reversible.

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References: Rook's Textbook of Dermatology 9th edition • 63.7, 63.8

Q129. Dermatology NEET 2025 ID: MH0162

5. A 48-year-old male presented with an asymptomatic lesion on the right cheek with central atrophy, which had been slowly progressive over the last 2 years. CXR reveals apical calcification. How will you confirm the diagnosis? CXR: Chest X-Ray PCR: Polymerase Chain Reaction

A Probe test

B ✓ PCR

C Slit smear

D Mantoux test

Correct Answer: PCR

PCR is used to confirm the diagnosis. The above clinical scenario of a slow, progressive, painless plaque over the cheek and side of the neck and the characteristic reddish-brown nodules, plaques, and ulcers with variable scarring and a serpiginous edge are suggestive of lupus vulgaris and chest X-ray showing apical calcification suggests healed pulmonary tuberculosis. Lupus vulgaris is a form of cutaneous tuberculosis that occurs in previously sensitised individuals, and polymerase chain reaction (PCR) can be used to confirm the diagnosis. Skin biopsy is usually helpful, but since the quantity of bacilli encountered in cutaneous lesions is small, stains for acid-fast bacilli are usually negative. A chest X-ray will help to analyse the pulmonary involvement. Lupus vulgaris originates from an underlying focus of tuberculosis, typically in a bone, joint, or lymph node, or maybe from the hematogenous and lymphatic spread. It can also occur over a primary site of inoculation, such as the site of scrofuloderma scar, BCG vaccine, and after a tattoo. It presents as a slowly enlarging plaque with an elevated border and central atrophy. The edges of the lesion are often serpiginous in shape and heal with scarring. In India, the buttocks and trunk are more frequently involved. In Europe, most of the lesions are present on the head, neck, and face. The diascopy of lesions reveals diagnostic apple jelly nodules. Polymerase chain reaction has been successfully used to confirm the diagnosis. Skin biopsy is usually helpful, but since the quantity of bacilli encountered in cutaneous lesions is small, stains for acid-fast bacilli are usually negative and hence, culture may be more helpful. Standard multidrug therapy for tuberculosis is used for the treatment. Isoniazid monotherapy is not given anymore due to the emergence of isoniazid-resistant Mycobacterium tuberculosis strains. The image given below shows lupus vulgaris in the axilla:

Other options: Option A: Probe test is generally used to detect bone involvement in osteomyelitis, not in lupus vulgaris. Option C: Slit smear is used for detecting Mycobacterium leprae in leprosy, and is usually negative in cutaneous tuberculosis. Option D: Mantoux test is often positive in tuberculosis, but cannot confirm active disease, as it may also be positive in healthy exposed individuals. Pearl PM1971: Classification of cutaneous tuberculosis The classification of cutaneous tuberculosis is based on the mode of acquisition.

Type	Pathophysiology	AFB
Exogenous		
Through broken skin or mucosa		
Tuberculous chancre	Occurs in individuals with no previous sensitization to TB	Multibacillary
Tuberculosis verrucosa cutis	Occurs in previously sensitized individuals with strong immunity	Paucibacillary
Endogenous		
Direct spread from a focus of infection		
Scrofuloderma	Contiguous spread from affected parts like lymph nodes, joints	Multibacillary
Periorificial tuberculosis	Autoinoculation at mucocutaneous borders from infected secretions	Multibacillary
Hematogenous		
Spread from a distant primary focus of TB		
Acute cutaneous miliary tuberculosis	Occurs in severe disseminated TB	Multibacillary
Lupus vulgaris	Occurs in previously sensitized individuals with strong immunity. Mostly hematogenous spread, sometimes exogenous.	Paucibacillary
Tuberculous gumma/ abscess	Occurs in immunocompromised patients	Multibacillary
Tuberculids		
Hypersensitivity reaction		
Lichen scrofulosorum (micropapular) Papulonecrotic tuberculid Erythema induratum of Bazin (nodular)	Hematogenous spread of *M. tuberculosis* antigens to the skin causes type III hypersensitivity which triggers type IV reaction with granuloma formation	Very rarely seen

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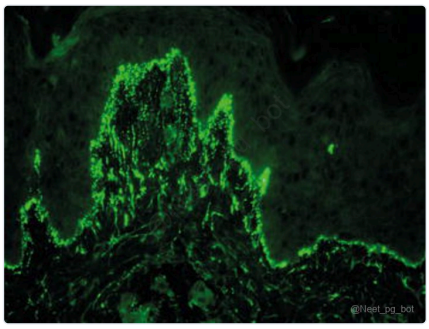
References: Rook's Textbook of Dermatology 10th edition • Page no 27.20-27.23
Online Resource • https://www.sciencedirect.com/science/article/pii/S0378603X14000850

Q130. Dermatology NEET 2025 ID: MH0159

6. A 45-year-old male presents with a complaint of pruritic rash over the elbows, knees and buttocks. He recently visited the gastroenterology OPD and was found to be sensitive to gluten. Skin biopsy showed IgA immunofluorescence. What is the most probable diagnosis?



A Epidermolysis bullosa

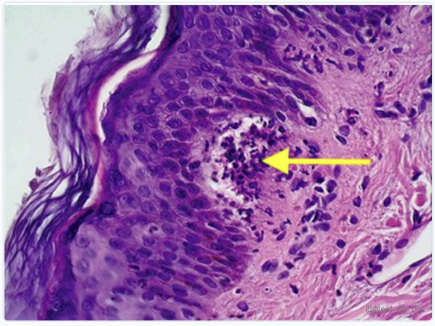
B Bullous pemphigoid

C ✓ Dermatitis herpetiformis

D Pemphigus vulgaris

Correct Answer: Dermatitis herpetiformis

The clinical history of intensely itchy lesions on the extensor aspect of both elbows and knees with direct immunofluorescence (DIF) showing focal granular deposits along the basement membrane is suggestive of dermatitis herpetiformis. Dermatitis herpetiformis, also known as Duhring's disease is an auto-immune disease associated with HLA DQ-2 and HLA DQ-8. The antibodies target epidermal transglutaminase-3, resulting in the collection of neutrophils at the dermal papilla. This condition is closely associated with Celiac disease (Gluten-sensitive enteropathy). The clinical manifestations include grouped vesicles and papules. These lesions are commonly located over the extensor aspects of knees, elbows, buttocks, back, and scalp with symmetrical distribution and are associated with intense pruritus. The characteristic histopathology finding is subepidermal blisters with papillary tip microabscesses containing neutrophils, which is shown in the image below:



A gluten-free diet is the mainstay of treatment, but it takes months to years to show clinical improvement. Hence, dapsone is used for short-term management in these patients. Pearl PM0224: Vesiculobullous Disorders

Disease	Clinical Manifestations	Histology	Immuno-pathology	Autoantigens
Pemphigus vulgaris	Flaccid blisters, denuded skin,oro-mucosal lesions	Acantholytic blister formed in **suprabasal layer** of epidermis	Cell surface deposits of IgG on keratinocytes- fish net pattern	**Dsg3, Dsg1** in patients with skin involvement
Pemphigus foliaceus	Crusts and shallow erosions on the scalp, central face, upper chest, and back	Acantholytic blister formed in **superficial layer** of epidermis	Cell surface deposits of IgG onkeratinocytes	**Dsg1**
Bullous pemphigoid	Large tense blisters on flexor surfaces and trunk	**Subepidermal** blister with **eosinophil-rich** infiltrate	Linear band of IgG and/or C3 in epidermal BMZ	**BPAG1, BPAG2**
Dermatitis herpetiformis	Extremely pruritic small papules and vesicles on elbows, knees, buttocks, and posterior neck	**Subepidermal** blister with **neutrophils** in dermal papillae	Granular deposits of IgA at the tips of dermal papillae	**Epidermal trans-glutaminase**
Linear IgA disease	Pruritic small papules on extensor surfaces; occasionally larger, arciform blisters	**Subepidermal** blister with **neutrophil**-rich infiltrate	Linear band of IgA in epidermalBMZ	**BPAG2**
Epidermo-lysis bullosa acquisita	Tense bullae and erosions	Subepidermal blister	IgG and C3 DEJ – Continous and linear	**Type VII collagen**

The images below shows clinical pictures of pemphigus patients. * Image A - oral erosions in a patient with pemphigus vulgaris * Image B - heavily clotted and crusted lesions around the mouth in a patient with paraneoplastic pemphigus * Image C - skin lesions found on the trunk in a patient with pemphigus vulgaris * Image D - skin lesions on the trunk in a patient with pemphigus foliaceus

Pearl PM0363: Classification of Vesiculobullous Disorders According to Pathology * Immunobullous disorders/ Primary blistering disease: occur due to autoimmunity

Intraepidermal	Subepidermal
Pemphigus vulgaris	Bullous pemphigoid
Pemphigus foliaceous	Dermatitis herpetiformis
Pemphigus vegetans	Epidermolysis bullous acquisita
Pemphigus erythematosus	

* Mechanobullous disorders: Due to defect in the structure of epidermis * Epidermolysis bullosa * Bullous ichthyosiform erythroderma * Other disorders * Infections - VZV, HSV, Bullous impetigo, SSSS * Metabolic - Porphyria, Diabetes, Hemodialysis * Drug eruptions - Fixed drug eruption, Steven Johnson Syndrome, Erythema Multiforme According to Site

Intraepidermal

1. Subcorneal (granular layer)

- Pemphigus foliaceous & erythematosus
- Staphylococcal scalded skin syndrome
- Miliaria crystalina
- Bullous impetigo
- Friction blister
- Subcorneal pustular dermatosis

2. Spinous layer

- Eczematous (atopic) dermatitis
- HSV/VZV infection
- Molluscum contagiosum
- Familial benign pemphigus (Hailey-hailey disease)

3. Suprabasal

- Pemphigus vulgaris
- Darrier's disease

4. Basal layer

- Erythema multiforme
- Epidermolysis bullosa simplex

Sub-epidermal (Dermo-epidermal)

1. Junctional (at basal lamina)

- Junctional epidermolysis bullosa
- Bullous pemphigoid
- Toxic epidermal necrolysis

2. Dermolytic (below basal lamina)

- Epidermolysis bullosa acquisita
- Epidermolysis bullosa dystrophicans
- Deep burns
- Porphyria cutanea tarda

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References: Rook's Textbook of Dermatology 9th edition • Page no 50.52-50.54

Q131. Dermatology NEET 2025 ID: MH0158

7. A female presented with patches in the oral cavity with complaints of burning sensation after taking a drug for 6 months. Symptoms worsened on eating spicy food. Which drug is most likely to cause this condition?

A ✓ Beta blocker

B Lithium

C Fluconazole

D Ciprofloxacin

Correct Answer: Beta blocker

Beta blocker is most likely to cause this condition. The image shows reticular white patches with erosions in the oral cavity, causing a burning sensation, especially with spicy food. The findings are characteristic of oral lichen planus, and among drugs that can trigger this condition, the most commonly associated drug is a beta blocker. Lichenoid drug eruption is a type of adverse reaction to certain medications that causes skin or oral lesions resembling lichen planus. The drugs most commonly implicated in lichenoid drug eruptions include antimalarials, gold, NSAIDs, penicillamine, thiazide diuretics, mercury amalgam, ACE inhibitors, beta blockers, lithium, methyldopa, and other sulfonylureas. Out of the given options, beta blockers are considered one of the most potent generators for the drug-induced forms of lichen planus, compared to lithium (Option B). Lichen planus-like drug reactions usually develop months or even years after starting the medication. They manifest as shiny purple papules or plaques, sometimes with the characteristic Wickham's striae (thin white lines on their surface), and can involve mucous membranes. Diagnosis relies on skin biopsy, which typically shows hyperkeratosis, hypergranulosis, saw-tooth acanthosis, and a band-like infiltrate in the upper dermis. Features that suggest a drug-related cause include necrotic keratinocytes, plasma cells, and eosinophils. Treatment centers on discontinuing the offending drug, which is crucial for resolution. Most cases improve with topical corticosteroids, while oral corticosteroids may be necessary for extensive or severe eruptions. Symptoms generally resolve within weeks to months after stopping the drug. Other options: Option C: Fluconazole is an antifungal, used to treat oral candidiasis and other fungal infections. It does not typically cause lichenoid drug eruption Option D: Ciprofloxacin is a fluoroquinolone and can cause tendon ruptures, joint pain, gastrointestinal upset, and neurological side effects. Oral lichenoid lesions are not a well-known adverse effect.

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References: Rook's Textbook of Dermatology 9th edition • Page 37.3, 118.9-118.11
Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 14th Edition • Page 1143-44
Online Resource • https://www.researchgate.net/publication/229324944_Lichen_Planus-Like_Drug_Eruptions_Due_to_b-Blockers_A_Case_Report_and_Literature_Review#:~:text=%5D.%20...-,...,effects%20%5B7%2C8%5D.
Online Resource • https://commons.ada.org/cgi/viewcontent.cgi?article=1075&context=nysdj
Online Resource • https://pubmed.ncbi.nlm.nih.gov/7997389/

Q132. Dermatology NEET 2021 ID: MF1282

A 10-year-old male child complains of mild painful swelling over the scalp for 3 months as shown in the image below. He also has a pet dog at his home. What is the diagnosis of the given condition?

A Furuncle

B Epidermoid cyst

C ✓ Kerion

D Folliculitis

Correct Answer: Kerion

The clinical vignette of a young boy in contact with a **zoophilic organism,** presenting with a **boggy inflammatory mass,** with **thick crusting** and **matting** of hair is suggestive of a **kerion**. Kerion is a severe **inflammatory** type of **tinea capitis.**

Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with **sinus** formation.

The clinical variants of tinea capitis are:

- **Endothrix** - a non-inflammatory type in which **multiple black dots** are present within the areas of alopecia. It is most commonly caused by **T. tonsurans and T. violaceum.**
- **Ectothrix** - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. **Microsporum audouinii**, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.
- **Kerion** - a severe inflammatory type that presents with an inflammatory mass with **thick crusting** and **matting of hair** with pus and sinus formation. It is commonly caused by **T. verrucosum and T. mentagrophytes**.
- **Favus** - is characterized by the presence of yellowish, cup-shaped crusts known as **scutula** that develop around hair follicles. It is caused by **T. schoenleinii**.

Treatment:

- **Terbinafine**: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.
- **Itraconazole**: 2-4 mg/kg/day for 4-6 weeks.

The image below shows a **kerion**:

The image below shows the **endothrix variant of tinea capitis**:

Option A: **Furuncle/ boil** is caused by Staphylococcus. It is painful and usually situated on the neck area, and the shedding of loosened hairs is generally much less evident than in kerion. It is pustular and later becomes necrotic and forms a scar.

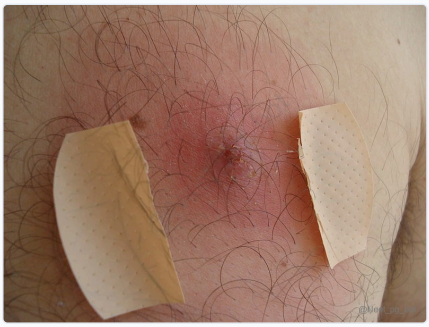
The image below shows a **furuncle**- pustule over the hair follicle with a necrotic base:

Option B: **Epidermoid cyst/ sebaceous cyst** results from squamous metaplasia in a damaged sebaceous gland. Many are the result of inflammation around a pilosebaceous follicle. The most common site of a sebaceous cyst is the **postauricular sulcus** or below and behind the ear lobule. They are fixed to the skin and usually have a central punctum. Treatment depends on the clinical state of the cyst. When inflamed or infected, they should be incised and drained initially, total surgical excision should be done once the inflammation has subsided.

The image below shows an **epidermoid cyst** of the post-auricular sulcus- cyst fixed to the skin, shows a central punctum:

Option D: **Folliculitis** would involve specific inflammation in the pilosebaceous units. It is caused by **Staphylococcus aureus.** They heal without a scar.

The image below shows **folliculitis**- inflammation of the pilosebaceous units leading to pustule formation:



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References: Rook's Textbook of Dermatology 9th edition, 26.22, 26.23, 26.24, 32.38, 32.39, 32.40, 90.27, 108.11, 134

Q133. Dermatology NEET 2021 ID: MF1287

A child with a history of night-blindness presented to you with the following findings. What is the diagnosis?

A ✓ Phrynoderma

B Keratosis pilaris

C Darier disease

D Follicular eczema

Correct Answer: Phrynoderma

A child having night-blindness, and multiple keratotic papules on elbows as shown in the image suggests a diagnosis of **phrynoderma** or toad-skin. This can occur due to the deficiency of **vitamin A.**

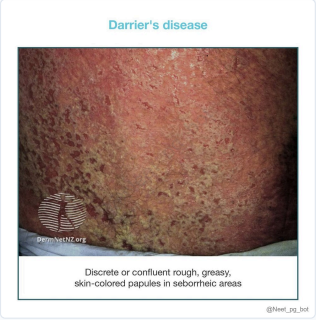
Phrynoderma is not specific to vitamin A deficiency. It may also be associated with deficiencies of B complex, riboflavin, vitamin C, vitamin E, and essential fatty acids. It manifests as **groups of papules** with a **central keratotic plug**. The commonest sites are **elbows and knees**. They are present around the follicles and give the skin a rough texture. Treatment involves the correction of poor diet and administration of a **multivitamin** preparation containing vitamin A.

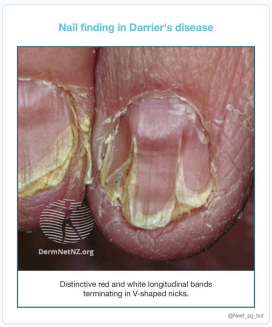
Option B: **Keratosis pilaris** is an inherited abnormality of keratinization. It affects the follicular orifices leading to **keratotic follicular plugging**, perifollicular erythema, and **follicular atrophy.** It usually presents during **puberty.** They present as small, keratotic papules on the extensor aspects of the limbs, particularly the arms which can lead to ingrowing of the hair. Treatment is purely reserved for cosmetic purposes.

Option C: **Darier disease** is an autosomal dominant genodermatosis characterized by keratotic papules that show **acantholysis** and dyskeratosis. **ATP2A2** gene mutations are present in Darier disease. Histologically, acantholysis (separation of keratinocytes) and blistering are seen in the suprabasal epidermis.

It presents with discrete or confluent rough, greasy, skin-colored papules in seborrheic areas (trunk, face scalp margins) and flexures (perineum, axillae, groin). There may also be isolated nipple hyperkeratosis. Nail findings include fragility, painful longitudinal splits, or distinctive red and white longitudinal bands terminating in **V-shaped nicks.**

Option D: **Follicular eczema** is a form of **atopic dermatitis** that presents with a discoid patch formed due to the coalescence of the follicular lesions with peripheral excoriation marks. The biopsy shows spongiotic dermatitis.





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References: Rook's Textbook of Dermatology 9th edition, Chapter 63, 66
Online Resource, https://pubmed.ncbi.nlm.nih.gov/22885452/

Q134. Dermatology NEET 2021 ID: MF1285

A 30-year-old presented with flaccid bullae on her skin which are easy to rupture. The biopsy revealed a suprabasal split. What is the most likely diagnosis?

A ✓ Pemphigus vulgaris

B Pemphigus foliaceus

C Pemphigus vegetans

D Erythema multiforme

Correct Answer: Pemphigus vulgaris

The given clinical vignette of a bullous disorder with the histologic picture pointing towards a **suprabasal blister** suggests the diagnosis of **pemphigus vulgaris.**

It is caused due to the production of antibodies mainly against **desmoglein 3.** Rarely antibodies against desmoglein 1 can also be found. The antibodies are commonly **IgG antibodies.** Desmoglein 3 is mainly present in the **basal and suprabasal layers** of the skin and mucosa.

The patients, therefore, present with **flaccid bullae** over the **skin** and mucosal ulcerations. **Oral mucosa** is commonly affected. Other areas where ulceration can be seen are conjunctiva, nasopharynx, larynx, etc. Cutaneous lesions are more common over the scalp, face, neck, upper chest, and back.

Direct immunofluorescence will show the deposition of **antibodies around the keratinocytes** in the **epidermis.** It presents with a characteristic **fish-net appearance** in the **immunofluorescence** assay It is treated with steroids, azathioprine, mycophenolate mofetil, rituximab, etc.

The images attached below depict the lesions, the suprabasal split, and the fish-net pattern respectively seen in a case of pemphigus vulgaris.

Option B: **Pemphigus foliaceus** is a benign form of pemphigus, an inflammatory **blistering** disorder. This is a form of a **subcorneal blister,** where the stratum corneum forms the roof of the bulla. It is caused by **autoantibodies** that result in the dissolution of intercellular attachments within the epidermis. Antibodies against **desmoglein 1** are mainly seen in pemphigus foliaceous**.** There is **no involvement** of the **mucosal epithelium** in pemphigus foliaceus.

Lesions are most common on the scalp, face, chest, and back. Bullae are superficial and mainly present as areas of erythema and crusting. These represent superficial erosions at sites of blister rupture.

Option C: **Pemphigus vegetans** is a **rare form** of the bullous disorder that presents with large, moist, verrucous, vegetating plaques studded with pustules on the groin, axillae, and **flexural** surfaces. It is associated with **eosinophilic microabscess.** It has two types, a severe Neumann type, and a milder Hallopeau type.

Option D: **Erythema multiforme** is a **self-limiting cytotoxic dermatitis** resulting from cell-mediated hypersensitivity most commonly to infection or drugs. The common triggers are

- **Herpes simplex infection**
- Mycoplasma infection
- Bacterial infections
- Fungal infections
- Vaccinations
- Malignancy: carcinoma, lymphoma, leukemia
- Drug reactions: sulfonamides, cotrimoxazole, penicillins
- Miscellaneous: Lupus erythematosus (Rowell Syndrome), Sarcoidosis, Polyarteritis Nodosa

It presents clinically with macular, papular, or urticarial lesions, along with the **classic acral iris** or **target lesions**. The lesions are distributed on the distal extremities, palms or trunk, and oral and genital mucous membranes.

Target lesions are less than 3 cm in diameter, rounded, and have **three** zones:

- The central area of **dusky erythema** or purpura
- Middle paler zone of **edema**
- The outer ring of **erythema** with a well-defined edge

Atypical lesions will have only 2 zones. They fade in **1-2 weeks** leaving dusky **discoloration.**

Pemphigus Foliaceus

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Pemphigus vegetans

Large, moist, verrucous, vegetating plaques studded with pustules over the genitals or flexural areas

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Eythema multiforme

Classical acral iris/ Target sign

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References: Rook's Textbook of Dermatology 9th edition, 50.2 - 50.10, 47.1 - 47.5
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3884912/
Online Resource, https://pubmed.ncbi.nlm.nih.gov/17619399/

Q135. Dermatology NEET 2021 ID: MF1394

What is the treatment of choice for a patient with paronychia of the index finger on the right hand, as shown in the image below?

A ✓ Amoxiclav

B Metronidazole

C Amikacin

D Norfloxacin

Correct Answer: Amoxiclav

The above image depicts **paronychia** associated with a **reddish erythematous** streak-like lesion on the hand suggestive of **paronychia** associated with **lymphangitis.** The treatment of choice for this patient is **amoxiclav.**

Acute paronychia is an **acute infection** of the **nail beds** and folds. It is most commonly caused by ***Staphylococcus aureus***. It presents as a **tender**, **erythematous** swelling around the nail.

Treatment of pyogenic paronychia:

- Protection against trauma
- Keep affected fingernails dry.
- Abscesses - **Incision** and **drainage**. (abscess is often opened by pushing the nail fold away from the nail plate)
- Antibiotics- A penicillin in combination with a β-lactamase inhibitor, e.g. amoxicillin with clavulanic acid (Amoxiclav)

Chronic paronychia is usually due to infection with *Candida albicans* and responds to topical antifungal therapy.

Lymphangitis:

Lymphangitis represents acute inflammation elicited by the spread of bacterial infections into lymphatics, with **group A β-hemolytic streptococci** being the most common causative agent. It is manifested by red, painful **subcutaneous streaks** (the inflamed lymphatics) and painful enlargement of the draining lymph nodes (lymphadenitis). It is managed by elevating the affected extremity and broad-spectrum antibiotics.

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References: Rook's Textbook of Dermatology 9th edition, 95.35, 95.36, 105.51
Bailey & Love's Short Practice of Surgery - 28th Edition, Page no: 56, 542
Sabiston Textbook of Surgery: The Biological Basis of Modern Surgical Practice, 20e, Page no: 2000

Q136. Dermatology NEET 2021 ID: MF1280

A patient taking multi-drug therapy (MDT) presents with worsening of existing lesions and nerve involvement. What will be your next best step of action?

A Stop MDT, start systemic corticosteroids

B ✓ Continue MDT, start systemic steroids

C Stop MDT give thalidomide

D Continue MDT, start thalidomide

Correct Answer: Continue MDT, start systemic steroids

The given scenario of **worsening** of **existing lesions** in the context of **leprosy treatment** with multi-drug therapy (MDT) is suggestive of a **type 1 lepra reaction.** The next best step in the management would be to **continue MDT** and **start systemic steroids.**

Type 1 lepra reactions present with **inflammation** of **existing skin lesions**. The lesions become red, painful, and edematous. They may rarely **ulcerate.** New lesions may occur but are very rare. It is associated with acute **neuritis,** with the tenderness and loss of sensory and motor function.

MDT is **never stopped** in patients with lepra reactions. For pain, NSAIDs can be supplemented along with oral steroids.

The use of **thalidomide** is restricted to severe forms of type-2 lepra reaction or **erythema nodosum leprosum.** These are characterized by **systemic effects** like **iritis, orchitis,** and **eruption of new skin lesions.**

The summary of lepra reactions is attached in the pearl below.

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References: Rook's Textbook of Dermatology: Volume 2 Edition 9, Page no: 28.5, 28.12, 28.16

Q137. Dermatology NEET 2021 ID: MD7381

A 30-year-old male patient presents to you with inflammatory alopecia. Which of the following variants of tinea capitis infection commonly result in this type of alopecia?

A Endothrix and Favus

B Ectothrix and Endothrix

C ✓ Favus and Kerion

D Kerion and Endothrix

Correct Answer: Favus and Kerion

Favus and **kerion** are the **inflammatory** variants of tinea capitis.

Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

The clinical variants of tinea capitis are:

- **Kerion** is a severe inflammatory type of **tinea capitis** which presents with an **inflammatory mass** with **thick crusting** and **matting of hair**.
- **Favus** is characterized by the presence of yellowish, cup-shaped crusts known as scutula that develop around hair follicles.
- **Endothrix** is a non inflammatory type in which **black dots** form around swollen hair shafts.
- **Ectothrix** is a mild inflammatory type in which dull **grey** coated broken hairs is seen.

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References: Rook's Textbook of Dermatology 9th edition, Page no 32.39-32.40, 89.35

Q138. Dermatology INI-CET 2025 ID: MG9291

1. A 42-year-old man noticed a slow, progressive, painless plaque on the side of the neck. He did not complain of any other systemic signs and symptoms. Which of the following tests are useful in this case? 1. HIV 2. Skin biopsy 3. KOH mount 4. Chest X-ray

A 1 and 2

B ✓ 2 and 4

C 2, 3 and 4

D 1, 2 and 3

Correct Answer: 2 and 4

Skin biopsy and Chest X-ray are useful in this case. The above clinical scenario of a slow, progressive, painless plaque on the side of the neck and the characteristic reddish-brown nodules, plaques, and ulcers with variable scarring and a serpiginous edge are suggestive of lupus vulgaris. Lupus vulgaris is a form of cutaneous tuberculosis that occurs in previously sensitised individuals. Skin biopsy is usually helpful, but since the quantity of bacilli encountered in cutaneous lesions is small, stains for acid-fast bacilli are usually negative. A chest X-ray will help to analyse the pulmonary involvement. Lupus vulgaris originates from an underlying focus of tuberculosis, typically in a bone, joint, or lymph node, or maybe from the hematogenous and lymphatic spread. It can also occur over a primary site of inoculation, such as the site of scrofuloderma scar, BCG vaccine, and after a tattoo. It presents as a slowly enlarging plaque with an elevated border and central atrophy. The edges of the lesion are often serpiginous in shape and heal with scarring. In India, the buttocks and trunk are more frequently involved. In Europe, most of the lesions are present on the head, neck, and face. The diascopy of lesions reveals diagnostic apple jelly nodules. Polymerase chain reaction has been successfully used to confirm the diagnosis. Standard multidrug therapy for tuberculosis is used for the treatment. Isoniazid monotherapy is not given anymore due to the emergence of isoniazid-resistant Mycobacterium tuberculosis strains. The image given below shows lupus vulgaris.

Pearl PM1971: Classification of cutaneous tuberculosis The classification of cutaneous tuberculosis is based on the mode of acquisition.

Type	Pathophysiology	AFB
Exogenous		
Through broken skin or mucosa		
Tuberculous chancre	Occurs in individuals with no previous sensitization to TB	Multibacillary
Tuberculosis verrucosa cutis	Occurs in previously sensitized individuals with strong immunity	Paucibacillary
Endogenous		
Direct spread from a focus of infection		
Scrofuloderma	Contiguous spread from affected parts like lymph nodes, joints	Multibacillary
Periorificial tuberculosis	Autoinoculation at mucocutaneous borders from infected secretions	Multibacillary
Hematogenous		
Spread from a distant primary focus of TB		
Acute cutaneous miliary tuberculosis	Occurs in severe disseminated TB	Multibacillary
Lupus vulgaris	Occurs in previously sensitized individuals with strong immunity. Mostly hematogenous spread, sometimes exogenous.	Paucibacillary
Tuberculous gumma/ abscess	Occurs in immunocompromised patients	Multibacillary
Tuberculids		
Hypersensitivity reaction		
Lichen scrofulosorum (micropapular) Papulonecrotic tuberculid Erythema induratum of Bazin (nodular)	Hematogenous spread of *M. tuberculosis* antigens to the skin causes type III hypersensitivity which triggers type IV reaction with granuloma formation	Very rarely seen

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References: Rook's Textbook of Dermatology 10th edition • Page no 27.20-27.23

Q139. Dermatology INI-CET 2025 ID: MG9289

2. Which of the following are side effects of topical corticosteroids? 1. Hypertrichosis 2. Acneiform eruption 3. Atrophy 4. Bluish-grey skin discolouration

A 1 and 4 only

B 4 only

C ✓ 1, 2 and 3 only

D 2 and 3 only

Correct Answer: 1, 2 and 3 only

Hypertrichosis, Acneiform eruption and Atrophy are side effects of topical corticosteroids. Side effects of topical corticosteroids

Local side effects	Systemic side effects (rare)
• **Cutaneous atrophy** • Skin fragility • Purpura • Striae • Telangiectasia (vasodilatation) • Perioral/periocular dermatitis • **Acneiform/rosaceiform eruptions** • Tachyphylaxis • Rebound • Contact allergy • Promotion of infection (herpes simplex, dermatophyte, candida) • Cataracts • Glaucoma • Impaired wound healing • **Hypertrichosis** • Pigment alteration	• Adrenal suppression • Cushingoid features • Hypertension • Hyperglycaemia • Low birth weight (with super-potent topical corticosteroids)

Other options: Bluish-grey skin discolouration is seen in inflammatory diseases and due to some drugs like phenothiazines and minocycline.

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References: Rook's Textbook of Dermatology 10th edition • Page no 4.16, 18.17

Q140. Dermatology INI-CET 2025 ID: MG9284

3. A 30-year-old man complained of painful lesions on his palms and soles. He also gave a history of sexual contact and a transient genital lesion one month ago. Which of the following tests is most appropriate for him?

A Tzanck smear

B ✓ Serological test

C KOH mount

D Biopsy

Correct Answer: Serological test

The given clinical scenario of a 30-year-old man complaining of painful lesions on his palms and soles, with a history of sexual contact and a transient genital lesion one month ago, suggests the most probable diagnosis to be secondary syphilis. A serological test is most appropriate for him. Syphilis is caused by Treponema pallidum, a spirochete bacterium. It is a sexually transmitted infection. A history of sexual contact and a transient genital lesion one month ago suggests a painless indurated chancre, which is usually single. This stage is primary syphilis, and the chancre heals spontaneously in 3–8 weeks. Secondary syphilis is characterised by generalised manifestations that occur on the skin and mucous membranes. Rashes in secondary syphilis are non-itchy, coppery red in colour and symmetrical in distribution. These first appear at around 8 weeks. Constitutional symptoms include fever, headache and bone and joint pains that are more evident at night. Serological testing is the most important screening test. It is done using a standard non-treponemal antibody test (VDRL, RPR) and a specific treponemal antibody test(TPHA, FTA-ABS). The VDRL slide test is widely used and requires the microscopic demonstration of antigen–antibody flocculation in heat-inactivated serum. Reactivity to non-treponemal antibody tests develops 1–4 weeks after the chancre appears in primary syphilis. Specific treponemal antibody tests are qualitative procedures and detect antibodies to antigenic determinants of treponemes. They do not help assess treatment responses. Treponema pallidum can be identified by dark-field microscopy. It shows corkscrew motility. Molecular amplification tests, like the Polymerase chain reaction, can also be used for identification, especially where interpretation of serological test results is difficult. Pearl PM2340: Stages of syphilis Primary syphilis: * Chancre - Painless, indurated, usually single. The chancre self heals in 1-4 months. * Lymphadenopathy - Painless, unilateral or bilateral, shotty and rubbery inguinal nodes * Dory flop sign - As the chancre is too hard to fold, it will flip over once the prepuce is retracted back. Secondary syphilis:

* A & B: Generalized non-itchy rash with no vesicles present on the palm and sole. This is the most common lesion. * C & D: Condyloma lata. * E: Moth-eaten non-scarring alopecia * F: Mucous patches are flat erosions on the tongue that can coalesce to form snail track ulcers. These are the most infectious lesions. * G: Pigmentary plaques * H: Psoriasiform syphilitic lesions * Generalized diffuse non-tender lymphadenopathy Tertiary syphilis: * Gummatous syphilis are deep punched-out ulcers, with gum-like secretions on the floor * Pseudo-chancre redux is a gumma occurring at the site of a previously healed ulcer * CVS - Aortitis and aneurysm of the ascending aorta * CNS - General paresis of the insane and tabes dorsalis Latent syphilis is the period between secondary and tertiary syphilis. In this period the patient is asymptomatic clinically and only serological evidence of the infection is present. It is classified into: * Early latent syphilis - Within 1 year of infection * Late latent syphilis - After 1 year of infection Pearl PM0026: Syphilis - Diagnostic Serology Nontreponemal tests (VDRL, RPR) VDRL- Venereal disease Research Laboratory test RPR- Rapid Plasma Reagin test * Quantitative tests(titres) * Reagin antibody tests; based on cardiolipin antigen * VDRL can be used to test CSF whereas RPR cannot be used * RPR does not require preheating of serum unlike VDRL * VDRL requires light microscope for interpretation whereas RPR can be read macroscopically * Decrease in titers confirms treatment * Sensitivity of RPR(86%)>VDRL(76%) in primary syphilis Treponemal Tests FTA-ABS- Fluorescent Treponemal Antibody Absorption test TPI-Treponema pallidum immobilization test TPPA-Treponema pallidum particle agglutination assay TPHA-Treponema pallidum haemagglutination test EIA-Enzyme-linked ImmunoAssay * Detect antibody to treponemal antigens * Qualitative tests * Greater sensitivity in early infection * Positive even after treatment hence cannot be used to detect reinfection. * First test to be positive: FTA-ABS * Most specific: TPHA,TPPA, EIA Pearl PM2031: Features of congenital syphilis

	Early feature	Late feature
Birth weight	Low birth weight may be <2500g	
Mucocutaneous	Snuffles Maculopapular rash Pustules Mucous patches Condyloma lata Vesiculobullous lesions Rhagades Desquamation Alopecia	Rhagades Gummata Palatal perforation
Dental		Mulberry molars **Hutchison's triad**: It is characteristic of congenital syphilis. The components are: Hutchinson's teeth, Interstitial keratitis and Sensorineural deafness

Laryngeal	Hoarse cry	
Ocular	Chorioretinitis Uveitis Glaucoma	Interstitial keratitis Uveitis Glaucoma
Gastro-intestinal	Hepatosplenomegaly Pancreatitis Enteritis	
Renal	Nephritis Oedema & ascites	
Lymphatics	Lymphadenopathy	
Hematological	Thrombocytopenia Anemia Disseminated intravascular coagulation	
CNS	Aseptic meningitis Chronic meningovascular syphilis Cranial nerve palsies Hydrocephalus	Mental delay VIII nerve deafness Convulsion disorders Paresis & paralysis Tabes Dorsalis
Skeletal	Periostitis Osteochondritis causing pseudoparalysis Osteitis	Periostitis causing frontal & parietal bossing Sabre tibia, scaphoid scapula Thickening of the medial part of the clavicle Short maxilla & protuberant mandible Saddle nose Clutton's joint

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References: Rook's Textbook of Dermatology 10th edition • Page no 29.6-29.8, 29.18

Q141. Dermatology INI-CET 2025 ID: MG9280

4. Which of the following statements are correct? 1. The epidermis is derived from the ectoderm. 2. The hypodermis and dermis are derived from the endoderm. 3. Stratum corneum is the outermost cellular layer. 4. The epidermis regenerates from deeper follicular cells, such as hair follicles and sebaceous glands.

A 2 and 3

B ✓ 1 and 4

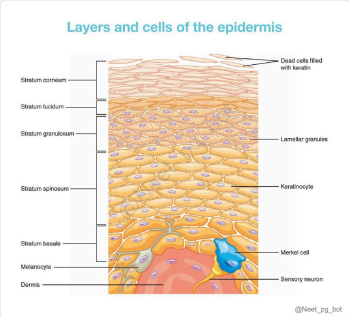
C 1, 3 and 4

D 1, 2 and 3

Correct Answer: 1 and 4

Statements 1 and 4 are the correct statements. The epidermis is derived from the surface ectoderm. During the 2nd month, it presents as two layers- a superficial layer of flat cells called periderm/epitrichium and a deep layer of cuboidal cells called basal/ germinative layer. The cells of the basal layer proliferate to form a third intermediate layer later. The epidermis regenerates from deeper follicular cells, such as hair follicles and sebaceous glands. This helps to maintain skin homeostasis. Other statements: Statement 2: The dermis and hypodermis are derived from mesoderm. The dermatomes give rise only to the dermis on the dorsal aspect of the head and trunk. The dermis of the limbs and that on the lateral and ventral aspects of the trunk arises from lateral plate mesoderm. The dermis over most of the head and the anterior aspect of the neck is derived from the neural crest. Statement 3: Stratum corneum is not the outermost cellular layer. It is the outermost layer of epidermis, where cells, known as corneocytes, have lost their nuclei and cytoplasmic organelles. Pearl PM1880: Cells of the epidermis

Cells of the epidermis	Origin and location	Characteristic features
Keratino-cytes	**Present in- All Layers of Epidermis** (90-95% of all epidermal cells) **Derived from-**Ectoderm Differentiated **in situ**	Rich in the **intermediate filaments** called **keratin** that provides mechanical strength to the cell. The keratinocytes in basal layer divide and generate the upper layers.
Melano-cytes	**Present in- Stratum basale** **Derived from-Neural Crest** **Migrate to the epidermis**	Synthesize **melanin** in granules called **melanosomes** that is transferred to keratinocytes of epidermis **through elongated processes called dendrites**. Each melanocyte transfers melanosomes to **36** keratinocytes. This is called as **Epidermal melanin unit** (36K:1M)
Merkel cells	**Present in-Stratum basale** (RETE RIDGES) **Derived from-Ectoderm/ Neural Crest** **Differentiates in-situ**	Also known as **Touch cells or Haascheiban cells**. They are **Type 1 slow adapting touch receptors**.
Langerhans cells	**Present in-Stratum spinosum** **Derived from-** Bone Marrow **Migrate** to Epidermis	They are dendritic **Antigen Presenting Cells (APCs)**. Characterized by presence of rod shaped or Tennis Racket shaped granules called **Birbeck granules** which are seen on electron microscopy. Immunohistochemical Markers are **CD1a**, **CD207(Langerin) and S100**.



@Neet_pg_bot

References: Rook's Textbook of Dermatology 10th edition • Page no 2.3, 2.43
Inderbir Singh's Human Embryology - 11th Edition • Page no 118-120

Q142. Dermatology INI-CET 2025 ID: MG9279

5. Which of the following are used to treat scabies in infants? 1. Permethrin 5% 2. Ivermectin 3. Crotamiton 10% 4. Benzyl benzoate 25%

A 1, 2 and 4

B 1, 2, 3 and 4

C 1, 2 and 3 only

D ✓ 1 and 3 only

Correct Answer: 1 and 3 only

Permethrin 5% and Crotamiton 10% are used to treat scabies in infants. Permethrin cream (5%) is the treatment of choice for scabies. It is applied topically twice, 1 week apart, and it is approved for infants ≥ 2 months of age. Crotamiton (10%) ointment is applied topically overnight on days 1, 2, 3, and 8. Other options: Ivermectin is contraindicated in children <15 kg. Benzyl benzoate (25%) ointment is contraindicated in children.

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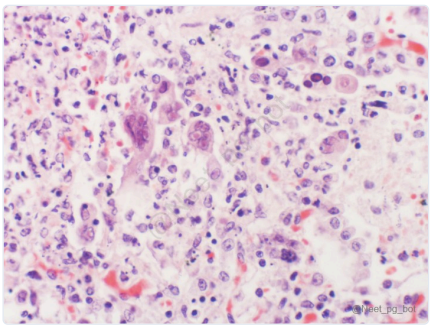
References: Nelson Textbook of Pediatrics, 22nd Edition • Page no 2140

Q143. Dermatology INI-CET 2025 ID: MG9278

6. A young man presented with painful vesicular lesions over the genitalia, with superficial erosions. Image as depicted. What is the drug of choice?



A Penicillin

B ✓ Acyclovir

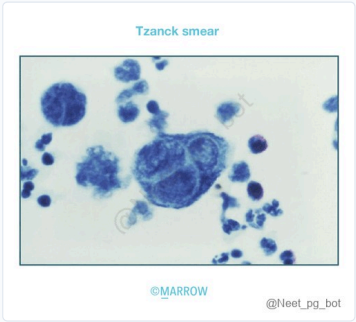
C Prednisone

D Ceftriaxone

Correct Answer: Acyclovir

The given clinical scenario of a young man complaining of painful genital vesicular lesions with superficial erosions, with histopathology showing multinucleated giant cells, suggests the most probable diagnosis to be Herpes genitalis (HSV-2), and the drug to be given is Acyclovir. Acyclovir is the most frequently used for the treatment of HSV infections and is available in IV, oral, and topical formulations. Other drugs that can be used are ganciclovir, famciclovir, and valacyclovir. All these drugs are found to be effective in shortening the duration of symptoms and lesions of mucocutaneous HSV infections in both immunocompromised and immunocompetent patients Herpes infection can be caused by the HSV-1 and HSV-2 viruses. HSV-1 causes orofacial lesions, i.e. herpes labialis, while HSV-2 mainly causes genital herpes which is characterised by vesiculoulcerative lesions of the penis of the male or the cervix, vulva, vagina, and perineum of the female. Genital herpes is often accompanied by symptoms like fever, headache, malaise, and myalgia, pain, itching, dysuria, vaginal and urethral discharge, and tender inguinal lymphadenopathy. Diagnosis of herpes simplex virus: * Most sensitive test- HSV DNA- PCR (can differentiate between HSV 1 and 2) * Virus isolation * Histopathology showing multinucleated giant cells * Microscopy - Tzanck smear, which is a rapid, diagnostic method, shows Tzanck cells. These are multinucleated giant cells with faceted nuclei and homogenously stained ground-glass chromatin. The following image shows the histology of HSV inclusion bodies.

The image given below shows Tzanck cells.



Pearl PM0008: Comparison of Genital Ulcers

	Genital herpes	Syphilitic chancre (Hard chancre)	Donovanosis/ granuloma inguinale	Chancroid (soft chancre)	LGV
Incubation period	2-7 days	10-90 days Average: 21days	7-84 days	1-14 days	3 days - 6 weeks
Causative agent	HSV-2>HSV-1	*Treponema pallidum*	*Klebsiella granulomatis*	*Hemophilus ducreyi*	*Chlamydia trachomatis* serovars L1,L2,L3
Clinical presentation	Cluster of vesicles on erythematous base. Painful and pruritic lymphadenopathy	Single painless ulcer with a clean base and a firm raised border. Painless lymphadenopathy	Painless papule or nodule erodes into beefy red granulomatous ulcer with rolled edges. No lymphadenopathy, pseudobubos are seen	Painful ulcer (ducreyi makes you cry). Painful inguinal lymphadenopathy	Painless genital ulcer. Painful inguinal lymphadenopathy
Diagnosis	Tzanck preparation shows multinucleated giant cells, PCR	Clinical diagnosis	Donovan bodies on biopsy	Clinical diagnosis	Nucleic acid amplification tests
Treatment	Acyclovir/famcyclovir or valacyclovir	Penicillin/Ceftriaxone	Azithromycin/Doxycycline	Azithromycin/ceftriaxone	Doxycycline/Azithromycin

Pearl PM2024: Syndromic Management of Sexually Transmitted Infections

Partner Management when asymptomatic is not required for Gardenella vaginalis, Candida and Herpes infections.

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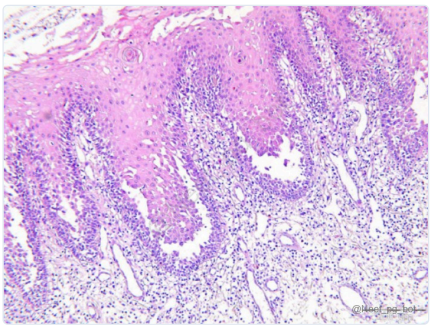
References: Rook's Textbook of Dermatology 10th edition • Page no: 3.30
Harrison's Principles of Internal Medicine, 21st Edition • Page no: 1461, 1473, 1477
Ananthanarayan and Paniker's Textbook of Microbiology Tenth edition • Page no: 476

Q144. Dermatology AIIMS 2017 ID: MA0630

The histopathological slide given below is suggestive of:



A Psoriasis vulgaris

B Leishmaniasis

C Mycosis fungoides

D ✓ Pemphigus vulgaris

Correct Answer: Pemphigus vulgaris

The given slide shows a **row of tombstones** appearance, which is classically seen in **pemphigus vulgaris.**

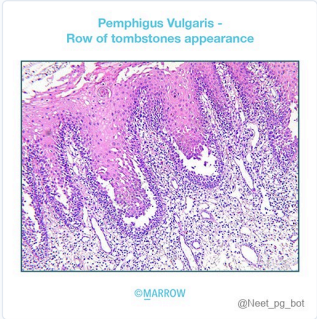
In this condition, there is a **suprabasal splitting** of the epidermis leading to blister formation. The **basal layer** remains **adherent** to the **basement membrane** and gives the resemblance to a row of tombstones as shown in the histology image below.

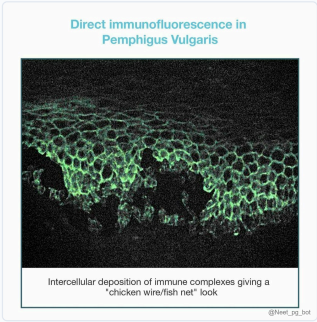
Pemphigus vulgaris is a **chronic immunobullous** disorder characterized by the presence of **antibodies** against **desmosomal proteins**, **desmogleins 1 and 3** expressed in the **skin** and **mucosa.**

Clinically, the patients present with ill-defined, **irregular, slow-to-heal** oral lesions and erosions. **Flaccid blisters** filled with **clear fluid** may arise on normal skin or an erythematous base. The rupture of such blisters produces **painful erosions** and heals **without scarring.**

Nikolsky's sign is positive which is elicited by applying firm **tangential** sliding pressure with a finger on lesional or perilesional skin. This leads to the separation of the normal-looking epidermis from the dermis, producing **erosion.**

Direct immunofluorescence is considered the most accurate **confirmatory test** of this condition. It shows intercellular **IgG** and **C3** deposition on the **surface** of **keratinocytes** throughout the epidermis in **normal, perilesional skin,** in a **characteristic fishnet/chickenwire appearance** as seen below.





@Neet_pg_bot

References: Robbins and Cotran Pathologic Basis of Disease, 10e, Page no. 1160
Rook's Textbook of Dermatology 9th edition, Page no. 50.1

Q145. Dermatology AIIMS 2017 ID: MA1188

A 23-year-old patient came with complaints of non-scarring alopecia in the vertex region of his head. Histological examination shows loss of hair with haemorrhage into the follicular areas and mild perinuclear lymphocytic infiltrate. What is the diagnosis?

A Alopecia areata

B Telogen effluvium

C Tinea infection

D ✓ Trichotillomania

Correct Answer: Trichotillomania

The clinical scenario depicting a young adult presenting with **non-scarring alopecia** in the **vertex** region and histology showing **perifollicular hemorrhage** is suggestive of **trichotillomania** (trichotillosis).

Trichotillomania has to be differentiated from **alopecia areata** (Option A). Trichotillomania presents with **incomplete hair loss** within a patch with varying lengths of hair whereas alopecia areata shows **smooth hairless patch** with **sparing of white hairs**. Regular **pitting** of nails and **exclamation mark hairs** are seen in alopecia areata. When the diagnosis is uncertain, a scalp biopsy can be done. **Perifollicular hemorrhage** is characteristic of trichotillomania while **peribulbar lymphocytic infiltrate** is diagnostic of alopecia areata.

Trichotillomania is classified under **obsessive-compulsive and related disorders**. It is characterized by recurrent body-focused repetitive behavior (hair pulling) and repeated attempts to decrease or stop the behavior. It is commonly seen in **females**. The hair loss is commonly seen over the vertex but can involve eyelashes, eyebrows, and beard.

The most effective treatment for trichotillosis is **habit-reversal training** (HRT).

Other options:

Option B: **Telogen effluvium** is diffuse hair loss resulting in **non-scarring alopecia** due to loss of telogen hairs due to **fever**, pregnancy, hypothyroidism or anemia.

Option C: **Tinea infection** shows patchy hair loss, resulting in non-scarring/scarring alopecia. Easy pluckability, broken hairs and wood's lamp positivity are features.

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References: Rook's Textbook of Dermatology 9th edition, Page no: 86.17, 89.31
Fitzpatrick's Dermatology, 9th Edition., Page no: 1695

Q146. Dermatology AIIMS 2017 ID: MB9258

Which of the following diseases is unlikely to present with skin lesions on the shin?

A Sarcoidosis

B Diabetes

C Hyperthyroidism

D ✓ Hypothyroidism

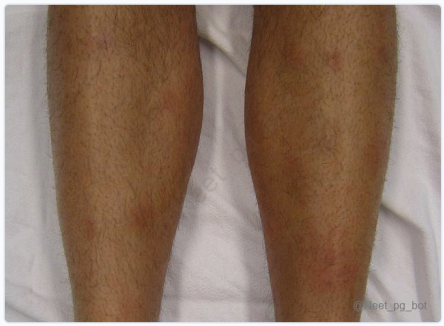
Correct Answer: Hypothyroidism

Hypothyroidism is unlikely to present with skin lesions on shin. **Myxedema** (severe **hypothyroidism**) and **pretibial myxedema** are different entities. **Pretibial myxedema** is the infiltrative dermopathy (mucin deposition) seen over shin in **Grave's disease**, a type of **hyperthyroidism**.

Other options :

Option B: **Diabetes**: Necrobiosis lipoidica diabeticorum is classically seen on the anterior leg.

Option A: **Sarcoidosis**: Erythema nodosum may be seen on the anterior leg.



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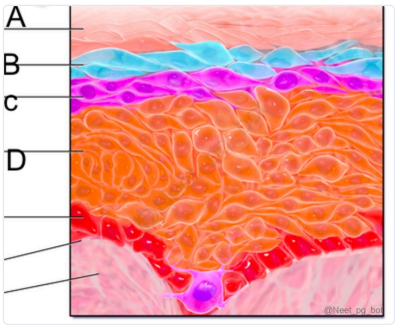
References: Rook's Textbook of Dermatology 9th edition, Page no. 59.11, 64.5, 98.6, 149.10

Q147. Dermatology AIIMS 2017 ID: MB9195

Identify the layer of skin (in the image given below) in which desmosomes are abundant.



A B

B A

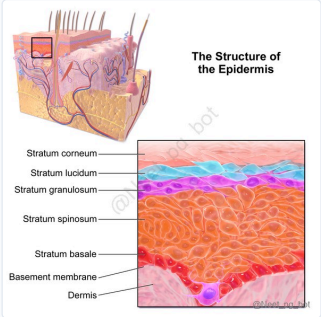
C ✓ D

D C

Correct Answer: D

Desmosomes are **abundant** in the **stratum spinosum** (Option C, label D in the image) layer of the epidermis of the skin.

Desmosomes are adhesion complexes in epidermis which link adjacent keratinocytes. They appear like spines on microscopy and hence this layer is also called the **prickle layer.** Defects of desmosomal proteins, inherited or acquired, result in blistering disorders.



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References: Rook's Textbook of Dermatology 9th edition, Page no: 2.1-2.2,2.18

Q148. Dermatology AIIMS 2017 ID: MB9230

The following image depicts:

A ✓ Scabies

B Ichthyosis

C Contact dermatitis

D Urticaria

Correct Answer: Scabies

The given image shows raised, brownish **thread-like** lesions in the **interdigital** distribution (**burrows**), typically seen in **scabies**.

Scabies is an ectoparasitic infection caused in humans by **Sarcoptes scabiei**. It is transmitted by **close personal contact**. **Pruritis** with **nocturnal exacerbation** and **contact cases** within a family should raise the suspicion of scabies. Typical locations of lesions are the finger webs, the flexor surfaces of the wrists, the elbows, the axillae, the buttocks and genitalia and the breasts of women (**Circle of Hebra**). In **adults**, the **face** is often **spared**. Pruritic papules and **nodules** on **male genitalia** and **breasts** serve as valuable diagnostic clues in the absence of burrows.

Dermoscopy of the burrows shows **jet-with-contrail** appearance. **Microscopy** of scraping from burrows can be done to demonstrate mites.

A single overnight application of **5% permethrin** cream from neck to toe in adults and from head to toe in children is the drug of choice for the treatment of scabies in all ages including infants (at least 2 months old) and pregnant women. **All family members**, even if asymptomatic, need to be treated simultaneously for this condition to prevent reinfection. Oral **ivermectin** can also be used.

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References: Fitzpatrick's Dermatology, 9th Edition., Page no: 3274
Rook's Textbook of Dermatology 9th edition, Page no: 34.39-34.43
Rook's Textbook of Dermatology 10th edition, Chapter 34, Page no: 34.41-34.48

Q149. Dermatology AIIMS 2017 ID: MB9180

A female came with complaints of bone and joint pain. NSAIDs were prescribed. After a few days, she presents again complaining vesicobullous lesions healing to give a brownish discoloration over the face (upper lip area) as shown in the image. This most probably suggests a diagnosis of

A Malaria

B Dengue

C Chikungunya

D ✓ Fixed drug eruption

Correct Answer: Fixed drug eruption

The patient has a history of **consumption of NSAIDs**, following which there is an appearance of **vesiculobullous lesions** on the **lips**, evolving to give a **brownish discolouration**. The most probable diagnosis is **fixed drug eruption** (FDE).

A strong differential diagnosis is **post-chikungunya pigmentation** (PCP) of the face seen on the **nose and cheeks (chik sign).** However, this is not the answer because there is no antecedent history of **fever**, **myalgia** or other symptoms suggestive of chikungunya. PCP often appears as the **fever subsides** and may persist for about 3-6 months. PCP mimics **melasma** and may be misdiagnosed if proper history is not taken.

The image below shows **chik sign**:

Fixed drug eruption (FDE) is a type of delayed hypersensitivity reaction. It is characterized by well-defined brown, **hyperpigmented plaques** that may follow vesiculobullous lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation. They usually present **30 min to 8 h** after drug exposure and **recur at the same location** on re-exposure. NSAIDs, Paracetamol and Co-trimoxazole are the most common drugs causing FDE.

@Neet_pg_bot

References: Harrison's Principles of Internal Medicine, 21st Edition, Pages 141-142, 1633
Rook's Textbook of Dermatology 9th edition, 118.11

Q150. Dermatology AIIMS 2017 ID: MB9229

Dense irregular collagen is found in:

A Ligaments

B ✓ Dermis

C Tendons

D Bones

Correct Answer: Dermis

Dense irregular collagen is found in **dermis**.

Irregular connective tissue is of three types: Loose, dense and adipose. Dense connective tissue is located in regions which are under **consistent mechanical stress** and where **protection** is required for the ensheathed organs. The abundant collagen fibres in the dermis run in a haphazard manner. This protects the skin.

It is usually present in :

- Reticular layer of the **dermis**
- Superficial **connective tissue sheaths** of the muscle and nerves
- **Adventitia** of large vessels
- **Capsules** of various organs and glands (e.g, testis, sclera, periosteum, perichondrium)

On the other hand, **dense regular collagen** fibres are found in **tendons.**

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References: Gray's anatomy: The anatomical basis of clinical practice - 42nd edition, Page no 40, 41

Q151. Dermatology AIIMS 2017 ID: MC9585

A patient presented with hair loss and itching of the scalp along with mild discharge. Which of the following is useful for diagnosis?

A Tzanck smear

B Gram stain

C Slit skin smear

D ✓ KOH mount

Correct Answer: KOH mount

The patient has a history of **itching** and **mild discharge** and examination shows a **well-demarcated** patch of **alopecia**, with **scaling** and **broken hair shafts.** This is characteristic of **tinea capitis.** Though the diagnosis is clinical, a **KOH mount** is used for confirmation.

Scrapings from the edge of a lesion are transferred to a slide, to which KOH is added. It is then examined under the microscope for the presence of **hyphae**, as shown below.

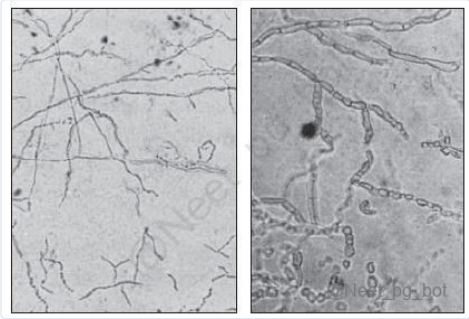
Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with **sinus** formation.

The clinical variants of tinea capitis are:

- **Endothrix** - a non-inflammatory type in which **multiple black dots** are present within the areas of alopecia. It is most commonly caused by **T. tonsurans and T. violaceum.**
- **Ectothrix** - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. **Microsporum audouinii**, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.
- **Kerion** - a severe inflammatory type that presents with an inflammatory mass with **thick crusting** and **matting of hair** with pus and sinus formation. It is commonly caused by **T. verrucosum and T. mentagrophytes**.
- **Favus** - is characterized by the presence of yellowish, cup-shaped crusts known as **scutula** that develop around hair follicles. It is caused by **T. schoenleinii**

Treatment:

- **Terbinafine**: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.
- **Itraconazole**: 2-4 mg/kg/day for 4-6 weeks.



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References: Rook's Textbook of Dermatology 9th edition, Page no: 32.8, 32.38 - 32.40

Q152. Dermatology AIIMS 2017 ID: MC9587

Identify the condition shown below:

A Epidermal verrucous nevus

B ✓ Congenital melanocytic nevus

C Melanoacanthoma

D Malignant melanoma

Correct Answer: Congenital melanocytic nevus

The image shows a child with a **well-circumscribed**, **black**, **heterogenous** lesion **covered with hair**. This is characteristic of a **congenital melanocytic nevus (CMN).** They may be present at birth or develop shortly thereafter. They are **benign** and can be brown, black, purplish, or red, and are usually a **palpable lesion.**

They can be **classified** based on size: small (<1.5 cm), medium (1.5-19.9 cm), and large or giant (>20 cm). **Larger CMN have a higher risk of developing malignant melanoma.**

Management of CMN depends on the size and location of the lesion. Smaller nevi may be observed, while larger nevi or those at a higher risk for complications may require surgical excision, particularly if there is a concern for malignant transformation.

Complications of CMN include neurological abnormalities (**intraparenchymal melanosis** is the commonest finding), **malignant melanoma**, and other tumors like **rhabdomyosarcoma.**

The images below show different melanotic skin lesions:

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References: Rook's Textbook of Dermatology 9th edition, Page no.75.9 - 75.15